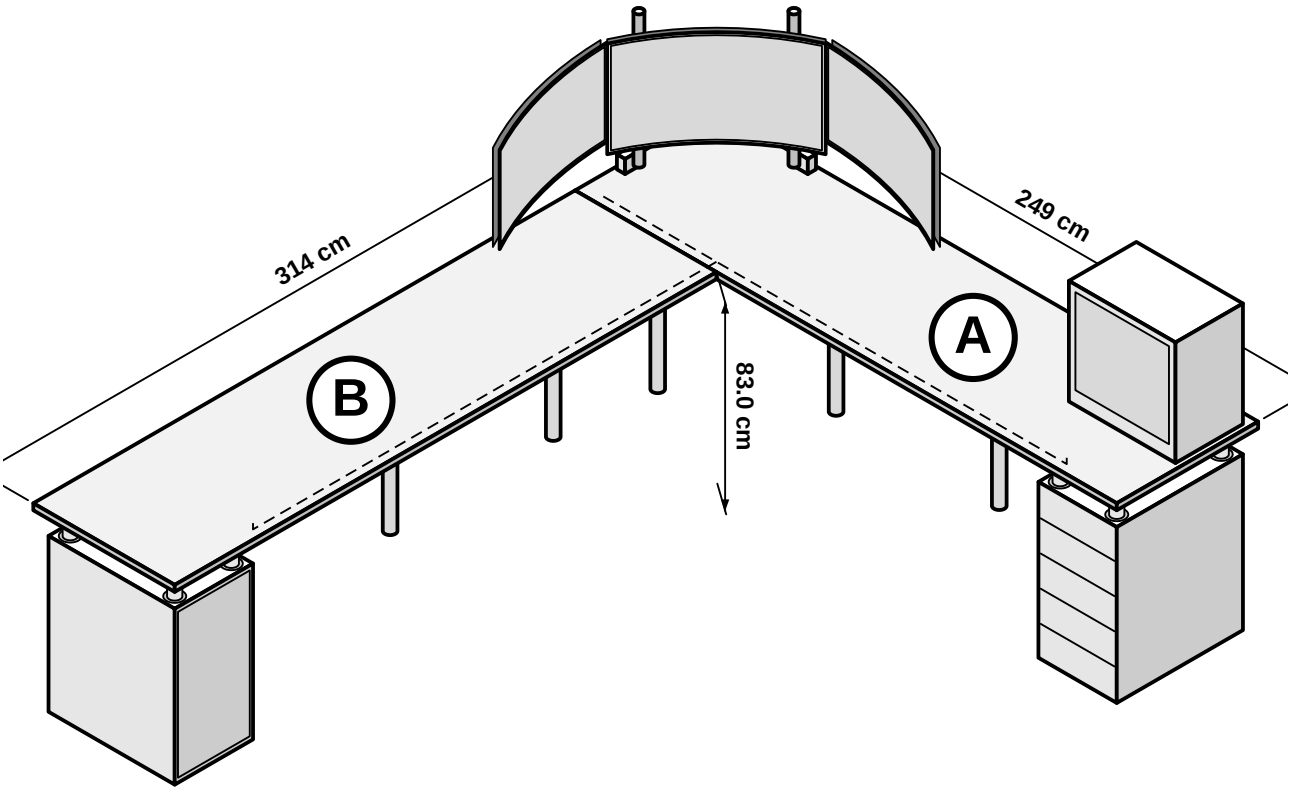
L-Shaped Corner Desk

Assembly manual



Footprint	249 × 314 cm
Surface height	83.0 cm
Tops	A along the back wall, B along the side wall (two IKEA EKBACKEN, each 249 × 65 cm, 28 mm thick)
Floor needed	252 × 317 cm, including the 3 cm wall gaps

Tape-measure sizes in cm. The rails are hidden under the front edges and are drawn dashed.

Contents

A · Finished assembly 3
B · Before you begin 5
B · Printed tools: how each one is used 8
C · Complete parts inventory 14
D · Component preparation 17
E · Steel rail fabrication and drilling 27
F · Steel corner assembly 37
G · Legs and cabinet support 41
H · Joining and positioning the countertops 51
H · 8b Wall-gap trim (optional) 56
I · Monitor mount and accessories 59
J · Final inspection and load testing 64
K · Maintenance and troubleshooting 70
Appendix · Outstanding measurements 72

A · Finished assembly

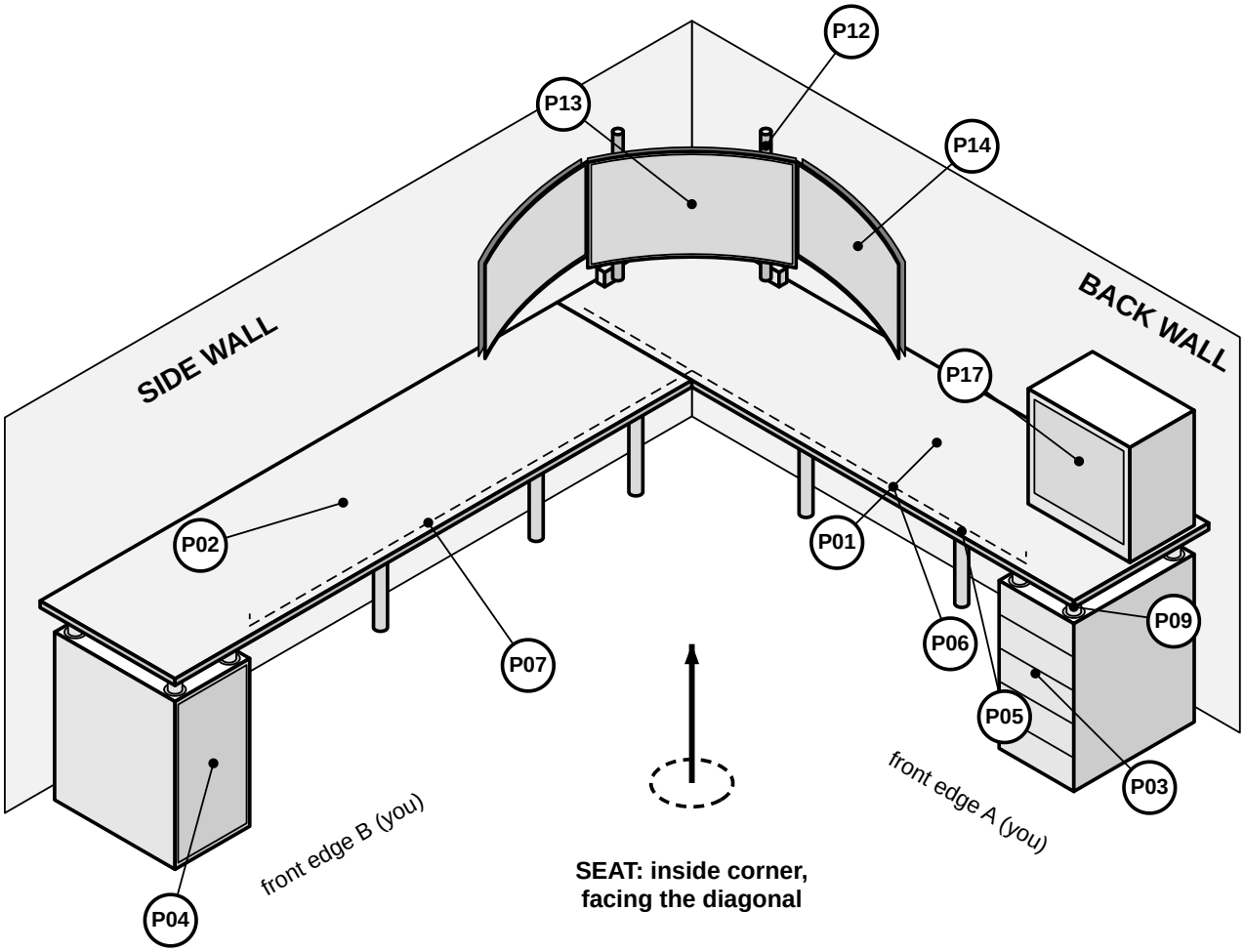


Fig. A-1 The finished desk seen from the room. Dashed lines under the front edges: the two steel rails, hidden by the tops. Left and right are as you sit at the inside corner facing the diagonal: top A is on your right, top B on your left.

No.	Part	No.	Part
P01	Top A, EKBACKEN 249 × 65 cm, 28 mm thick	**P07**	Rail B, steel angle 213 cm
P02	Top B, EKBACKEN 249 × 65 cm, 28 mm thick	**P09**	Round adjustable foot 4 in x 8
P03	ALEX 1, drawer unit 36 × 58 × 70 cm	**P12**	VIVO STAND-V300CU
P04	ALEX 2, storage unit with door 36 × 58 × 70 cm	**P13**	Samsung Odyssey G5 32 in
P05	OLOV leg x 6, 105 mm plate	**P14**	Acer XZ320QU x 2
P06	Rail A, steel angle 213 cm	**P17**	HYTE Y60 on printed pedestal

What this desk is

Two IKEA kitchen tops make an L. Top A lies along the back wall and keeps the corner square; top B lies along the side wall and its end butts against top A's front edge. Each top stands 3 cm off its wall, with its square, unfinished long edge toward the wall. The far end of each top rests on an ALEX cabinet through four adjustable feet: the feet are screwed to the top, and their pads drop into printed rings on the cabinet, so the desk cannot slide on the cabinets and the two heavy boxes hold the whole frame square. Six OLOV legs stand along the two wall edges, none at the front, so the space under the inside corner is open for your knees.

That leaves the two front edges with nothing under them for more than two metres, so each one is carried by a steel angle rail screwed to the underside 12 cm back from the front edge. Under the inside corner the two rails are bolted together with two gusseted steel fittings, so the corner of top B hangs from rail A. The rails never touch a cabinet: their ends load the top, and the top loads the nearest feet. The monitor stand clamps to top A's two wall edges, 40 cm from the corner; the surface is 83.0 cm above the floor.

A · Plan of the finished desk

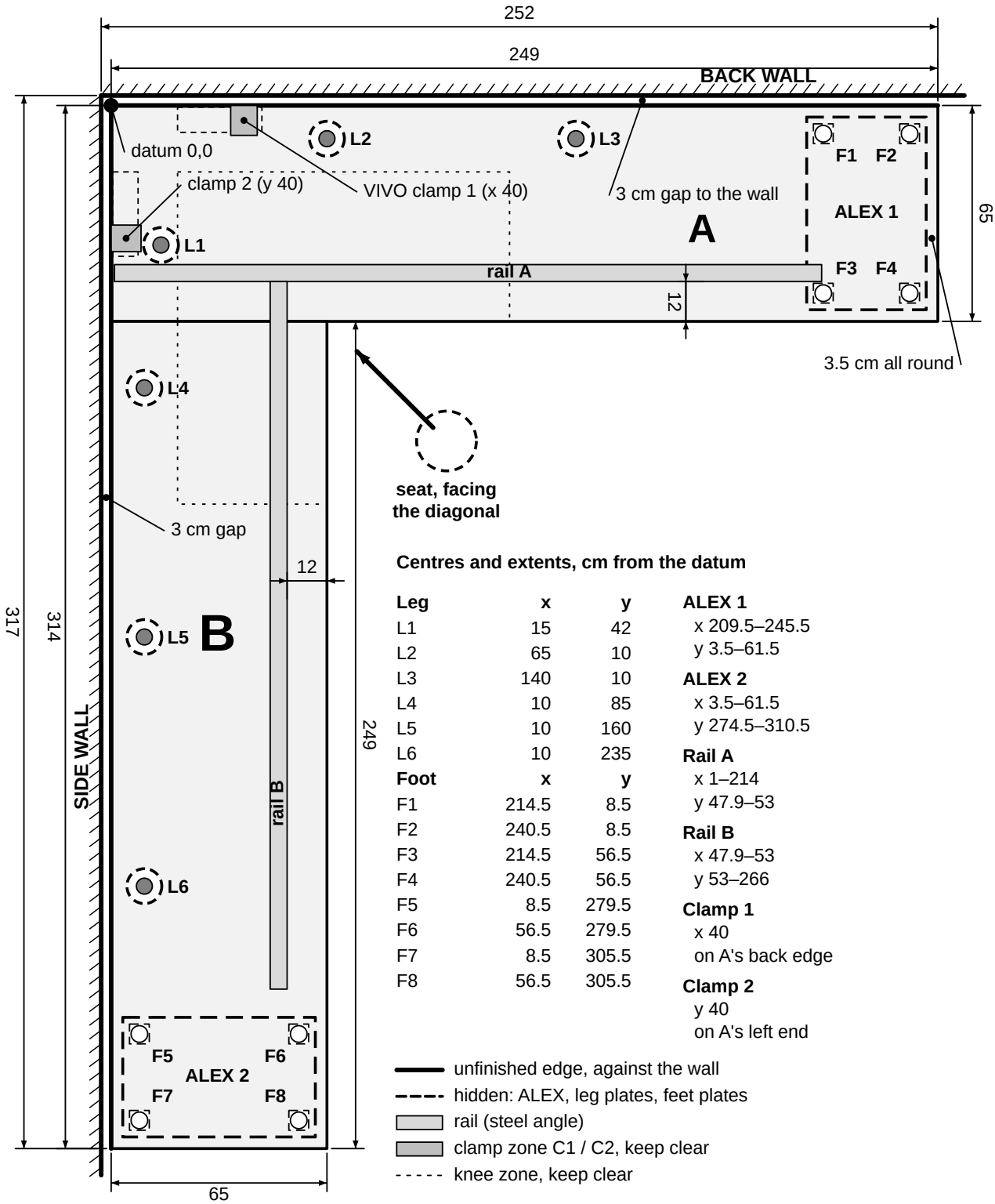


Fig. A-2 Top-down plan, seen from above. Datum: top A's back-left corner (0, 0); x runs along the back wall, y along the side wall, all in cm. Coordinates SOURCE-DEFINED (docs/build-manual.md). ALEX 1 drawers and ALEX 2 door face you.

Height above the floor	cm	Confidence
Top surface	83.0	SOURCE-DEFINED
Underside of the tops (OLOV setting)	80.2	SOURCE-DEFINED
Rail lower edge	75.1	SOURCE-DEFINED
ALEX top (the feet stand on it)	70.0	ASSUMPTION / FIELD CHECK measure your units

B · Before you begin

Read this chapter once before you pick up a drill. Budget a weekend: the steel on day one, the assembly on day two.

Safety

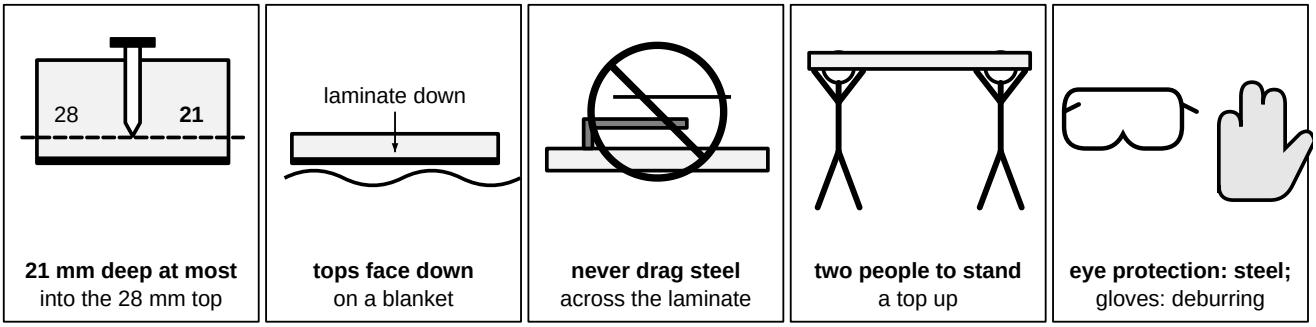
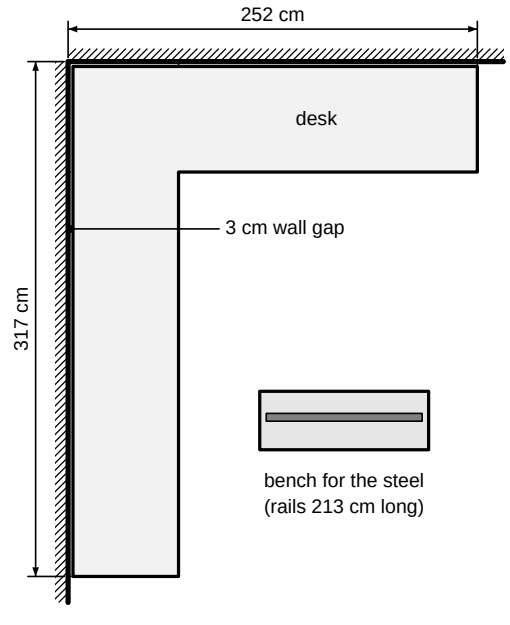


Fig. B-1 · The five rules that protect the tops and you.

> **NEVER A FASTENER DEEPER THAN 21 MM INTO A TOP**
> The EKBACKEN tops are 28 mm particleboard with a thin laminate skin. A screw or pilot that reaches the skin lifts it into a bump you will see and feel for ever. Every screw in this manual stops at least 7 mm short of the top face (28 − 21). If your tops measure 27 mm instead of 28, shorten every screw into the top by 1 mm. If you ever substitute a screw, keep this rule. Chapter D shows every depth.

- **Laminate protection.** Spread a blanket on the floor first. The tops lie face down on it, laminate on the blanket, for every marking and drilling step. Never slide or drag steel (rails, fittings, the clamp plates) across the laminate: lift it on and off.
- **Eye protection** for every hole drilled in steel; drill at low speed with cutting oil. **Gloves** while you deburr and file the steel: drilled edges are sharp.
- **Two people** to turn a top over and stand it on its legs (249 cm long). A helper is also needed to slide the finished desk off and back to the walls.

Work area



Plan, 1:40. Both walls MEASURE.

Fig. B-2 · Floor needed: 252 × 317 cm, including the 3 cm gaps to both walls.

The finished desk needs **252 × 317 cm** of clear floor in the corner: each top is 249 cm long and stands 3 cm off its wall. MEASURE ON HARDWARE Measure both walls (wall to wall) and the baseboard thickness before you start.

You also need floor space for one top at a time lying face down on the blanket (249 × 65 cm plus room to kneel all round), and a **bench** with two clamps for drilling the two 213 cm steel rails. All steel is drilled on the bench, never under the desk.

Clear the corner before chapter G: the ALEX units are placed by the walls before the tops go on.

Workshop tools

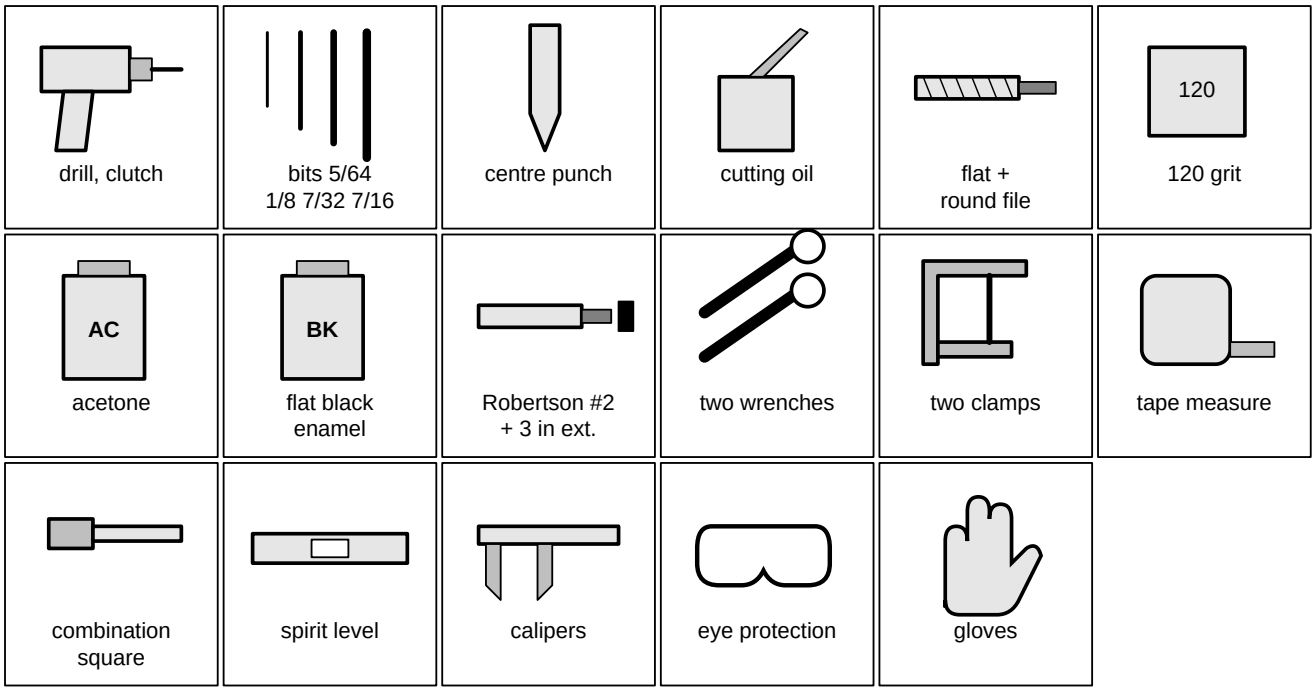


Fig. B-3 · Tools and protective equipment. Not to scale.

Tool	For
Corded or 18 V drill, clutch set low	all holes and screws
Bits: 5/64, 1/8, 7/32 in HSS; 7/16 in cobalt	ring and trim pilots; top and steel pilots; rail screw holes; fitting bolt holes
Centre punch (or a nail)	only for a mark the jig or plug cannot reach
Cutting oil or 3-in-1	every hole in steel
Flat file, round file, 120 grit, acetone, flat black enamel	deburr, ease a tight hole, scuff, degrease, paint the rails
Robertson #2 bit + 3 in magnetic extension	rail screws beside the vertical leg (the chuck would hit the leg otherwise)
Two wrenches or sockets for the 3/8-16 bolts (one for the head, one for the nut)	the corner fittings
Two clamps, 6 in or larger	hold each rail on its line; hold a fitting as a drilling template
Tape measure, combination square, masking tape, pencil	layout and the tape labels on the tops and rails
Spirit level, 60 cm or longer	setting the legs and feet
Calipers	top thickness, pad diameter, ALEX panel (the measurements on the last page of this chapter)

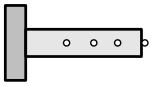
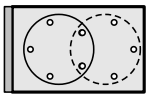
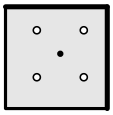
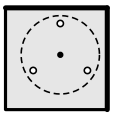
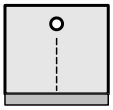
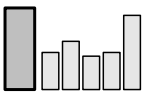
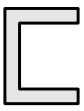
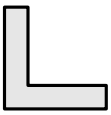
Protective equipment

- Safety glasses or goggles: every hole in steel, and opening the 7/16 in holes.
- Work gloves: deburring, filing and handling the cut rail ends.
- A blanket large enough for a whole top (249 × 65 cm).

Printed tools (P18): print these before you mark anything

All of them are in `parts/tools/`; `parts/README.md` has print settings. Print as saved, no supports. **Never scale a printed tool in the slicer: its distances are the tool.** The lipped and fenced tools print lip-up and are turned over to use.

> **EVERY JIG HOLE IS BIT-SIZED**
> Hold the jig on, start each pilot 3–4 mm through the jig's hole, lift the jig, then finish the hole to depth with the drill guide's puck. The small centre holes are for a scriber only; nothing is drilled through them.

Tool	What it does	Qty
	**Edge T-square.** Lip over any edge; holes named for their distance from the edge in mm: 50 (ring centres on the ALEX, 5 cm in), 85 (feet, 8.5 cm in, with the plate's four 1/8 in holes 46.8 mm apart round it), 100 (legs L2–L6, 10 cm in) and 150 (leg L1, 15 cm in).	1
	**Leg jig.** Lip over the wall edge; an OLOV's centre and its five 1/8 in holes in one placement: 100 pattern for L2–L6 (solid circle), 150 pattern for L1 (dashed circle, ringed holes).	1
	**Feet corner jig.** Fences on a far corner of a face-down top; a foot's centre 8.5 cm from both edges and its four 1/8 in holes 46.8 mm apart: F2, F4, F7, F8.	1
	**ALEX corner template.** Fences on an ALEX corner; ring centre 5 cm from both edges, the ring's three 5/64 in holes (radius 33 mm) and the three optional ALEX-foot holes (radius 14.5 mm). Top and bottom corners.	1
	**Rail marking gauge.** Lip over the angle's edge; pencil notches at 19 mm (screw line on the flat leg) and 29 mm (bolt line on the vertical leg).	1
	**Rail drill jig.** Lip over the flat leg's free edge; a 1/8 in bore on the 19 line; sight grooves on the station line. PETG.	1
	**Slot plug** (fitting-centring-plug). Drops into a fitting's 21 × 12 slot; a 1/8 in bore dead centre, so the pilot cannot wander in the slot's play.	2
	**Pilot drill guide.** SET tower (45 mm for the 1/8 in bit, 35 mm for the 5/64 in) and five depth pucks: 12 PLATE, 15 LEG, 10 EDGE, 12 ALEX, 25 RAIL. Re-set the bit every time you change it.	1 set
	**Foot height gauge.** A C with 102 mm inside: all eight feet wound to the same height without a ruler.	1
	**Wall gauge.** An L: the 65 leg (6.5 cm, ALEX back to wall) and the 30 leg (3 cm, a top's raw edge to wall).	1

Also print the eight locating rings (P10) now: `parts/structure/foot-ring`, PETG or ABS, flange down. Their 51 mm bore depends on the pad diameter you measure (last page of this chapter).

Printed tools: how each one is used

1 · Edge T-square

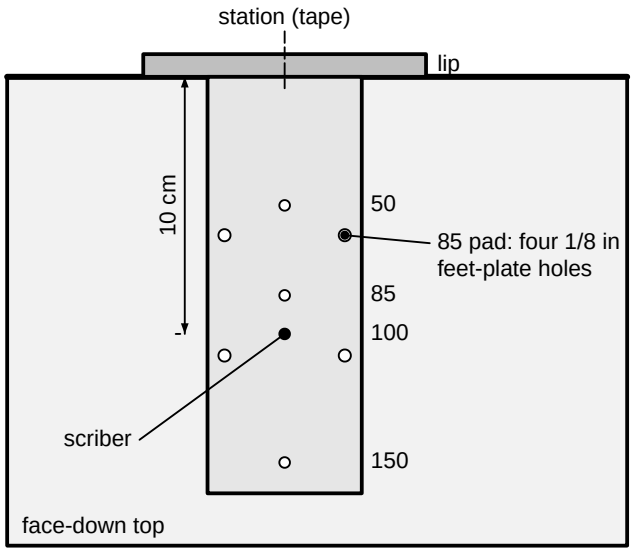


Fig. B-4 · The T-square on a face-down top, lip over the edge, at its station. Hole names are mm from the edge.

1. Turn it over so the lip hangs down. Hook the lip over the edge named in the marking table.
2. Slide the head along the edge until the blade's centre line sits on the station you ticked with the tape.
3. Put the scriber through the hole the table names: **50** ring centres on the ALEX (5 cm in), **85** feet (8.5 cm in), **100** legs L2–L6 (10 cm in), **150** leg L1 (15 cm in). Each hole is named for its distance from the edge in mm.
4. The four holes round **85** are 1/8 in drill holes for a feet plate (46.8 mm apart): start the four pilots through them, lift, finish with the 15 LEG puck.
5. **First-print check:** hook it over a top's edge: the 100 hole must sit 10.0 cm from the edge. If not, re-print it, never scaled.

2 · Leg jig

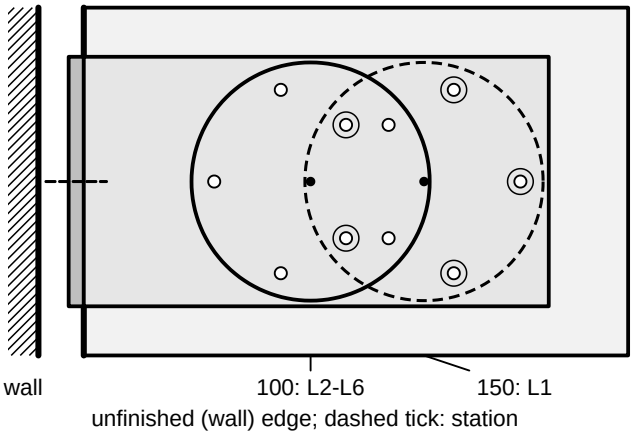


Fig. B-5 · The leg jig hooked over the wall (unfinished) edge.

1. Turn it over (lip down). Hook the lip over the wall edge and slide it to the leg's station.
2. **L2–L6:** the **100** pattern: plain holes, solid circle, one hole toward the wall. **L1:** the **150** pattern: ringed holes, dashed circle.
3. Start the five 1/8 in pilots 3–4 mm deep through the pattern's holes.
4. The small centre hole is for a scriber only; never drill it.
5. Lift the jig and finish each pilot with the **15 LEG** puck.

3 · Feet corner jig

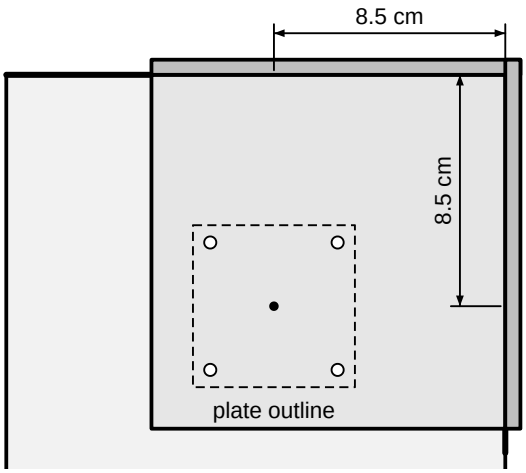


Fig. B-6 · The feet corner jig on a far corner of a face-down top, fences against both edges.

1. Turn it over (fences down). Drop it over a **far corner** of the top so both fences bear on the two edges.
2. It gives the foot centre 8.5 cm from both edges and the plate's four holes, 46.8 mm apart, square to the edges.
3. Start the four 1/8 in pilots through it, lift it, finish with the **15 LEG** puck.
4. It does **F2, F4, F7, F8**. The other four feet (F1, F3, F5, F6) are 34.5 cm from the end: the T-square's 85 pad does those, at the tape stations in chapter D.

4 · ALEX corner template

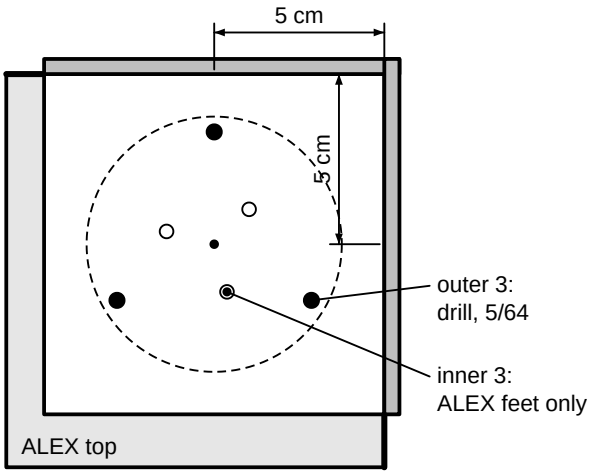


Fig. B-7 · The template on an ALEX top corner. Black: the three ring holes you drill. White: the ALEX-foot holes.

1. Turn it over (fences down). Drop it over a corner of a cabinet's **top**, fences against both edges: ring centre 5 cm from both edges.
2. Start the **three outer** pilots (5/64 in, radius 33 mm) through it. Lift it, finish with the **12 ALEX** puck.
3. The **inner three** (radius 14.5 mm) are only for the optional ALEX feet, marked on the **bottom** corners. Do not drill them on the top.
4. Set the ring on the corner and turn it until its three countersunk holes sit on the three pilots; it only fits one way round. Same template for all eight corners.

5 · Rail marking gauge

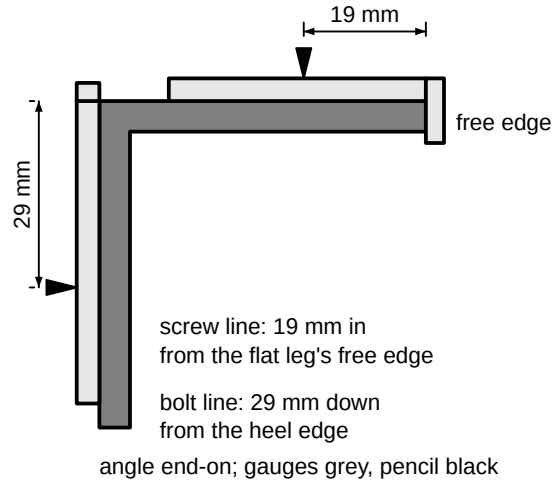


Fig. B-8 · The angle end-on with the gauge in its two positions.

1. Turn it over (lip down). Lay it on a face of the angle with the lip over an edge; the lip's inner face is the zero.
2. **Screw line:** lip over the flat leg's free edge, pencil in the **19** mm notch, slide the full length.
3. **Bolt line:** lip over the vertical leg's heel edge, pencil in the **29** mm notch, slide.
4. Use a 0.8 mm pencil point: the notches put it on exactly 19.0 and 29.0 mm.

6 · Rail drill jig

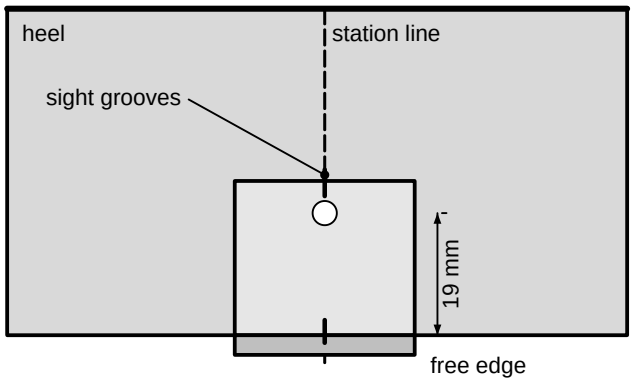


Fig. B-9 · The rail drill jig on the flat leg (seen from above), lip under the free edge.

1. Turn it over (lip down). Hook the lip over the flat leg's free edge.
2. Slide it until the sight grooves on both sides sit on the station line.
3. Hold it down and drill the 1/8 in pilot through the bore: it is on the 19 mm screw line, square, and cannot wander.
4. Pull the jig off. Open the hole to 7/32 in without the jig; the bit follows the pilot. Nineteen holes, one jig.

7 · Slot plug (fitting-centring-plug)

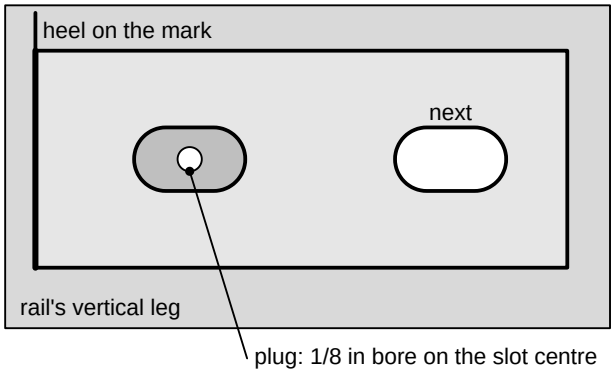


Fig. B-10 · A fitting clamped on the rail's vertical leg, heel on its mark, plug in the first slot.

1. Clamp the fitting to the rail with its heel on the mark from chapter E.
2. Drop a plug into the first slot. Drill the 1/8 in pilot through its bore. Move the plug to the next slot and repeat.
3. Take the fitting off. Open each pilot to 7/16 in; the bit follows the pilot.
4. **Why:** the slots are 21 × 12 mm, so a bolt or a punch could sit anywhere along the slot. The plug fills it and puts every pilot on the slot centre.

8 · Pilot drill guide

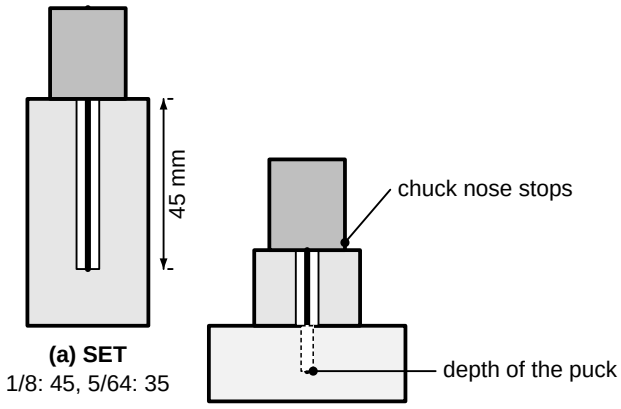
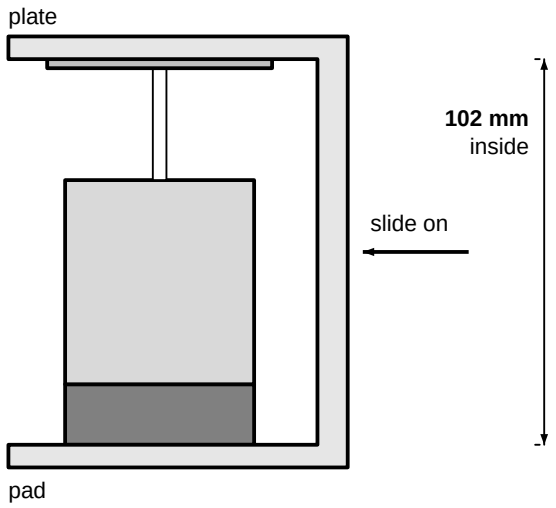


Fig. B-11 · (a) Setting the bit in the SET tower. (b) Drilling through a puck.

1. **Set the bit:** loosen the chuck, drop the bit tip-first into the SET tower's hole for it (45 mm for 1/8 in, 35 mm for 5/64 in), rest the chuck nose on the tower, tighten.
2. **Drill:** stand the puck for that hole flat on the surface (or on the plate, or on the rail's flat leg), bit down its bore, drill until the chuck nose lands on the puck.
3. **Pucks:** the number is the hole depth in mm. 12 PLATE (1/8 in, joining plates) · 15 LEG (1/8 in, OLOV and feet plates) · 10 EDGE (5/64 in, trim screws sideways into the raw edge, bit horizontal) · 12 ALEX (5/64 in, rings and ALEX feet) · 25 RAIL (1/8 in, through the angle, 20 mm into the wood).
4. Re-set the bit every time you change it. A domed chuck nose: drill one test hole in scrap and measure it.

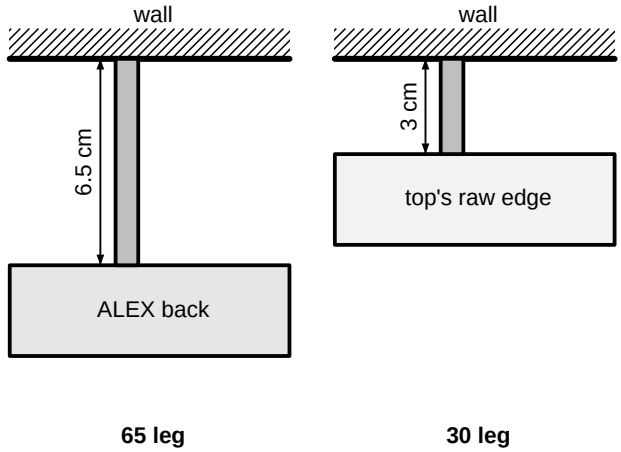
9 · Foot height gauge



1. Stand the foot on its pad.
2. Slide the C over it from the side.
3. Wind the pad until the gauge just slides on with no rock: 102.0 mm inside.
4. All eight feet are then identical, no ruler.

Fig. B-12 · The C gauge on a round foot: plate face to pad bottom.

10 · Wall gauge



1. Hold the **65** leg (6.5 cm) between the wall and a cabinet's back while you place it, and again when you slide the desk back after the trim.
2. Hold the **30** leg (3 cm) between the wall and a top's raw edge to check the 3 cm gap.
3. No tape needed: it just fits, or it does not.

Fig. B-13 · The two legs of the wall gauge in the gap.

THE RULE FOR EVERY PRINTED TOOL
Every jig hole is bit-sized: start the pilot 3–4 mm through the jig, lift the jig, finish the hole to depth with the puck. Centre holes are for a scriber only. Never scale a tool in the slicer: its distances are the tool.

How sure is each number?

Every dimension in this manual carries one of three confidence labels, or a fourth label that is an instruction to you:

VERIFIED	Measured on the purchased part, or taken from the part's own geometry file (the printed parts).
SOURCE-DEFINED	Defined by the project documents (the build manual): a design position, not a measurement. It is right by definition; you put the part there.
ASSUMPTION / FIELD CHECK	An engineering estimate or a check done on site. Confirm it before you rely on it.
MEASURE ON HARDWARE	Not yet measured on your parts. Measure it yourself before the step that uses it, and stop if it differs.

Confirm before you drill anything

> **MEASURE BEFORE DRILLING**
> These numbers are not yet confirmed on the parts you have. Each one is checked again where it is used; if one differs, stop and correct the layout before any hole is drilled.

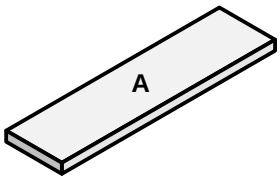
Measure	Expected	Label	Used in
EKBACKEN thickness, both tops (calipers)	28 mm	MEASURE ON HARDWARE	D: every screw depth into the tops. At 27 mm, shorten every screw into the top by 1 mm.
Delivered rail lengths, both pieces, and the angle thickness	213 cm layout length (bought as 84 in = 213.4 cm); 4.8 mm thick	MEASURE ON HARDWARE	E: hole stations. 213 cm is the layout length; the far end may carry the extra.
Strut fitting slots, positions from the heel	21 × 12 mm slots, centres 29.15 and 78.45 mm from the heel (legs 100.5 × 41 mm)	VERIFIED	E, F: measured 23 Oct 2026; the fitting with the slot plug is still the drilling template.
Feet pad diameter (calipers)	50 mm	MEASURE ON HARDWARE	G: the rings are bored 51 mm for a 50 mm pad; re-print them if yours differs.
ALEX top panel thickness, at an edge	16 mm or more for the 5/8 in screws	MEASURE ON HARDWARE	G: ring screws, 13 mm into the ALEX top.
ALEX height, with and without its stock feet	70.0 cm	ASSUMPTION / FIELD CHECK	G: with the feet it sets the underside at 80.2 cm.
VIVO top arm reach; maximum clamp opening	reach about 89 mm ASSUMPTION / FIELD CHECK; opening 31 mm or more	MEASURE ON HARDWARE	I: the clamps on the 28 mm top plus a pad plate.
Room: wall to wall along both walls; baseboard thickness	252 cm and 317 cm	MEASURE ON HARDWARE	B (work area), G (placing the ALEX units by the walls).
Feet load rating	unknown; the design wants 50 kg or more each	ASSUMPTION / FIELD CHECK	C (P09); check the listing or the maker.

Already measured and printed as VERIFIED: the OLOV plate (ø105 mm, five 5.5 mm holes on an 85 mm circle), the feet plate (59.5 mm, four 5.2 mm holes 46.8 mm apart), the strut fittings (21 × 12 mm slots at 29.15 and 78.45 mm from the heel), the VIVO crossbar (50.2 cm centre to centre), pole (35 mm), pole inset (43.7 mm), clamp back (80 × 17.5 mm) and lower pad.

C · Complete parts inventory

Check every part against this list before you start. Part numbers P01–P21 and hardware numbers H01–H11 are used in every callout balloon in this manual. Lengths a tape measures are in cm; thicknesses, holes, plates and screws in mm. REQUIRED parts are needed for the desk to stand; OPTIONAL parts can be left out.

thickness drawn ×4



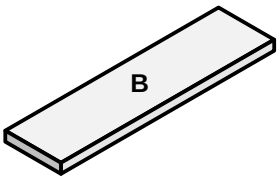
P01 **× 1**

Top A, IKEA EKBACKEN brown walnut, 249 × 65 cm, 28 mm thick. Front long edge rounded and laminated; back long edge square and unfinished (wall side).

MEASURE ON HARDWARE thickness

REQUIRED

thickness drawn ×4

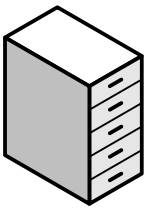


P02 **× 1**

Top B, identical to P01: 249 × 65 cm, 28 mm thick. Becomes B by being laid along the side wall (chapter D).

MEASURE ON HARDWARE thickness

REQUIRED



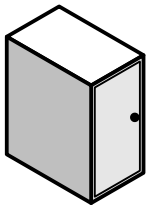
front

P03 **× 1**

ALEX 1, IKEA drawer unit, 36 × 58 × 70 cm. Under top A's far end, drawers toward you.

ASSUMPTION / FIELD CHECK height 70 cm

REQUIRED



front

P04 **× 1**

ALEX 2, IKEA storage unit with door, 36 × 58 × 70 cm. Under top B's far end, door toward you.

ASSUMPTION / FIELD CHECK height 70 cm

REQUIRED

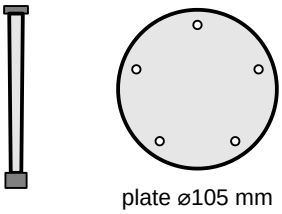
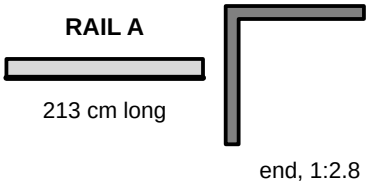


plate ⌀105 mm

P05 **× 6**

IKEA OLOV leg, adjustable 60–90 cm, set to 80.2 cm. Plate ⌀105 mm, five ⌀5.5 mm holes on an ⌀85 mm circle. Max 13 kg each. VERIFIED plate

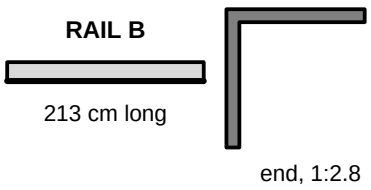
REQUIRED



P06 **× 1**

Rail A, 2 × 2 × 3/16 in hot-rolled steel angle (50.8 × 50.8 × 4.76 mm), 213 cm layout length (bought as 84 in = 213.4 cm). MEASURE ON HARDWARE length

REQUIRED

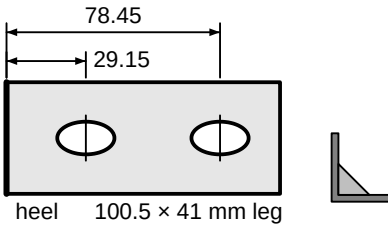


P07 **× 1**

Rail B, identical to P06 until its corner end is taped (chapter D).

MEASURE ON HARDWARE length

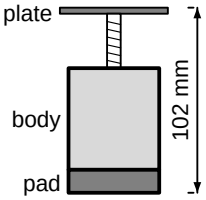
REQUIRED



P08 **× 2**

Gusseted 90° strut fitting, 5 mm steel, for 1-5/8 in channel. Legs 100.5 × 41 mm; two 21 × 12 mm slots per leg, centres 29.15 and 78.45 mm from the heel. VERIFIED

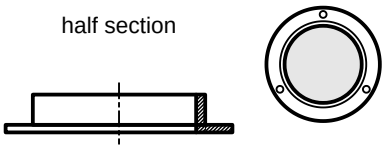
REQUIRED



P09 **× 8 (+4)**

Round adjustable foot, 4 in, wound to the 102 mm gauge. Body and pad ⌀50 mm; plate 59.5 mm square, four ⌀5.2 mm holes 46.8 mm apart. 12 on hand: 8 used, 4 SPARE. VERIFIED plate · MEASURE ON HARDWARE pad · load ASSUMPTION / FIELD CHECK

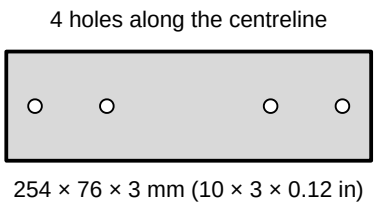
REQUIRED



P10 **× 8**

Locating ring, printed PETG or ABS: bore 51 mm, OD 57 mm, ring 10 mm high on a ⌀75 × 2.5 mm flange, three countersunk ⌀3.8 mm holes on radius 33 mm. **VERIFIED** geometry file

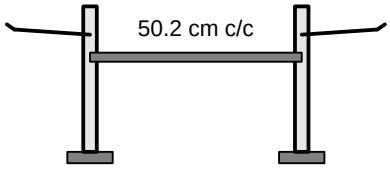
REQUIRED



P11 **× 4**

Mending plate, 10 × 3 × 0.12 in (254 × 76 × 3 mm), holes at 2.0, 7.0, 18.4, 23.4 cm from one end. Two join the tops (P1, P2); two pad the VIVO clamp jaws. **SOURCE-DEFINED**

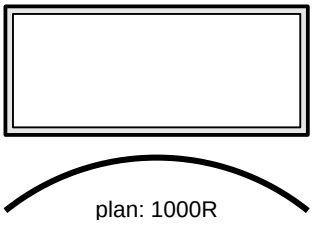
REQUIRED



P12 **× 1**

VIVO STAND-V300CU: two clamps, two ⌀35 mm poles 61 cm (24 in) tall, crossbar 50.2 cm centre to centre, two side arms, centre bracket. Clamp back 80 mm along the edge, 17.5 mm behind it. **VERIFIED**

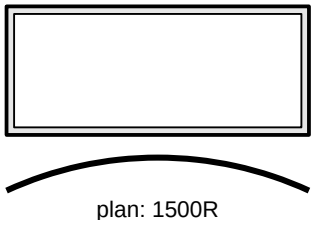
REQUIRED



P13 **× 1**

Samsung Odyssey G5 32 in, 1000R, VESA 75, about 5 kg. Centre screen, landscape, bottom edge 23 cm above the top. **SOURCE-DEFINED**

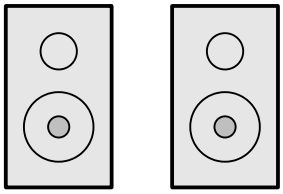
REQUIRED



P14 **× 2**

Acer XZ320QU 31.5 in, 1500R, VESA 100, 5.6 kg with stand (about 5 kg on the arm). Side screens, landscape. **SOURCE-DEFINED**

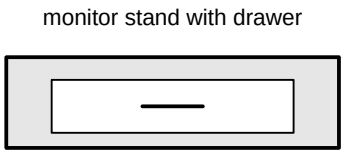
REQUIRED



P15 **× 2**

Edifier R1280DBs speakers, 23.4 cm tall, one under each Acer. **SOURCE-DEFINED**

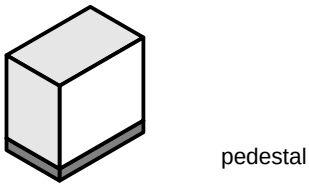
REQUIRED



P16 **× 1**

IKEA ELLOVEN monitor stand with drawer, 10 cm tall, under the G5. **SOURCE-DEFINED**

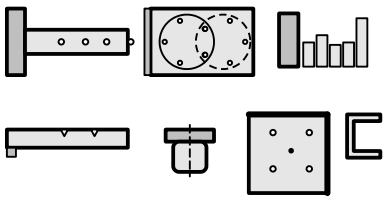
REQUIRED



P17 **× 1**

HYTE Y60 on its printed pedestal (MakerWorld 679012): footprint 49 × 31 cm, 48 cm tall with the pedestal, over ALEX 1. **SOURCE-DEFINED**

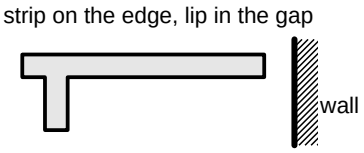
REQUIRED



P18 **1 set**

Printed tools: T-square, leg jig, feet corner jig, ALEX corner template, rail gauge, rail drill jig, slot plug × 2, drill guide (tower + 5 pucks), foot height gauge, wall gauge. List in chapter B.

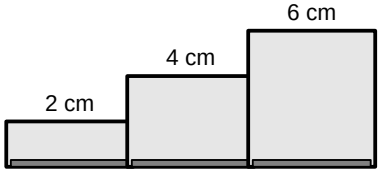
REQUIRED



P19 **1 set**

Wall-gap trim, printed: corner, fillers, clamp pieces, segments, connector tiles. Cut list in `parts/README.md`. Fixed with H07.

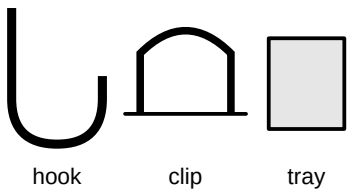
OPTIONAL



P20 **× 8**

ALEX feet, printed: ⌀56 mm, one height for all (2, 4 or 6 cm), with TPU pads. Raise the whole desk by their height.

OPTIONAL



P21 **1 set**

Cable parts, printed: rail cable hooks (magnets H11), cable clips, power-bar cradles, cable trays.

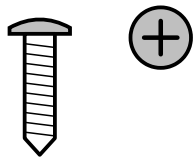
OPTIONAL

Drawings are silhouettes for recognition, not to scale unless a scale is written on them.

Hardware H01–H11

Screws and bolts are drawn **1:1** (true length, nominal diameter): lay a screw on the drawing to check it. Washers and the nut are symbols. All hardware was ordered 22 Oct 2026 SOURCE-DEFINED. Buy a few spares of each screw.

H01 × 30

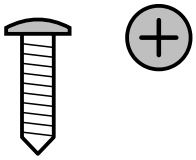


1:1 · head as supplied

M4 × 16 machine screw, fully threaded, IKEA 108443 (with the legs). OLOV plates, 5 per leg.

REQUIRED

H02 × 32

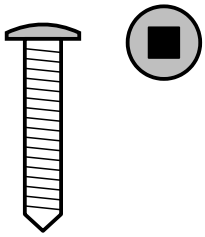


1:1 · pan head

#8 × 5/8 in pan head. Feet plates, 4 each.

REQUIRED

H03 × 19

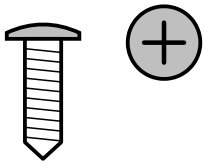


1:1 · pan head, Robertson

#10 × 1 in pan head, Robertson. Rails into the tops: 9 in rail A, 10 in rail B.

REQUIRED

H04 × 8

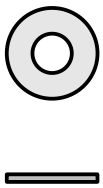


1:1 · pan head

#10 × 5/8 in pan head. Joining plates P1, P2, 4 each.

REQUIRED

H05 × 8

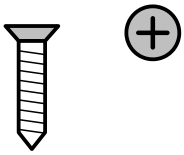


symbol, not to scale

#10 flat washer, under each H04.

REQUIRED

H06 × 24

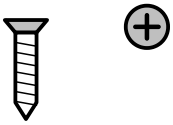


1:1 · flat head

#6 × 5/8 in flat head. Rings, 3 each. +24 for ALEX feet, more for cable parts.

REQUIRED

H07 × 48

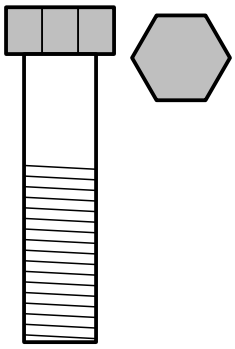


1:1 · flat head

#4 × 1/2 in flat head. Wall-gap trim, two per piece.

OPTIONAL

H08 × 6

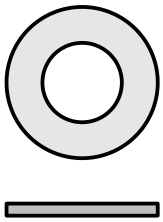


1:1 · hex head

3/8-16 × 1-1/2 in hex bolt, grade 5. Corner fittings.

REQUIRED

H09 × 12

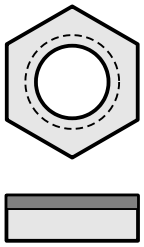


symbol, not to scale

3/8 flat washer, under each bolt head and each nut.

REQUIRED

H10 × 6

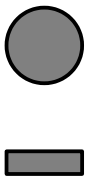


symbol, not to scale

3/8-16 nylon-insert lock nut. Stiff to turn; that is normal.

REQUIRED

H11 × 8



1:1

10 × 3 mm neodymium disc magnet, two per rail cable hook.

OPTIONAL

D · Component preparation

This chapter marks both tops face down and labels the two rails. Nothing is fixed yet. Positions are in **cm**, as the tape reads them; thicknesses, holes and screws in **mm**. Every position is measured from one **datum**: **top A's back-left corner**. **x** runs along the back wall, **y** along the side wall. Top A covers x 0–249 cm, y 0–65 cm; top B covers x 0–65 cm, y 65–314 cm.

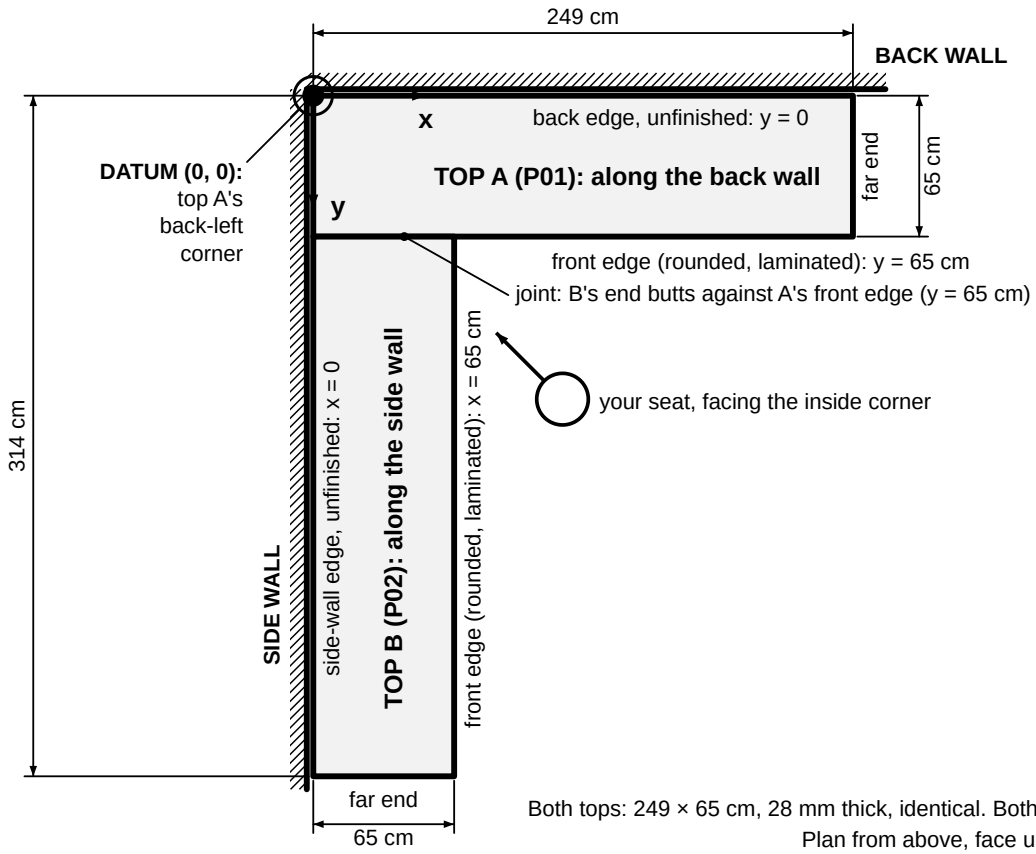


Fig. D-1 · The datum and the two tops, seen from above as installed. SOURCE-DEFINED

Which edge is which

Each top has one **rounded, laminated** long edge: the **front**. The other long edge is **square and unfinished**: it goes against the wall. Either short end can become the side-wall end.

Which top is A

The two tops (P01, P02) are identical. **A** is the one you lay along the back wall; its unfinished edge is y = 0. **B** lies along the side wall; its unfinished edge is x = 0 and one short end, the **joint end**, butts against A's front edge. Top B is top A turned 90°, never mirrored.

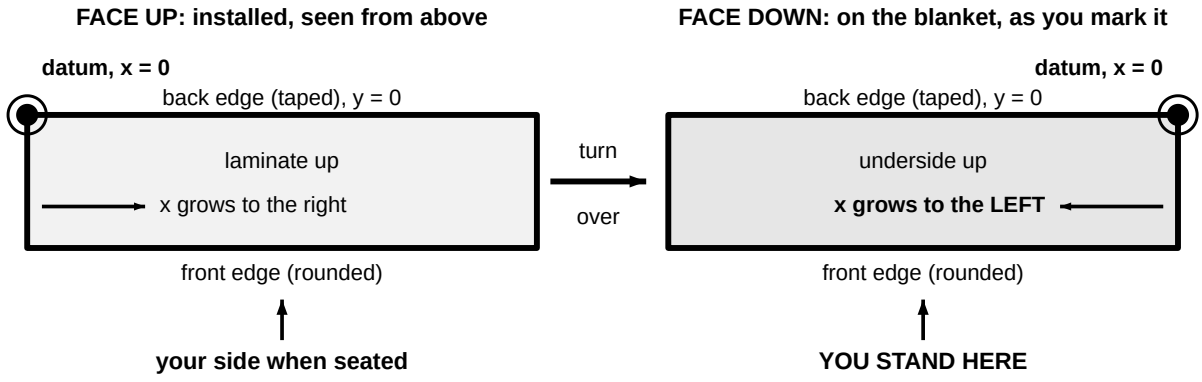
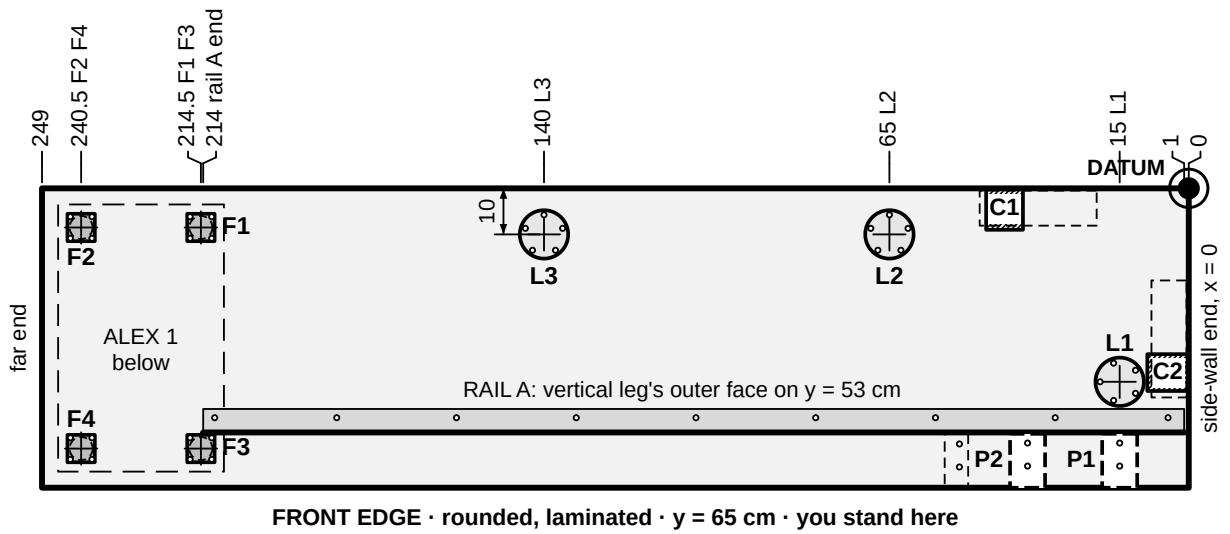


Fig. D-2 · Face-down convention used on every marking drawing. Stand opposite the taped reference edge (top A: at its front edge; top B: at its far end, joint end away from you): the underside is the plan mirrored left to right, so **x grows to the LEFT and the datum side (x = 0) is on your RIGHT**. Top B uses the same mapping.

Top A, face down: overview

1

Calipers on top A: 28 mm **MEASURE ON HARDWARE**. If it is 27 mm, shorten every screw into it by 1 mm. Tape the **unfinished long edge** ("A · BACK EDGE") and the short end you choose as the **side-wall end** ("A · SIDE-WALL END"). Lay top A face down on the blanket and stand at the front edge, back edge away from you.



Scale 1:16. y: tape hooked on the back edge. Details: Figs D-4, D-5.

Fig. D-3 · Top A underside as you see it, all cm. Datum at the far right corner. Circles: OLOV plates (P05); squares: feet plates (P09); grey band: rail A (P06); hatched C1, C2: clamp zones; dashed: pad plates, joining plates P1, P2 (P11), rail B's corner end and ALEX 1 (reference).

Mark	Hook the tape on	Tape reads (cm)	Then
L1	the back edge, tape along the side-wall end	**42**	leg jig, **150** pattern, lip over the side-wall end. Plate ends 6.5 mm short of rail A
L2	the side-wall end, tape along the back edge	**65**	leg jig, 100 pattern, lip over the back edge
L3	the side-wall end, tape along the back edge	**140**	leg jig, 100 pattern, lip over the back edge
F1	the side-wall end, tape along the back edge	**214.5**	T-square, lip over the back edge, hole 85 and its four plate holes
F3	the side-wall end, tape along the **front** edge	**214.5**	T-square, lip over the front (rounded) edge, hole 85 and its four plate holes
F2, **F4**	no tape: feet corner jig on the far back and far front corners		8.5 cm in from both edges
Rail A line	the back edge, at both ends of the line and the middle	**53**	line from 1 to 214 on the side-wall-end tape; the vertical leg's outer face goes on it (12 cm from the front edge)
C1	the side-wall end, along the back edge	**36 to 44**	keep clear, 9 cm in from the back edge
C2	the back edge, along the side-wall end	**36 to 44**	keep clear, 9 cm in from the side-wall end
Pad plate 1	the side-wall end, along the back edge	**20.0 to 45.4**	outline only, 0.5 to 8.1 cm in from the back edge
Pad plate 2	the back edge, along the side-wall end	**20.0 to 45.4**	outline only, 0.5 to 8.1 cm in from the side-wall end
P1 / **P2**	the side-wall end, along the front edge	**11.2–18.8 / 31.2–38.8**	outline only, from 53.3 cm (back-edge tape) to the front edge

All positions **SOURCE-DEFINED** (docs/build-manual.md step 2). Pilots: 1/8 in, started 3–4 mm through the jig; finishing them to depth and fitting the legs and feet follows in the next chapter.

Top A: the corner

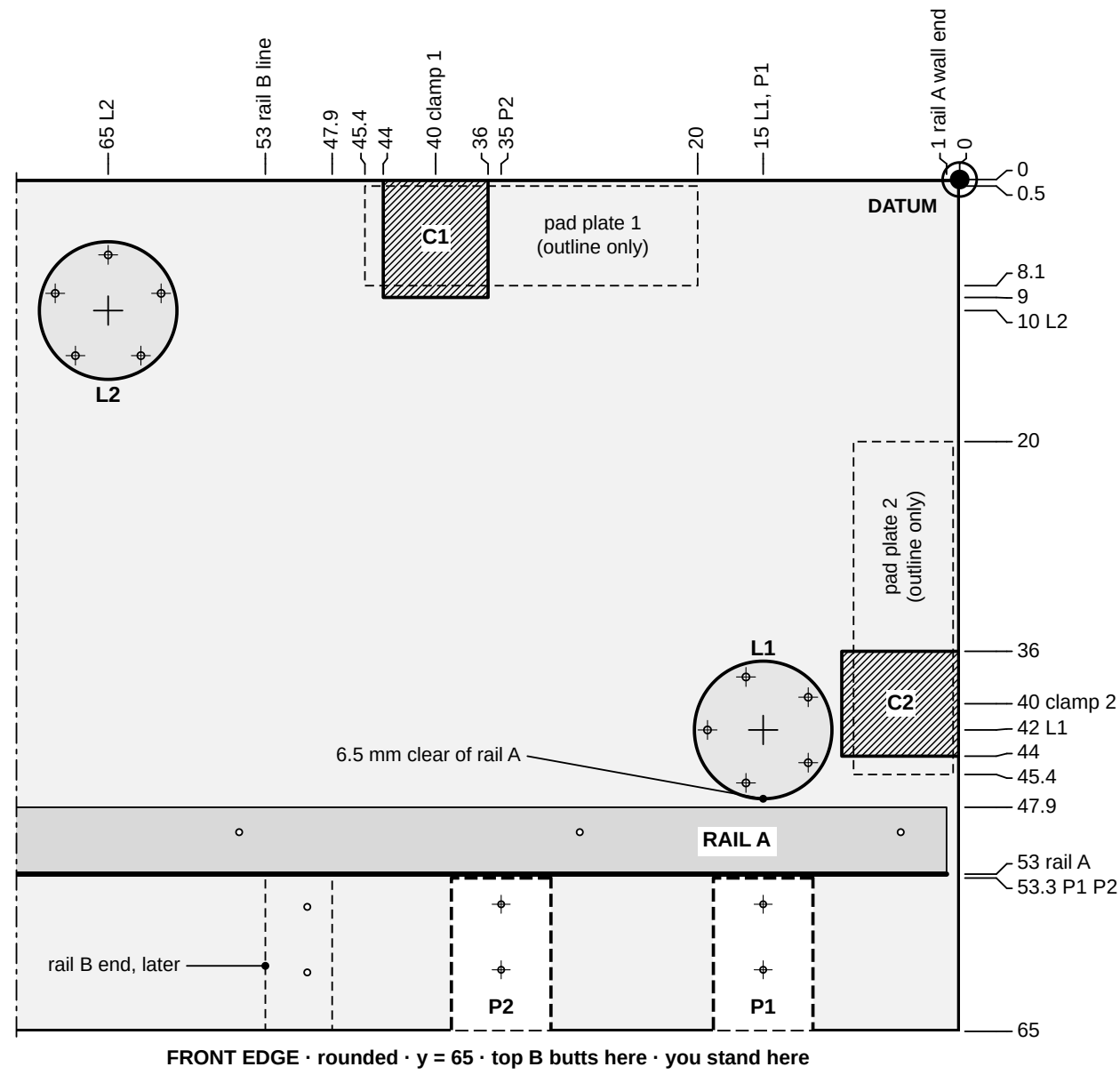
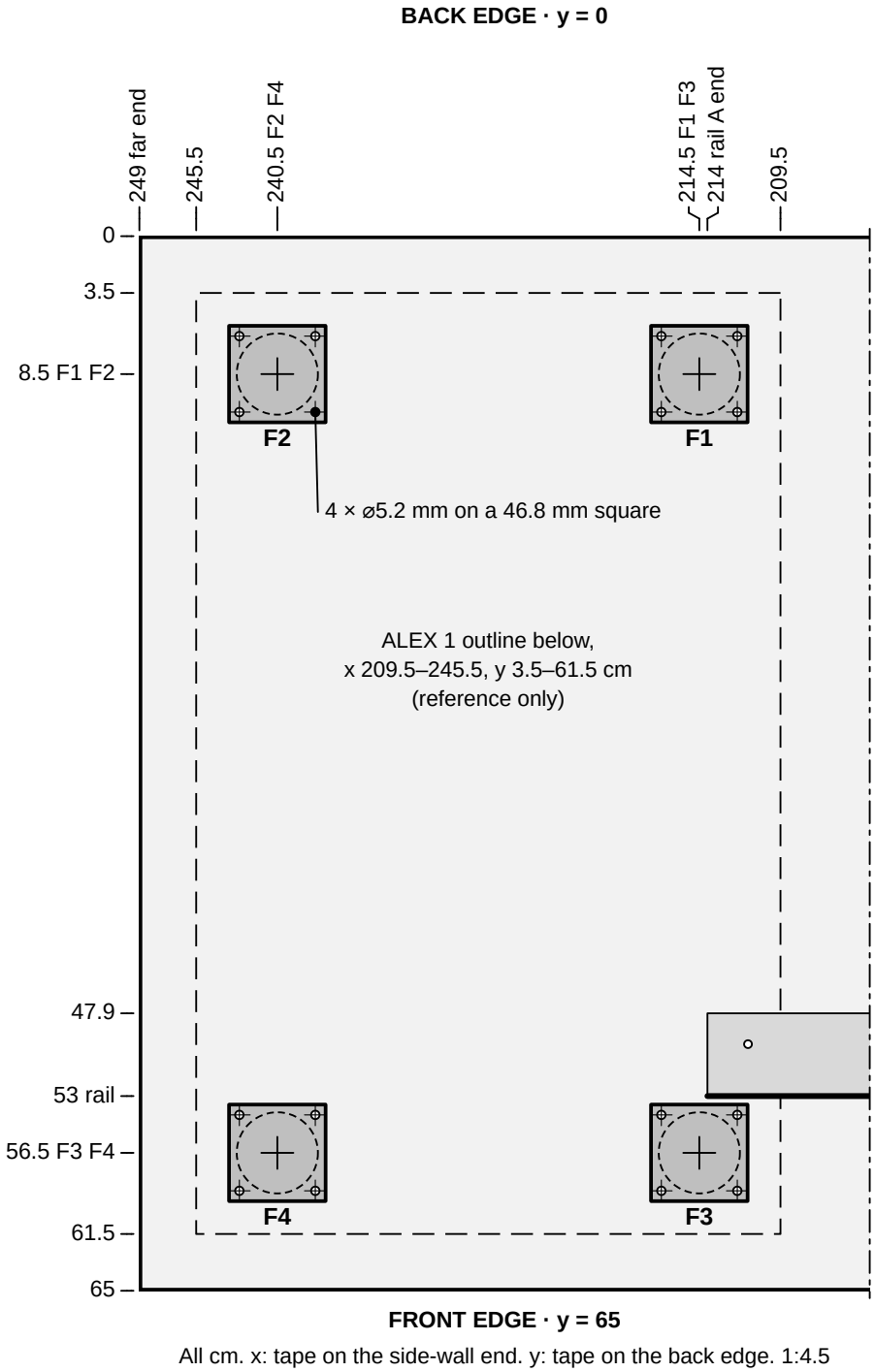


Fig. D-4 · Top A corner, face down, 1:5, all cm. The numbers along the top are the tape hooked on the side-wall end; the numbers on the right are the tape hooked on the back edge.

2 **L1:** hook the tape on the back edge and tick **42 cm** along the side-wall end. Hook the leg jig's lip over the **side-wall end** at that tick and use its **150** pattern (dashed circle, ringed holes): start the five 1/8 in pilots. **L2:** tape on the side-wall end, tick **65 cm** along the back edge; lip over the back edge, **100** pattern (solid circle).

3 Rail A line: tape hooked on the back edge, tick **53 cm** near each end and in the middle; join the ticks from **1 cm** to **214 cm** (side-wall-end tape). Hatch C1 and C2 with pencil. Draw the two pad-plate outlines and the P1, P2 outlines. Nothing is drilled inside C1, C2 or the outlines.

Top A: the far end (feet for ALEX 1)



All cm. x: tape on the side-wall end. y: tape on the back edge. 1:4.5

Fig. D-5 · Top A far end, face down, 1:4.5, all cm.

4 **F2, F4:** no tape. Drop the feet corner jig over the far back corner, then the far front corner, fences against both edges. Start the four 1/8 in pilots through it.

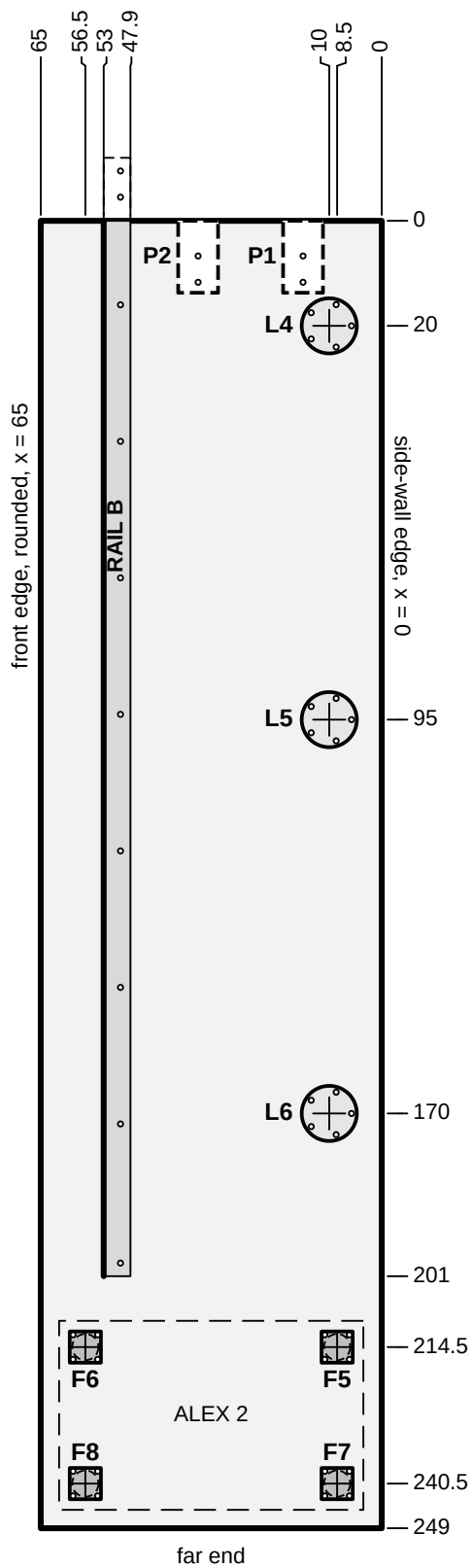
5 **F1:** tape hooked on the side-wall end reads **214.5 cm** along the back edge: tick it. T-square lip over the back edge on the tick, scriber through hole 85 to check, then start the four pilots round it. **F3:** the same, **214.5 cm** along the **front** edge, lip over the front edge.

6 **L3:** tape on the side-wall end reads **140 cm** along the back edge; leg jig, 100 pattern (Fig. D-3).

Feet plate: 59.5 mm square, four ⌀5.2 mm holes 46.8 mm apart **VERIFIED**. Plates square with the top's edges.

The ALEX 1 outline is for orientation only; nothing is marked for it.

Top B, face down: overview



All cm. Right: tape hooked on the joint end.
Top: tape on the side-wall edge. 1:14

Fig. D-6 · Top B underside, joint end away from you, 1:14, all cm.

7

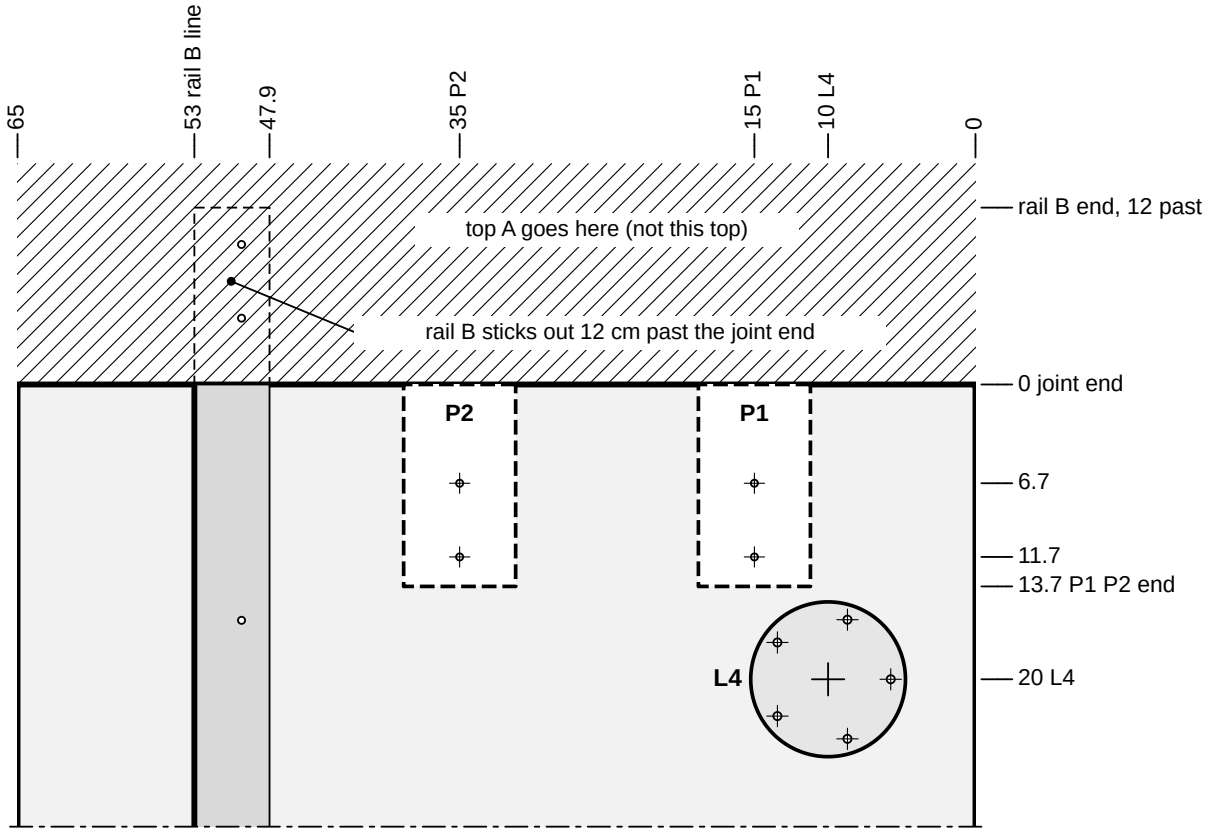
Calipers on top B: 28 mm MEASURE ON HARDWARE. Tape its **unfinished long edge** ("B · SIDE-WALL EDGE") and the short end that will butt against top A ("B · JOINT END"). Lay it face down and stand at the far end, joint end away from you: the side-wall edge is on your right.

> On top B every station along the length is read with the tape **hooked on the joint end**. The numbers below are what the tape shows.

Mark	Tape on the joint end reads (cm)	Then
L4	**20**	along the side-wall edge; leg jig, 100 pattern, lip over the side-wall edge
L5	**95**	
L6	**170**	
F5	**214.5**	along the side-wall edge; T-square hole 85, lip over the side-wall edge
F6	**214.5**	along the **front** edge; T-square hole 85, lip over the front edge
F7, **F8**	no tape	feet corner jig on the two far corners
Rail B line	**0 to 201**	at 53 cm from the side-wall edge (12 cm from the front edge); the vertical leg's outer face goes on it; the rail's corner end sticks out 12 cm past the joint end
P1 / **P2**	**0 to 13.7**	outline only, 11.2–18.8 / 31.2–38.8 cm from the side-wall edge; screws later at 6.7 and 11.7 cm

All positions SOURCE-DEFINED (docs/build-manual.md step 3). L4–L6 have one pilot pointing at the side wall, like L2 and L3.

Top B: the joint end



All cm. Top: tape hooked on the side-wall edge. Right: tape hooked on the joint end.
Joint end away from you, x grows to the left. Scale 1:5

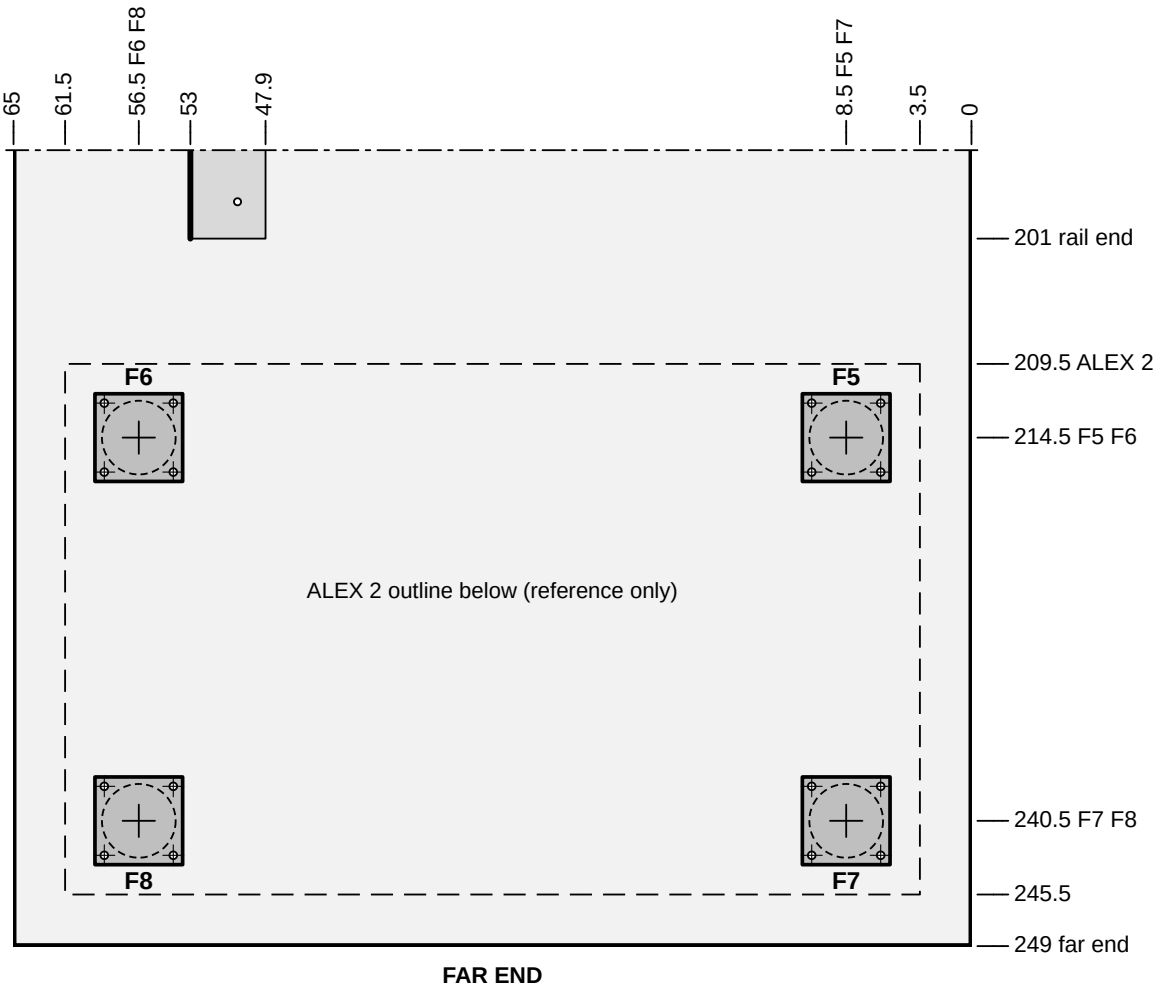
Fig. D-7 · Top B joint end, face down, 1:5, all cm. The hatched area beyond the joint end is where top A will be; rail B's corner end lies there.

8

L4: tape hooked on the joint end reads **20 cm** along the side-wall edge; leg jig, 100 pattern, lip over the side-wall edge. Draw the **rail B line**: tape hooked on the side-wall edge, tick **53 cm** near each end and in the middle; join the ticks from the joint end to **201 cm** on the joint-end tape. Draw the P1 and P2 outlines from the joint end to **13.7 cm**.

Rail B's holes 1 and 2 (shown beyond the joint end) go into top A when the tops are joined; on top B only its holes 3 to 10 are used.

Top B: the far end (feet for ALEX 2)



All cm. Top: tape on the side-wall edge. Right: tape hooked on the joint end. 1:5

Fig. D-8 · Top B far end, face down, 1:5, all cm.

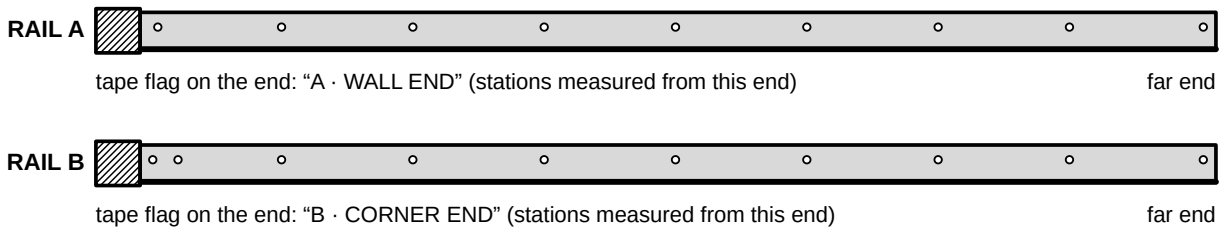
9

F7, F8: no tape: feet corner jig over the two far corners, fences against both edges; start the four 1/8 in pilots. **F5:** tape hooked on the joint end reads **214.5 cm** along the side-wall edge: T-square, lip over the side-wall edge. **F6: 214.5 cm** along the front edge, lip over the front edge. **L5, L6:** leg jig, 100 pattern, at **95 cm** and **170 cm** on the joint-end tape (Fig. D-6).

Rails: labels and orientation

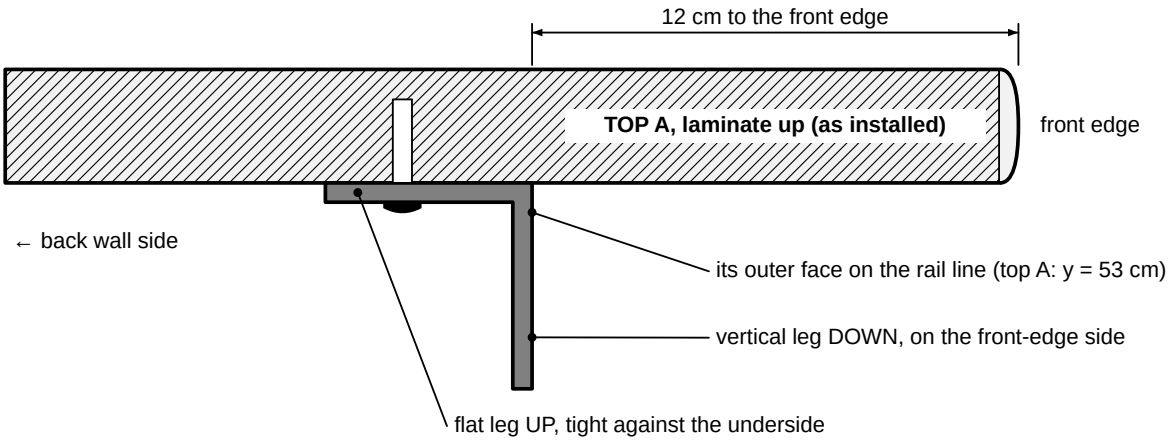
10

The two steel angles arrive identical; either can be rail A. Tape one end of each: **"A · WALL END"** and **"B · CORNER END"**. Every station on a rail is measured from its taped end. Measure both delivered lengths **MEASURE ON HARDWARE** (213 cm layout length; the far end may carry the extra).



Bench view, flat leg up. Holes shown for orientation; positions are in chapter E.

Fig. D-9 · The two rails on the bench, flat legs up, labelled ends left.



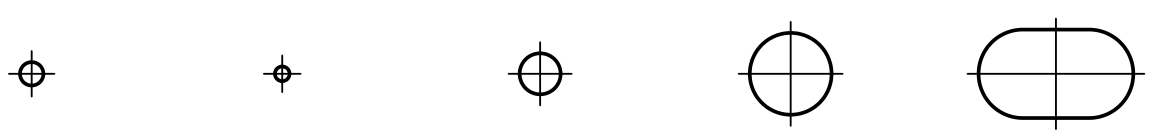
Section through top A and rail A, end-on, about 1:2. Rail B on top B: the same, outer face on x = 53 cm.

Fig. D-10 · How a rail sits under its top: **flat leg UP** against the underside, all screws through it; **vertical leg DOWN** on the front-edge side, its outer face on the rail line, 12 cm from the front edge. Rail A: outer face at 53 cm from the back edge, wall end 1 cm in from the side-wall end. Rail B: outer face at 53 cm from the side-wall edge, corner end 12 cm past top B's joint end.

The printed rail end caps are handed: cap A on rail A's far end and rail B's far end, cap B on rail A's wall end; rail B's corner end takes no cap.

Hole types and depths

The three hole types



pilot ⌀3.2 (1/8 in) into the tops: legs, feet, rails, joining plates

pilot ⌀2.0 (5/64 in) into the ALEX tops and the trim edge

clearance ⌀5.56 (7/32 in): rail flat legs, for #10 screws

through ⌀11.1 (7/16 in) rail vertical legs, 6 holes, 3/8 bolts

slot 21 × 12 in each fitting leg, bought, not drilled

Fig. D-11 · Diameters in mm, 1:1. Pilot holes go into wood; clearance and through holes go through steel. The fittings come with 21 × 12 slots (not drilled); the slot plug centres the rail's 7/16 in pilot in them.

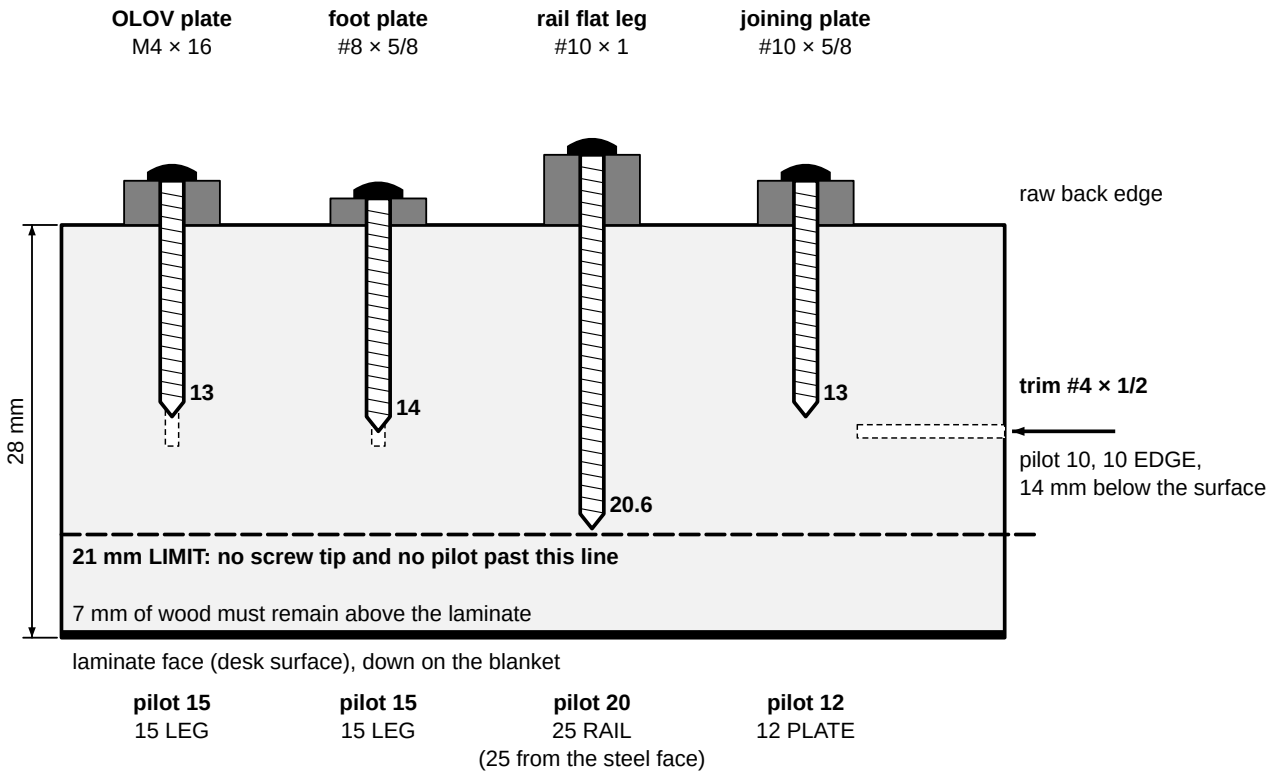
Every fastener, its pilot and its puck

Fastener	Qty	Goes through	Into the wood (mm)	Pilot	Puck
H01 · M4 × 16 (with the OLOV)	30	3 mm leg plate	13	1/8 in, 15 mm deep	15 LEG
H02 · #8 × 5/8 in pan head	32	feet plate (1.5–2 mm)	14	1/8 in, 15 mm deep	15 LEG
H03 · #10 × 1 in pan head, Robertson	19	4.8 mm angle	20.6	1/8 in: the 25 RAIL puck stands on the steel and stops the bit 20 mm into the wood	25 RAIL
H04 + H05 · #10 × 5/8 in with washer	8	3 mm joining plate	13	1/8 in, 12 mm deep	12 PLATE
H06 · #6 × 5/8 in flat head	24	2.5 mm ring flange	13, into the ALEX top	5/64 in, 12 mm deep	12 ALEX
H07 · #4 × 1/2 in flat head (optional trim)	48	trim lip	sideways into the raw back edge, 14 mm below the surface	5/64 in, 10 mm deep	10 EDGE
H08 + H09 + H10 · 3/8-16 bolts	6	fittings and rail vertical legs	none	7/16 in, bench-drilled	—

Depths SOURCE-DEFINED (guide.html fastener table, facts.json). The bolt holes and the 7/32 in screw holes in the steel are drilled on the bench (chapter E); everything into wood goes through a jig first and is finished with its puck.

Never through the laminate

> **NEVER THROUGH THE LAMINATE**
> No screw tip and no pilot goes deeper than **21 mm** into a 28 mm top: at least **7 mm** of wood must remain under the laminate face. Set every pilot depth with the drill guide's puck, never by eye. If a top measures 27 mm, shorten every screw into it by 1 mm.



Section through a 28 mm top, face down as you drill it (underside up), 2:1. Grey: the part being screwed on. Dashed: the pilot hole. Bold numbers: how far the screw goes into the wood, mm.

Fig. D-12 · Every fastener into the tops at its depth, 2:1, mm.

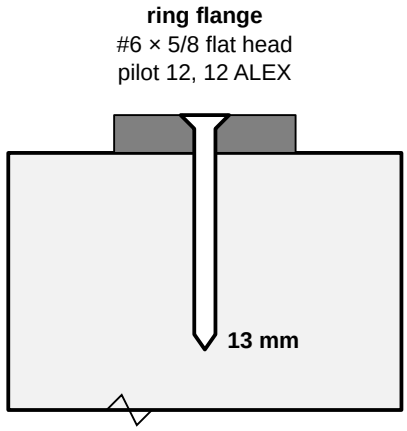


Fig. D-13 · Ring screws go into the ALEX top, not the desk. **MEASURE ON HARDWARE** panel thickness.

Setting a puck

11 Loosen the chuck. Drop the bit tip-first into the SET tower's hole for that bit (45 mm for 1/8 in, 35 mm for 5/64 in), rest the chuck nose on the tower, tighten. Re-set every time you change bits.

12 Stand the puck flat on the surface (on the rail's flat leg for 25 RAIL), bit down its bore, drill until the chuck nose lands on the puck. The hole is then the depth written on the puck.

If your chuck nose is domed, drill one test hole in scrap and measure it first.

E · Steel rail fabrication and drilling

All steel work happens on the bench, before either rail goes near a top. Under the desk you only drive screws and bolts; you never drill steel there. Two identical lengths of 2 × 2 × 3/16 in hot-rolled angle arrive; either can be rail A. Tape readings in cm, caliper and drill sizes in mm.

MEASURE BEFORE MARKING OR DRILLING MEASURE ON HARDWARE
1 · Tape each rail from end to end and write the length on the tape label. Design length **213 cm**; bought as 84 in (2133.6 mm, so about 213.4 cm). If a rail is under 213 cm, stop and check before drilling.
2 · Calipers on each angle: legs 50.8 mm, thickness 4.76 mm (the docs expect about 4.8 mm).
3 · Every station is measured from the **labelled end**: rail A's **wall end**, rail B's **corner end**. The **far end carries any extra** length. Never centre the hole pattern on the bar and never shift a hole to share out the extra.

1 **Label the datum ends.** Wrap tape round one end of each rail and write on it: **RAIL A · WALL END** on one, **RAIL B · CORNER END** on the other. Measure both rails from those ends (Fig. E-1). File both cut ends of each rail smooth.

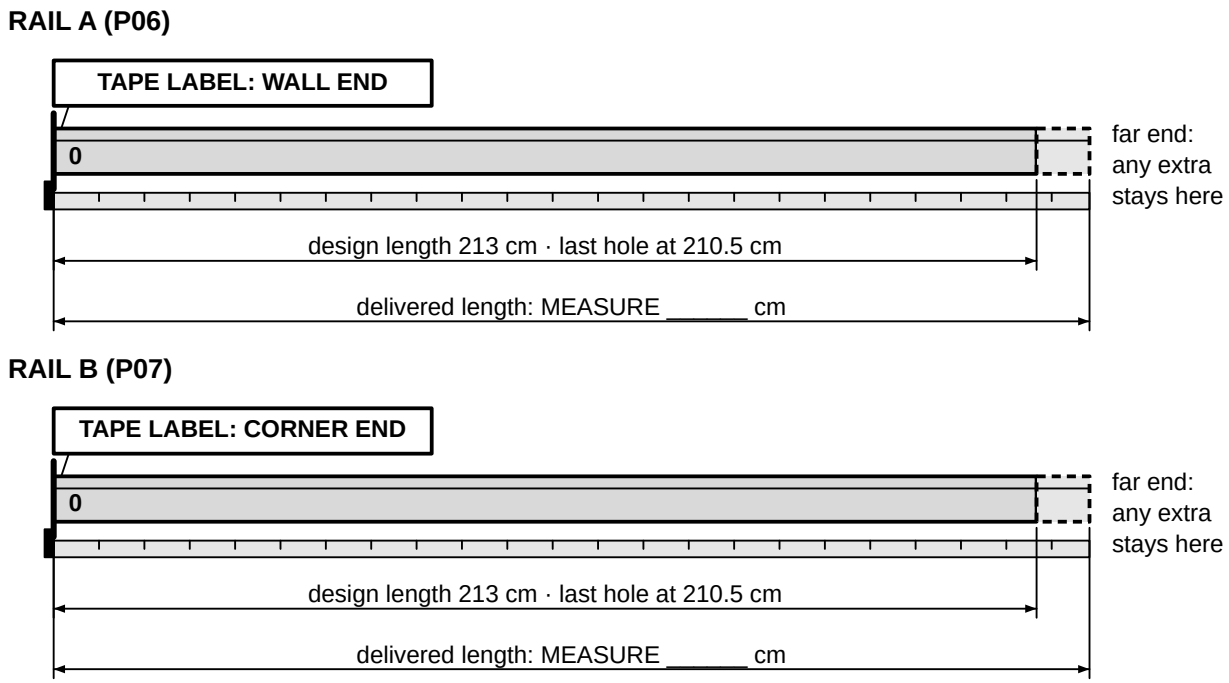


Fig. E-1 · The labelled end is 0 for every station. The last hole is at 210.5 cm on both rails, so a longer rail only lengthens the far end beyond it. SOURCE-DEFINED design length; MEASURE ON HARDWARE delivered length.

You need	For
P06 rail A, P07 rail B; one P08 fitting as the template	the work pieces
P18 printed tools: edge T-square, rail marking gauge, rail drill jig (PETG), 2 fitting slot plugs	stations, the 19 and 29 lines, pilots, bolt-hole transfer
Bits 1/8, 7/32 and 7/16 in (the 7/16 cobalt), cutting oil, flat file, 2 clamps, tape, combination square, pencil	drilling and deburring; a drill press if you have one
Acetone or mineral spirits, 120 grit, flat black enamel	finish (step 9)

Which way up: the section and the two lines

FLAT LEG
lies against the top's underside

VERTICAL LEG
hangs down, toward you; outer face at
y = 53 cm (rail A)
x = 53 cm (rail B)

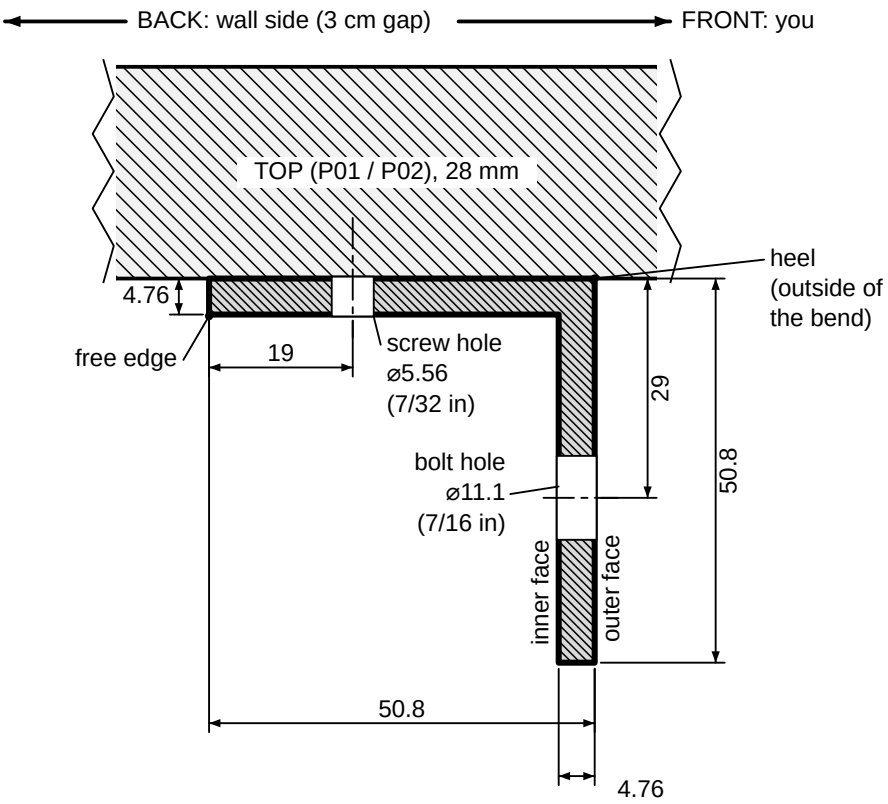
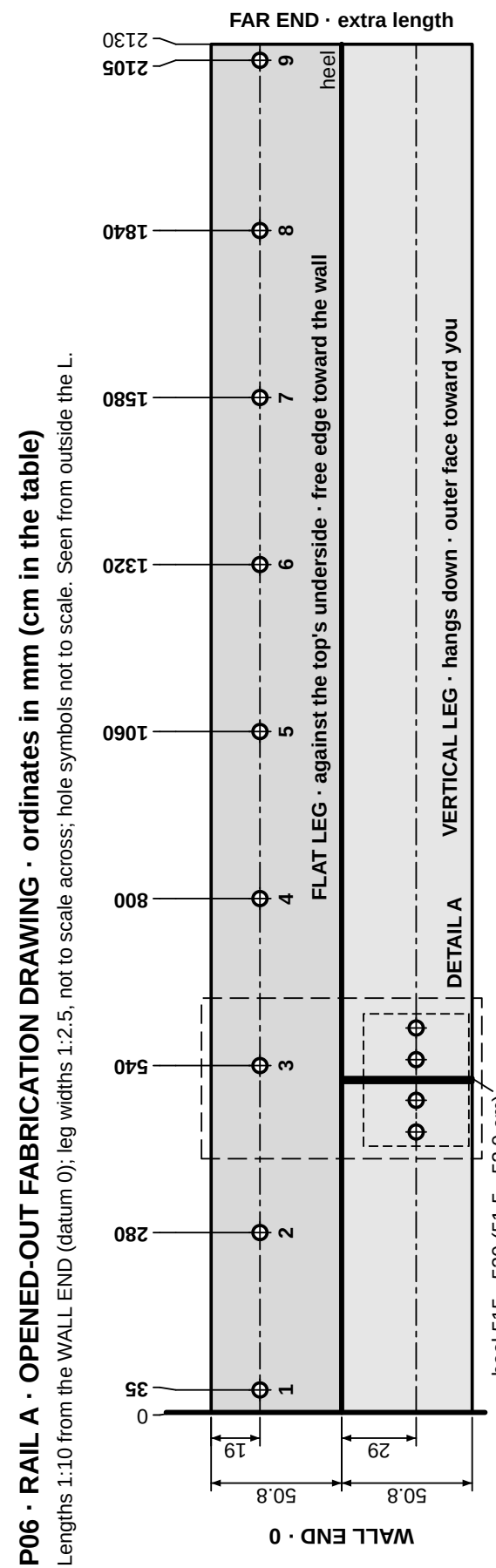


Fig. E-2 · Rail A seen end-on from its far end, as it hangs under top A (rail B is the same section turned 90°). Datum: top A's back-left corner, x along the back wall, y along the side wall. The vertical leg's outer face is at y = 53 cm on rail A and x = 53 cm on rail B, 12 cm back from the front edge (65 cm). Section dimensions in mm.

Item	mm	What it means	Confidence
Section	50.8 × 50.8 × 4.76	2 × 2 × 3/16 in hot-rolled angle	SOURCE-DEFINED MEASURE ON HARDWARE thickness
Flat leg		lies against the top's underside; its free edge points to the wall	SOURCE-DEFINED
Vertical leg		hangs down on the side nearer you; outer face y = 53 cm (A), x = 53 cm (B)	SOURCE-DEFINED
Screw line	19	from the flat leg's free edge; centres of the ⌀5.56 (7/32 in) screw holes	SOURCE-DEFINED
Bolt line	29	below the heel (the top's underside); centres of the ⌀11.1 (7/16 in) bolt holes	SOURCE-DEFINED
Fitting slots	21 × 12	two per P08 leg, centres 29.15 and 78.45 from the heel; legs 100.5 × 41, 5 thick (all mm)	VERIFIED 23 Oct 2026

Bench set-up. Clamp the rail with the flat leg's face up and the vertical leg hanging down over the bench edge. That is how it is marked and how the flat-leg holes are drilled. For the vertical leg (steps 6 and 7) turn it so the vertical leg's outer face is up and the flat leg hangs over the edge.

The two fabrication drawings that follow are full pages, turned sideways: turn the booklet a quarter turn clockwise to read them. Every hole is given from the labelled end, never from the hole before: the drawings carry mm ordinates, the hole table beside each drawing gives the same stations in cm for the tape.

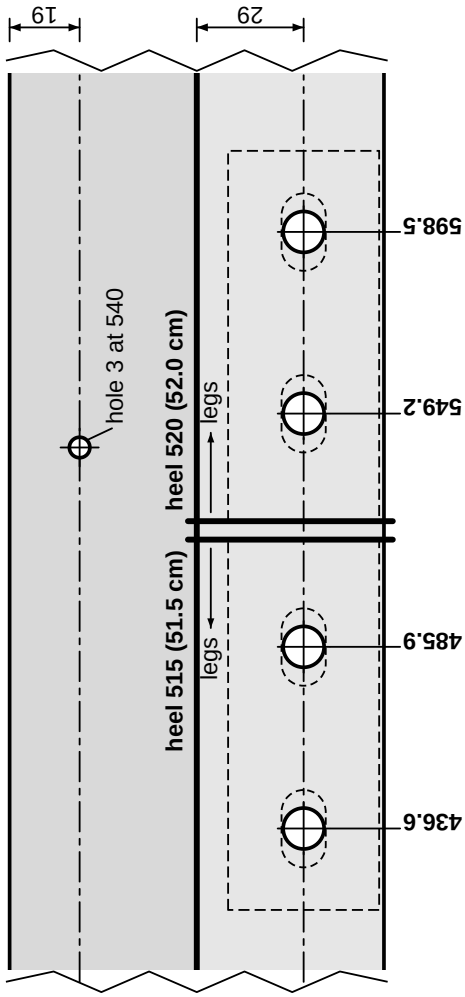


9× ø5.56 (7/32 in) THRU in the flat leg, centres on the screw line 19 from the free edge, at the ordinates above.
4× ø11.1 (7/16 in) THRU in the vertical leg on the bolt line 29 below the heel, at 43.7, 48.6, 54.9, 59.8 cm (Detail A, mm).
Positions VERIFIED from the measured fitting; drill them THROUGH the clamped fitting with the slot plug (step 6).

Holes from the wall end

No.	mm	cm	hole	leg, line
1	35	3.5	ø5.56 (7/32 in)	flat, 19 line
2	280	28	ø5.56 (7/32 in)	flat, 19 line
3	540	54	ø5.56 (7/32 in)	flat, 19 line
4	800	80	ø5.56 (7/32 in)	flat, 19 line
5	1060	106	ø5.56 (7/32 in)	flat, 19 line
6	1320	132	ø5.56 (7/32 in)	flat, 19 line
7	1580	158	ø5.56 (7/32 in)	flat, 19 line
8	1840	184	ø5.56 (7/32 in)	flat, 19 line
9	2105	210.5	ø5.56 (7/32 in)	flat, 19 line
I2	436.6	43.7	ø11.1 (7/16 in)	vertical, 29
I1	485.9	48.6	ø11.1 (7/16 in)	vertical, 29
O1	549.2	54.9	ø11.1 (7/16 in)	vertical, 29
O2	598.5	59.8	ø11.1 (7/16 in)	vertical, 29

DETAIL A · 1:2 · dashed: the fitting clamped as the template



P06 Rail A · 2 × 2 × 3/16 in hot-rolled angle
50.8 × 50.8 × 4.76 mm, 213 cm: MEASURE first
On the desk: wall end x = 1 cm, outer face y = 53 cm
(datum: top A's back-left corner)

Fig. E-3 · P06 rail A, opened out: the flat leg (upper strip) and the vertical leg (lower strip) unfolded about the heel. Nine screw holes from the wall end; four bolt holes at 43.7, 48.6, 54.9 and 59.8 cm, drilled through the fitting (heels on the 51.5 and 52.0 cm marks). Ordinates in mm, cm in the table. SOURCE-DEFINED stations; VERIFIED bolt holes.

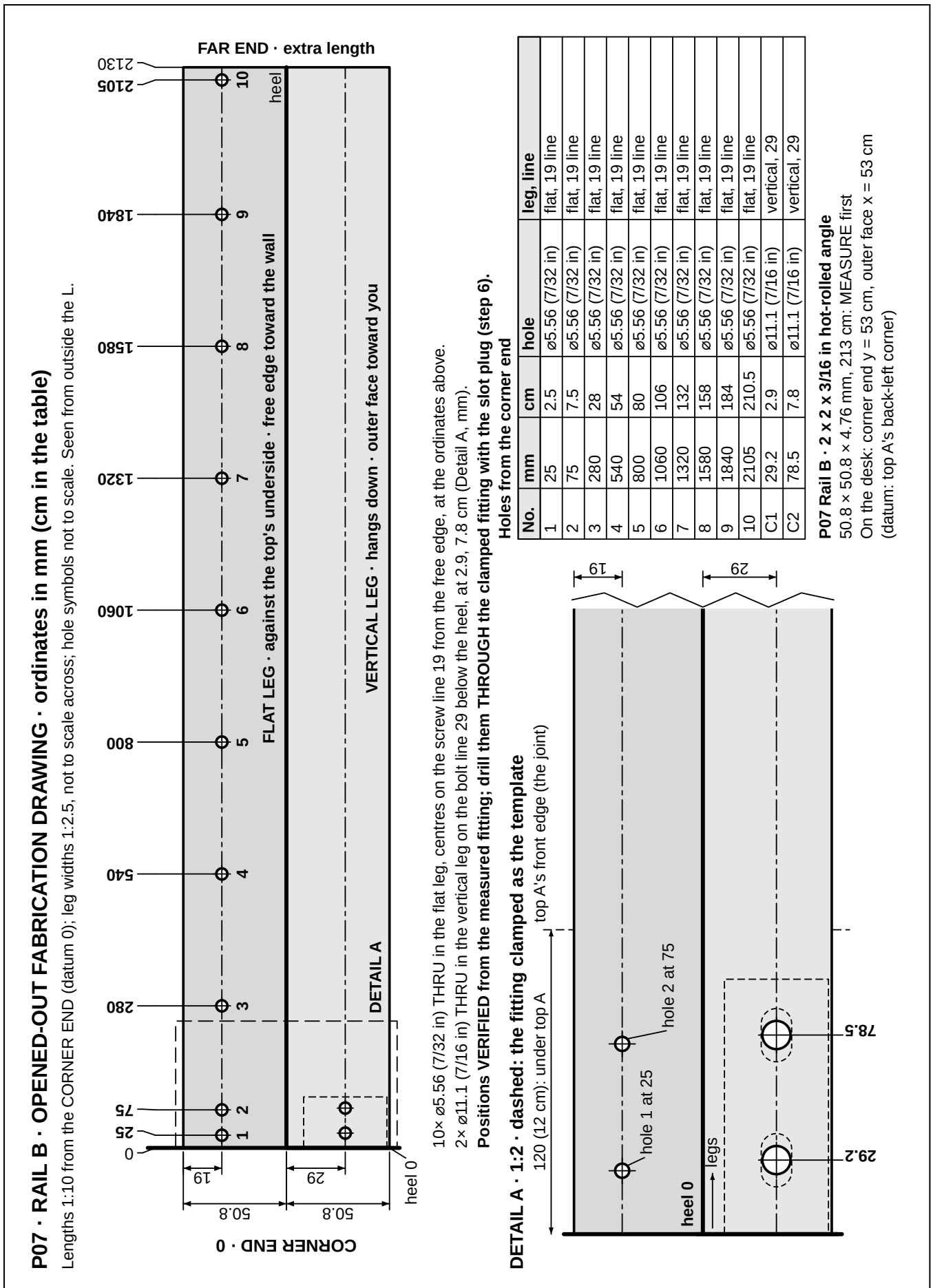
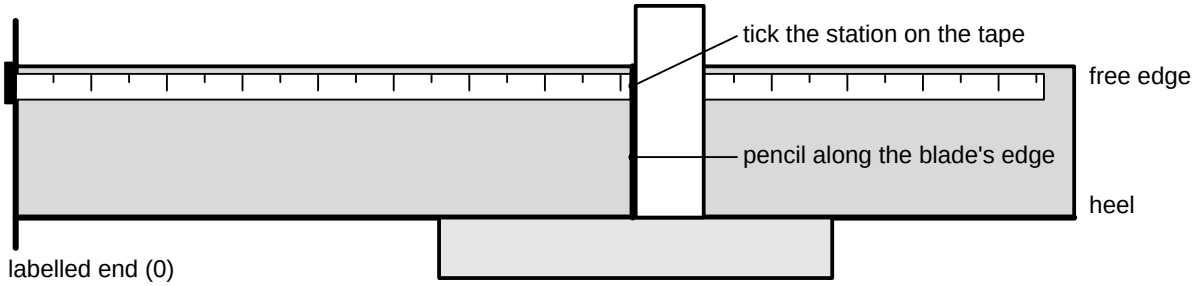


Fig. E-4 · P07 rail B, opened out. Ten screw holes from the corner end: holes 1 and 2 go into top A (the first 12 cm sits under top A), holes 3 to 10 into top B. Two bolt holes at 2.9 and 7.8 cm, shared by both fittings, drilled through the fitting with its heel flush with the corner end. SOURCE-DEFINED stations; VERIFIED bolt holes.

2 **Mark the stations.** Hook the tape over the labelled end and tick every station on the flat leg from that one hook: rail A **3.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm**; rail B **2.5, 7.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm**. Square each tick across the leg along the T-square's blade, its lip hooked over the heel edge.

Hook the tape over the labelled end once and tick every station from that one hook;
never measure from one hole to the next.



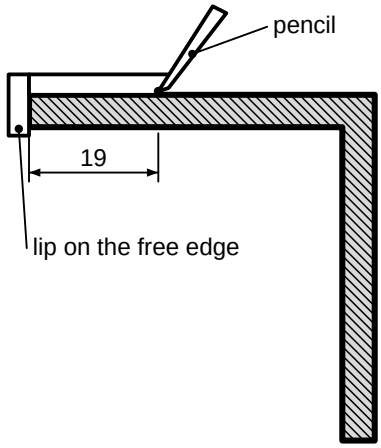
T-square (P18): lip over the heel edge, blade across the leg

Plan: flat leg face up, vertical leg over the bench edge, labelled end at the left.

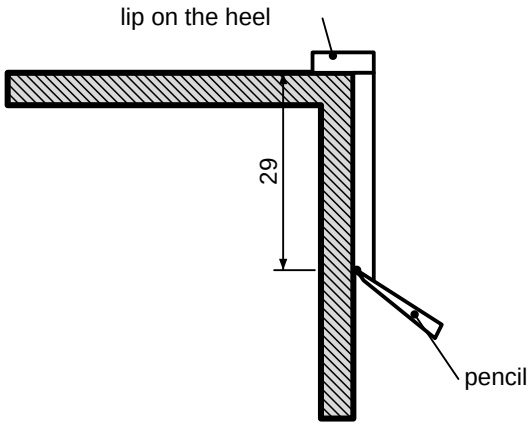
Fig. E-5 · One hook, every station. The station line only has to cross the leg; the jig finds the 19 mm.

3 **Pencil the two lines with the rail marking gauge, full length.** Screw line: lip over the flat leg's free edge, pencil in the **19** notch, slide. Bolt line: gauge on the vertical leg's outer face, lip over the heel, pencil in the **29** notch, slide. The bolt line is only a check: the fitting's holes must sit on it in step 6.

a SCREW LINE on the flat leg
lip over the free edge,
pencil in the 19 notch, slide



b BOLT LINE on the vertical leg
gauge on the outer face, lip over the heel,
pencil in the 29 notch, slide



Sections, marking gauge (P18) in use. Run each line the full length of both rails.

Fig. E-6 · The gauge's lip is the zero; the V notches put the pencil point on 19.0 and 29.0. SOURCE-DEFINED

4 **Pilot the screw holes through the drill jig.** Jig lip down over the flat leg's free edge, sight grooves on the station line, hold it down and drill 1/8 in through its bore: square, on the 19 mm line, no punch needed. All 19 stations (9 on rail A, 10 on rail B). Use a drill press if you have one.

5 **Open and deburr.** Jig off. Open every pilot to **⌀5.56 (7/32 in)** at low speed with a drop of oil on each; the bit follows the pilot. Deburr both faces with the file or a larger bit turned by hand.

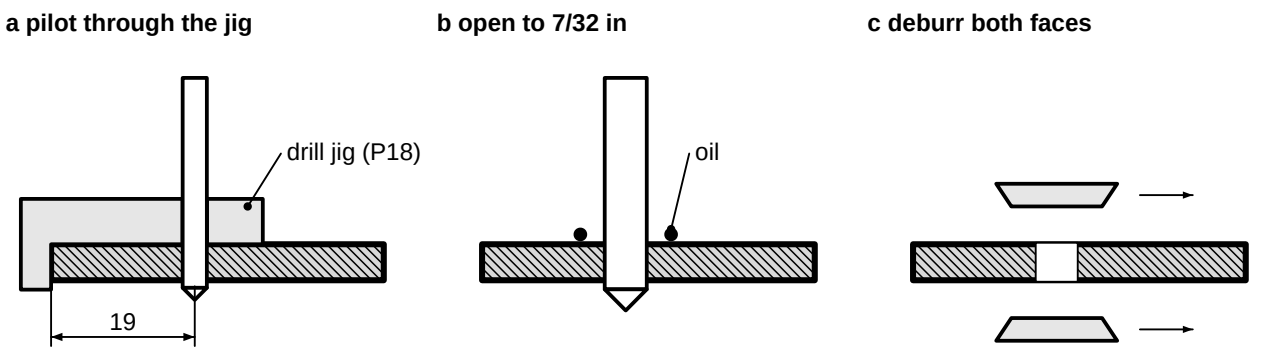


Fig. E-7 · Flat leg in section. The jig's bore is 1/8 in plus a running clearance; it keeps the bit square and on the 19 line.

THE FITTING IS THE TEMPLATE VERIFIED
The fittings were measured on 23 Oct 2026: legs 100.5 × 41, two 21 × 12 slots per leg with centres 29.15 and 78.45 mm from the heel. That puts the six bolt holes at the positions below. Even so, do not lay them out with the tape: clamp the fitting on its heel mark and drill through it with the printed **slot plug** (steps 6 and 7), so each hole lands on its slot's centre. Check first that a plug drops into each slot without rattle. The heel of a fitting is the **outside of its bend**.

Clamp the fitting	Heel on	Legs toward	Holes it makes (cm from the labelled end)
Rail A, outside fitting	the 52.0 cm mark	the far end	O1 54.9, O2 59.8
Rail A, inside fitting	the 51.5 cm mark	the wall end	I1 48.6, I2 43.7
Rail B, one fitting (both share the holes)	the corner end itself (0)	the far end	C1 2.9, C2 7.8

Heel stations SOURCE-DEFINED; hole positions from the measured slot centres VERIFIED. Both fittings are identical, so holes copied from one fit the other.

6 **Clamp the fitting and pilot through the plugs.** Rail with the vertical leg's outer face up. Clamp one fitting to the vertical leg with its heel exactly on the mark and its holes centred on the 29 mm line, legs the way Fig. E-8 shows. Drop a slot plug into each slot and drill the 1/8 in pilot through each plug's bore into the rail. Repeat for each of the three placements.

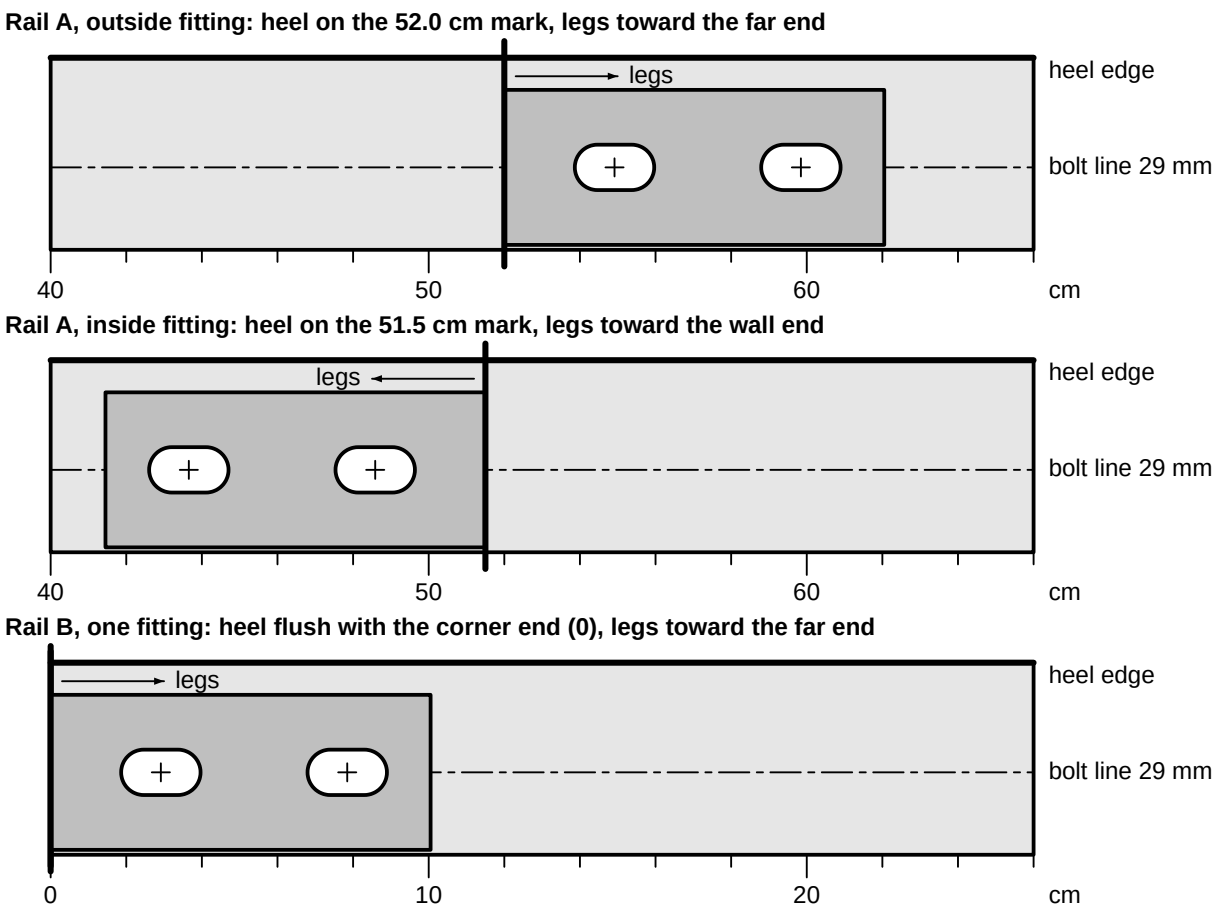


Fig. E-8 · The three clamp positions, vertical leg seen face on (1:2). Heavy line: the heel mark. Slots and their centres as measured. **VERIFIED**

7 **Fitting off, open to ⌀11.1 (7/16 in).** Cobalt bit, slow, oil on each. The bit follows the pilot, so the hole lands on the slot's centre, not anywhere along its 21 mm. Deburr. Six holes: four in rail A, two in rail B.

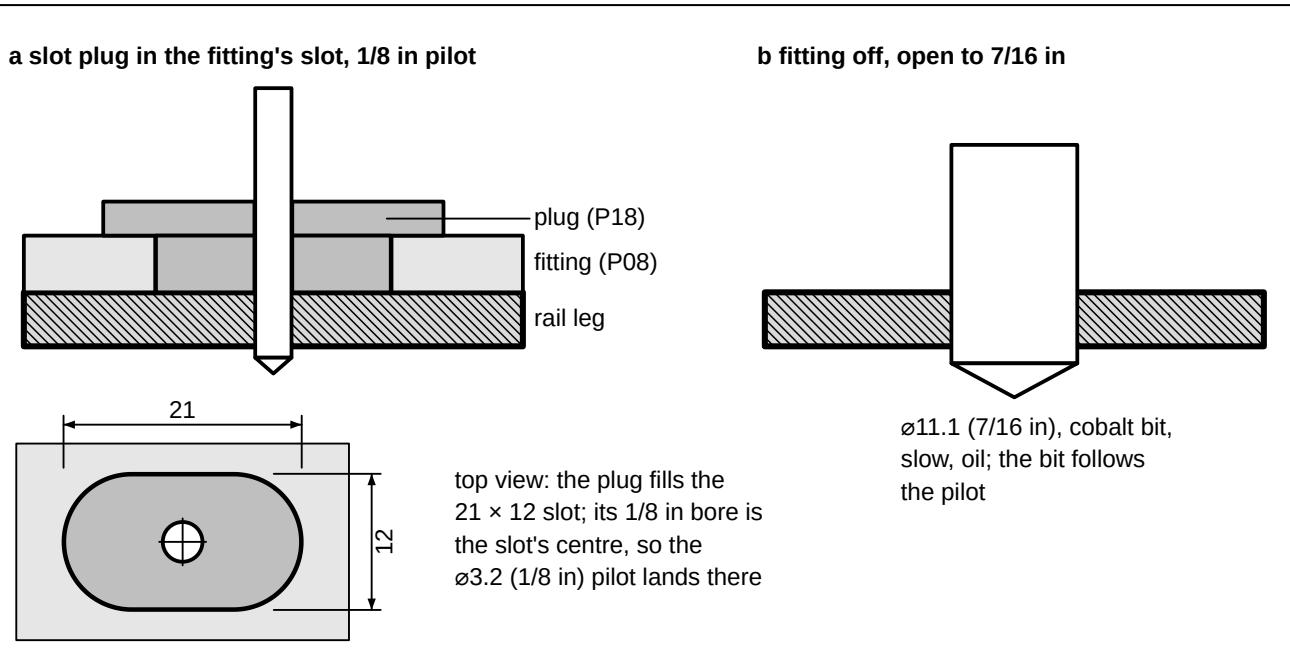
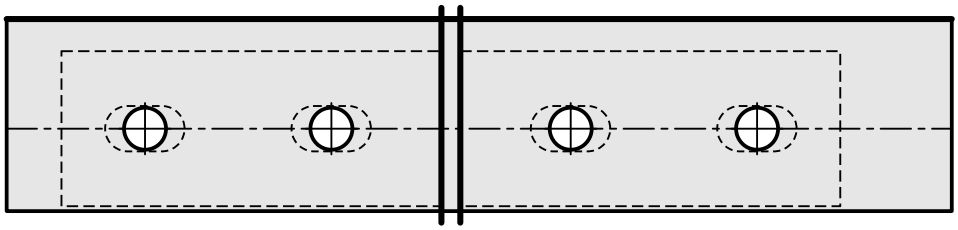


Fig. E-9 · The slot plug: a stadium that fills the 21 × 12 slot with a running clearance, 1/8 in bore at its centre. Without it a punch could land anywhere along the slot.

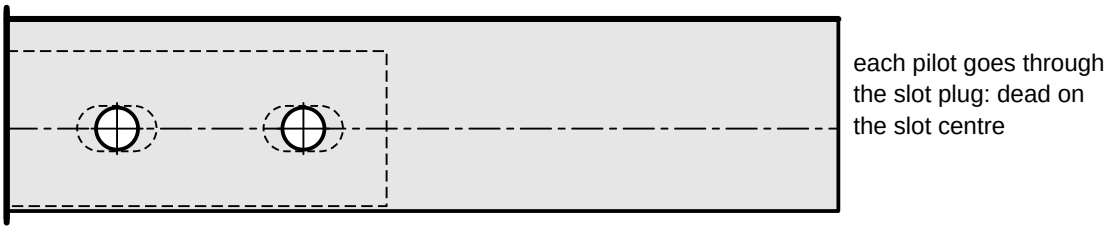
WHY THE HOLES ON BOTH RAILS MEET WITHOUT A JIG VERIFIED

1 · The same fitting is the template for all three placements: rail A heel on the 52.0 cm mark legs toward the far end, rail A heel on the 51.5 cm mark legs toward the wall end, rail B heel flush with the corner end.
2 · The slot plug puts every pilot on a slot centre, 29.15 and 78.45 mm from wherever the heel sat. So when a fitting is bolted on, its slots sit over the holes it made.
3 · Both fittings are identical, so the inside one fits the holes the outside one made.
4 · What makes the two rails meet at the corner is rail B's end against rail A's vertical leg with B's outer face on the 52.0 cm mark. The floor test (step 8) proves it before painting: every bolt must slide in by hand. If one won't, open that rail hole with a round file now, never under the desk.
The rail drill jig is only for the 19-line screw holes, on either rail; it plays no part in the bolt holes.

1 One fitting, three placements. Rail A, vertical leg:



Rail B, vertical leg:



2 Bolted: each slot sits over the hole it made (rail A from the knee space, rail B cut)

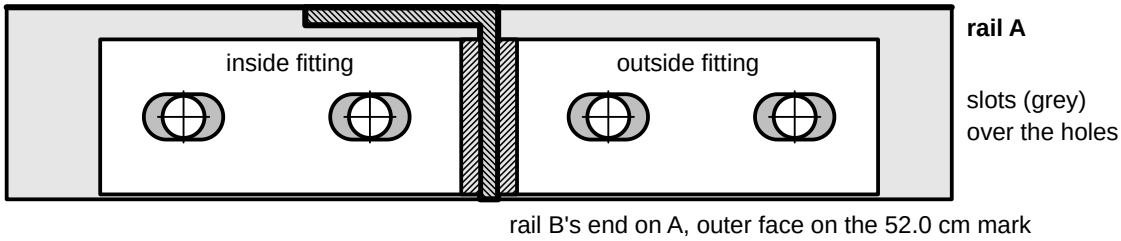
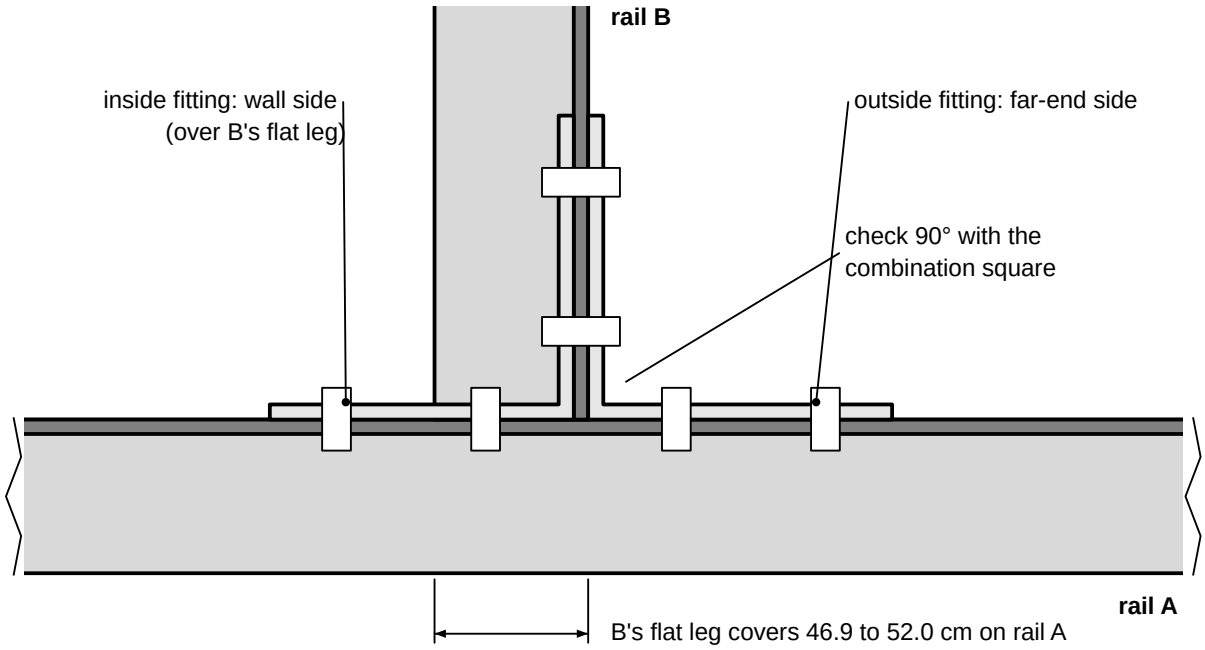


Fig. E-10 · Top: the one fitting in its three placements, heel marks heavy, slots dashed. Bottom: bolted at the corner, each slot over the hole it made. VERIFIED slot centres; SOURCE-DEFINED heel marks.

8 **Floor test.** Lay both rails upside down on the floor, flat legs down, vertical legs up. Butt rail B's corner end against the outer face of rail A's vertical leg so that rail B's outer face is on rail A's **52.0 cm** mark; rail B's flat leg then covers 46.9 to 52.0 cm on rail A's tape. Bolt both fittings on loosely: the outside one on the far-end side of rail B, the inside one on the wall side. Every bolt must slide in by hand. Check the corner with the combination square. If a bolt will not go in, open that hole with a round file now, not under the desk.

Plan from above. Both rails upside down on the floor: flat legs down, vertical legs up.
Bolts in loosely; every one must slide in by hand.



B's outer face on rail A's 52.0 cm mark; B's end touching A: no gap

Fig. E-11 · Floor test, plan from above. Bolts drawn as boxes. SOURCE-DEFINED

9 **Finish.** Take the fittings off. Degrease both rails, scuff with 120 grit, two coats of flat black enamel, 24 h to cure. The holes need no masking.

The rails are now ready for the tops: rail A goes on top A with its wall end at x = 1 cm (1 cm in from the taped end), rail B on top B with its corner end projecting 12 cm past the joint end. Chapter F bolts the corner once both tops stand.

F · Steel corner assembly

The two rails are joined at the inside corner by two gusseted strut fittings, one each side of rail B's end, bolted to the vertical legs. This bolted joint, not the joining plates, carries the inside corner of top B. Both tops are standing (build manual step 7); you work from below.

> **SEQUENCE: THE FITTINGS GO ON LAST** SOURCE-DEFINED
> Before the first fitting goes on, all of these must be done:
> 1 · The two surfaces are flush at the joint, leveled with the feet (never packed).
> 2 · Rail B's end touches the outer face of rail A's vertical leg with no gap, and rail B's outer face is on the 52.0 cm mark on rail A.
> 3 · Joining plates P1 and P2 are screwed across the joint. They align the tops; they do not carry the corner.
> 4 · Rail B's screws 1 and 2 are driven into top A. The inside fitting sits right under them and would block the driver.

	Part	Qty		Hardware	Qty
P06	Rail A, angle 213 cm, drilled (E)	1	**H08**	3/8-16 × 1½ in hex bolt, grade 5	6
P07	Rail B, angle 213 cm, drilled (E)	1	**H09**	3/8 flat washer, under each head and nut	12
P08	Gusseted 90° strut fitting, 5 mm, legs 100.5 × 41 mm, two 21 × 12 mm slots per leg VERIFIED	2	**H10**	3/8-16 nylon-insert lock nut	6

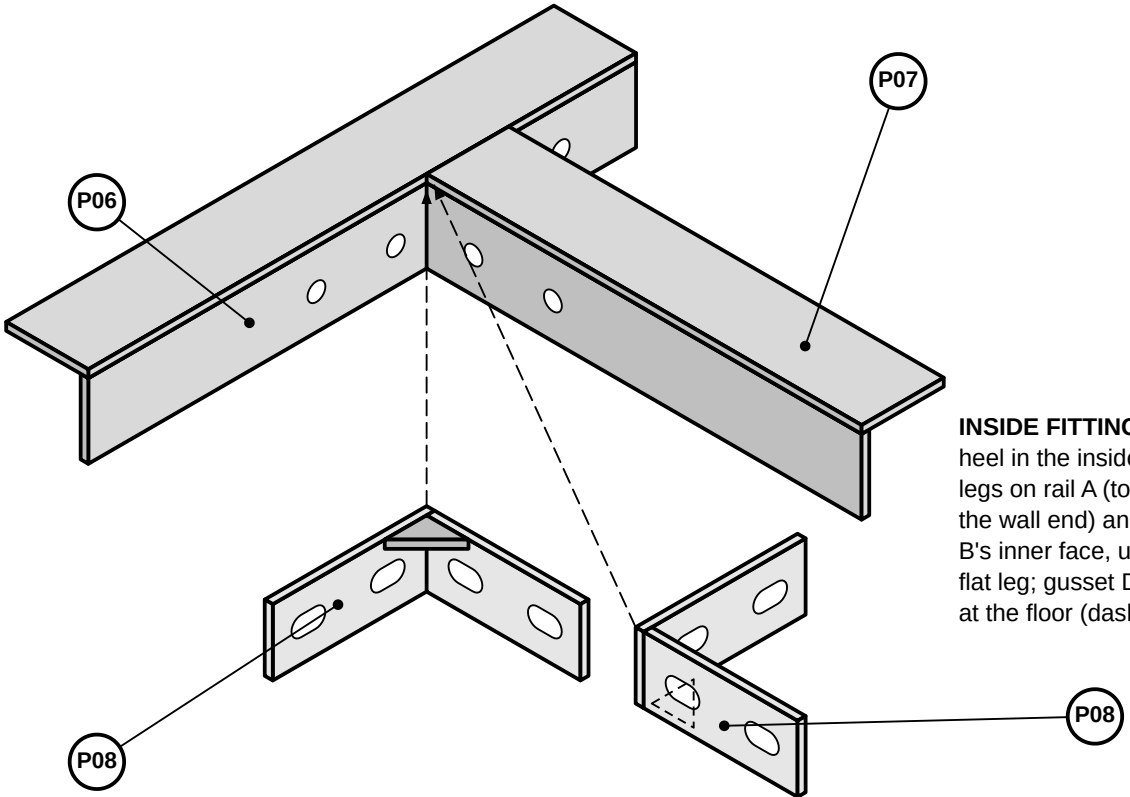


Fig. F-1 · The corner seen from the knee space with the tops lifted away. Both fittings are drawn pulled out of place; the dashed lines run from each heel to where it goes. The outside fitting fills the outside corner, the inside one the pocket under rail B's flat leg, gusset down. Neither is a mirror image: they are the same part, the inside one turned gusset-down. SOURCE-DEFINED placement.

Which bolt goes where

Bolts	Through	Head and washer	Washer and nut
O1, O2	outside fitting + rail A's vertical leg	fitting side, in the knee space	rail A's inner face, under its flat leg
I1, I2	inside fitting + rail A's vertical leg	fitting side, in the pocket under rail B	rail A's inner face, under its flat leg
C1, C2	outside fitting + rail B's vertical leg + inside fitting	outside fitting, knee side	inside fitting, in the pocket

Every bolt goes in **from the fitting side**: four through rail A's leg with one fitting each, two through rail B's leg with a fitting on each side.

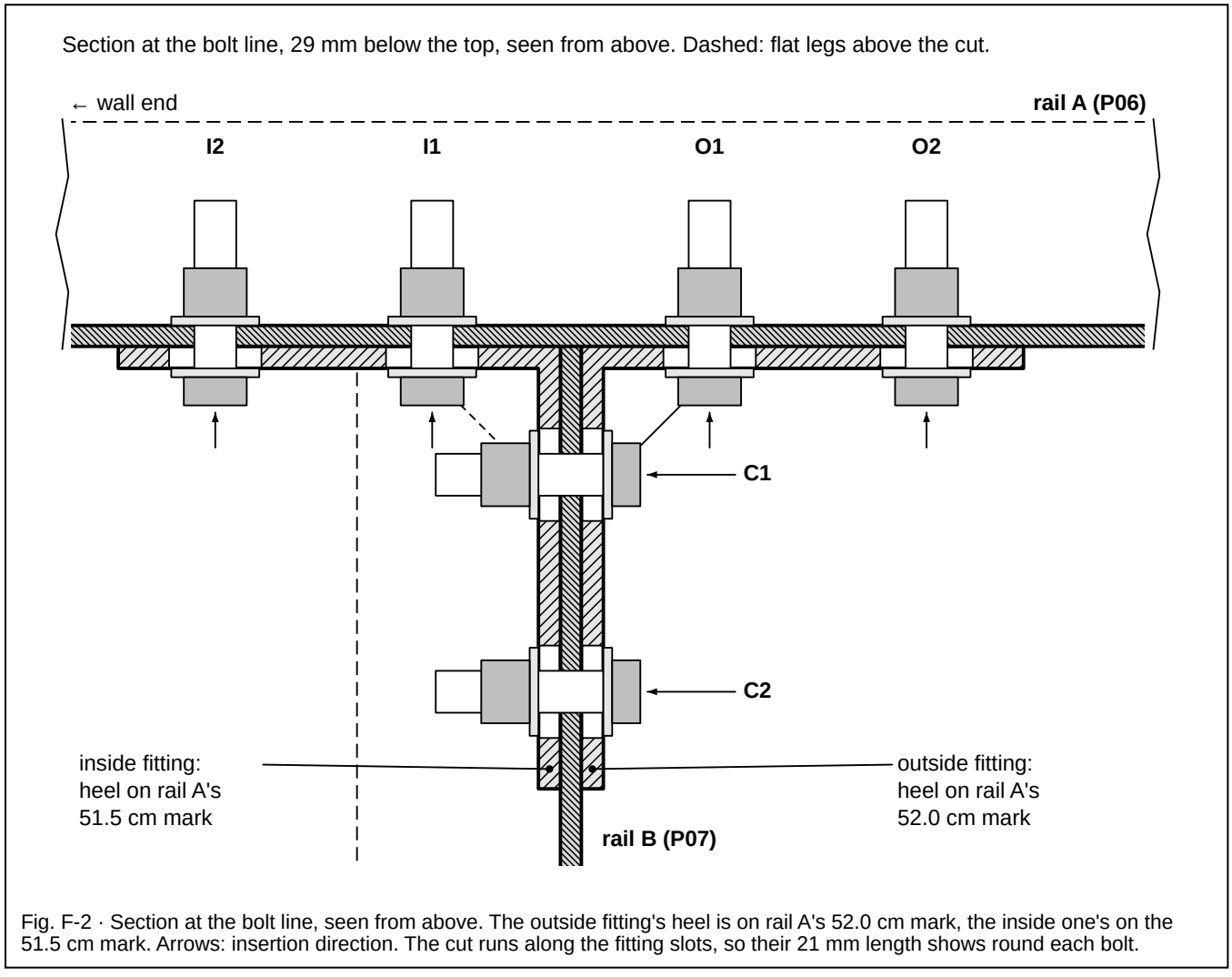


Fig. F-2 · Section at the bolt line, seen from above. The outside fitting's heel is on rail A's 52.0 cm mark, the inside one's on the 51.5 cm mark. Arrows: insertion direction. The cut runs along the fitting slots, so their 21 mm length shows round each bolt.

1

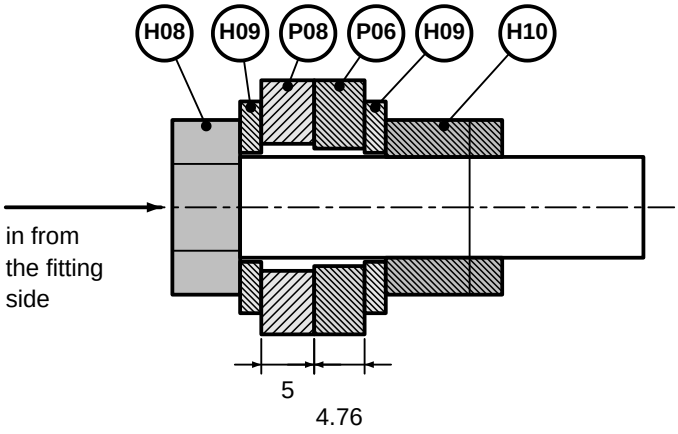
Outside fitting. Heel into the outside corner, one leg flat on rail A's vertical leg toward the far end, the other on rail B's outer face, gusset up, facing the knee space. Push O1 and O2 through from the fitting side with a washer under each head; washer and nut on rail A's inner face. Finger tight.

2

Inside fitting. Into the pocket: heel in the inside corner, one leg on rail A's vertical leg toward the wall end, the other on the inner face of rail B's vertical leg, **gusset pointing at the floor**. Gusset-up it would sit 1 mm under the head of rail B's screw 1. I1 and I2 from the fitting side, washers, nuts on rail A's inner face. Finger tight.

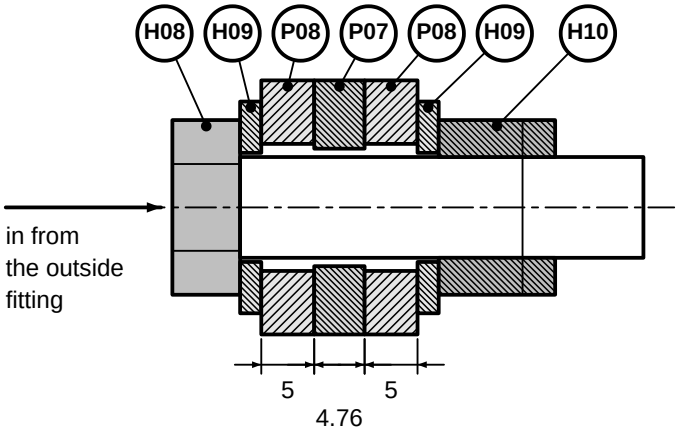
3 **Through bolts.** C1 and C2 from the outside fitting: through the outside fitting, rail B's vertical leg and the inside fitting, washer under the head, washer and nut in the pocket. Finger tight. All six bolts are now in; none is tight.

Rail A: 4 bolts, one fitting each (O1, O2, I1, I2)



fitting 21 × 12 slot (cut across): 2.5 mm play across, 11.5 mm along rail hole ⌀11.1 (7/16 in): 1.6 mm of play
so the joint holds by friction: tighten fully, 25 N·m

Rail B: 2 bolts through outside fitting, rail B and inside fitting (C1, C2)



head and washer on the outside fitting (knee side); washer and nut on the inside fitting, in the pocket under rail B's flat leg

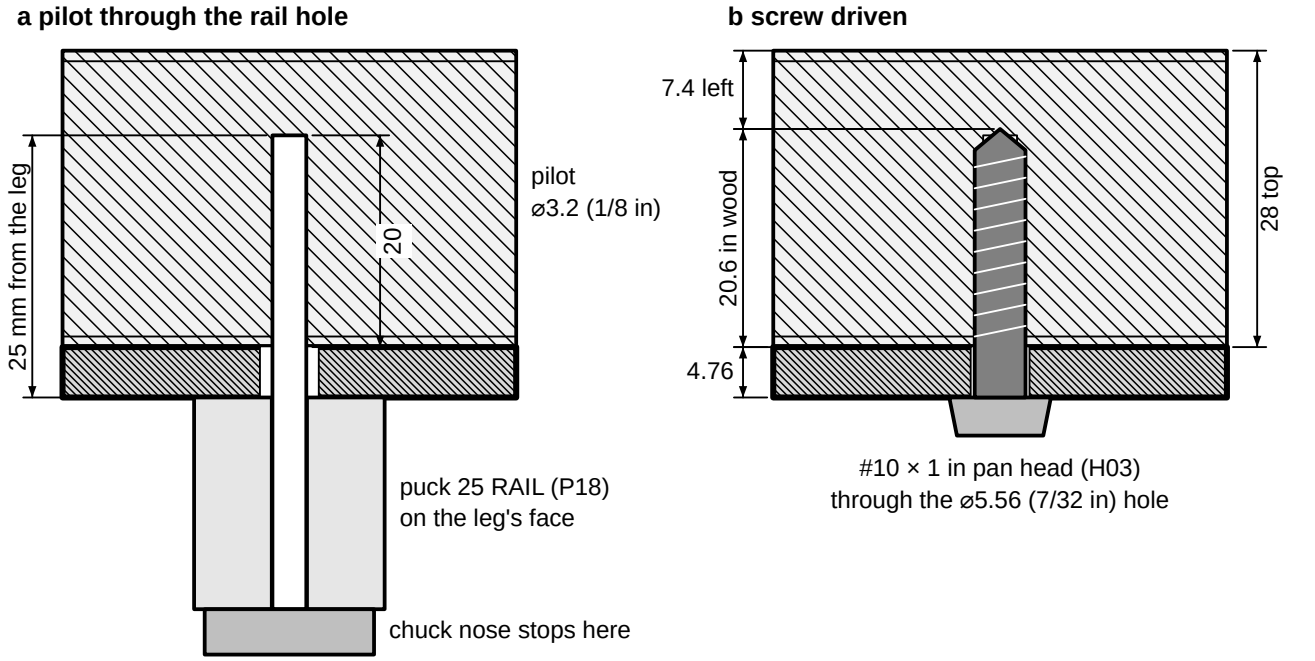
Fig. F-3 · Bolt stack-ups, cut along the bolt. All mm. Plates drawn cut off. The fitting's slots are 21 × 12; the cut shows their 12 width. VERIFIED fitting; SOURCE-DEFINED bolts.

4 **Check again, then tighten all six.** Fingertip across the joint: no step. Rail B's end still touching rail A. Then tighten every bolt fully:

! **TIGHTENING: 25 N·m (18 ft·lb)** SOURCE-DEFINED
With a torque wrench, **25 N·m (18 ft·lb)**, under the standard dry torque even for a grade 2 bolt. Without one, pull as hard as one hand can on a 15 cm wrench, then stop.
Why tight, not snug: this is a friction joint. The fittings' 21 × 12 slots leave 2.5 mm of play across and 11.5 mm along each bolt, the 7/16 in rail holes 1.6 mm, so only the clamping force keeps rail B's corner from settling.
When: only after the two surfaces are flush and rail B's end touches rail A.
Later: if the corner ever drops a millimetre, retighten these six bolts.

How the rails hold the tops

Each rail is screwed to its top through the flat leg: nine #10 × 1 in pan-head Robertson screws (H03) in rail A, ten in rail B, 19 in all. Rail B's holes 1 and 2 land in top A (its first 12 cm lies under top A) and are driven in build-manual step 7.4, before the fittings; the rest go in with the tops face down.



Never more than 21 mm into the 28 mm top. Drawn upright, as for rail B's holes 1 and 2.

Fig. F-4 · One rail screw in section. All mm. The pilot is 1/8 in, 25 mm from the flat leg's face (the face the puck stands on), which is 20 mm into the wood; the screw passes the 4.76 mm steel and goes 20.6 mm into the 28 mm top, leaving 7.4 mm under the laminate. SOURCE-DEFINED

Item	mm	Source
Top thickness	28	SOURCE-DEFINED MEASURE ON HARDWARE calipers (if it reads 27 mm, use screws 1 mm shorter everywhere in the top)
Flat leg	4.76	SOURCE-DEFINED
Screw into the top	20.6	SOURCE-DEFINED; limit 21 mm into any top
Pilot depth from the leg's face	25	printed drill guide, puck 25 RAIL SOURCE-DEFINED

Settling check. Lay a straightedge across the joint at the inside corner and write down any step. Check again after two weeks and after the first month; if the corner has settled, retighten the six fitting bolts.

G · Legs and cabinet support

Six OLOV legs and eight feet plates go on the tops while they lie face down on a blanket; the cabinets stand by the walls; then each top is stood on its bare cabinet once so its pads can mark where the eight locating rings go.

No.	Part	No.	Hardware	Qty
P01 **P02**	Top A and top B, EKBACKEN 249 × 65 cm, 28 mm thick	**H01**	M4 x 16 mm machine screw, IKEA 108443 (with the OLOV legs)	30
P03 **P04**	ALEX 1, drawer unit; ALEX 2, storage unit with door (36 × 58 × 70 cm)	**H02**	#8 x 5/8 in pan head screw (feet plates, 4 each)	32
P05	OLOV leg x 6, 105 mm plate	**H06**	#6 x 5/8 in flat head (rings, 3 each)	24
P09 **P10**	Round adjustable foot 4 in x 8; printed locating ring x 8	**P18** printed tools: leg jig, feet corner jig, edge T-square, ALEX corner template, drill guide (tower + pucks), foot height gauge, wall gauge		

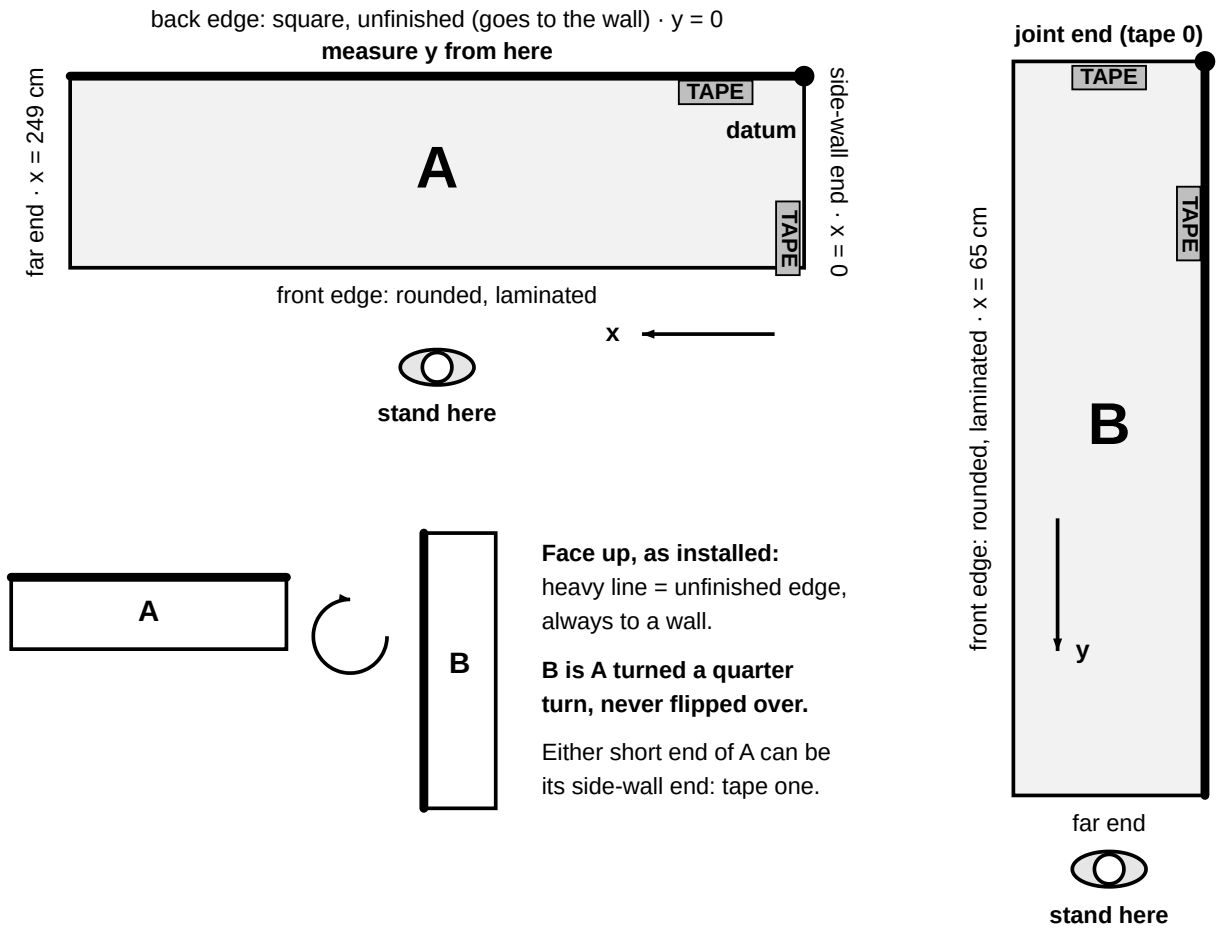


Fig. G-1 How each top lies while you mark it: face down, laminate on the blanket. Heavy line: the square, unfinished long edge, which always goes to a wall (y = 0 on A, x = 0 on B). Tape the unfinished edge and the end you measure from, and stand where the person symbol is; the drawings in this chapter then match what you see, with x growing to the left. Datum: top A's back-left corner (0, 0) as installed.

Decide the desk height first

The legs are pre-set in step 1, so choose now. The desk is built at 83.0 cm as planned. Optional printed ALEX feet (2, 4 or 6 cm tall, ⌀56 mm, `parts/accessories/alex-foot.scad`) raise everything by their height; nothing on the desk moves. SOURCE-DEFINED

ALEX feet	Underside of the tops (cm)	Top surface (cm)	Every OLOV set to (cm)	Rail lower edge (cm)
none (as planned)	80.2	83.0	80.2	75.1
2 cm	82.2	85.0	82.2	77.1
4 cm	84.2	87.0	84.2	79.1
6 cm	86.2	89.0	86.2 (OLOV max 90)	81.1

G · Underside support layout

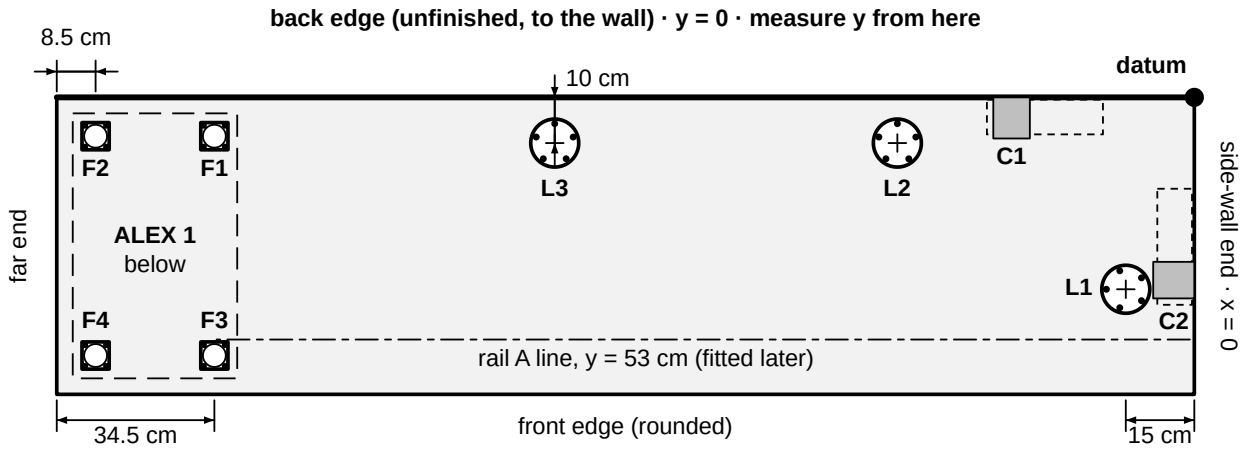


Fig. G-2 Top A face down, back edge away from you: x grows to the left from the side-wall end. Circles: OLOV plates with their five holes, one pointing at the wall (L1's pattern is turned). Squares: feet plates. Grey: clamp zones C1, C2, keep clear; thin dashed: the clamp pad outlines; dash-dot: rail A's line. Datum (0, 0) at the top-right corner of this view. Drawing labels in cm.

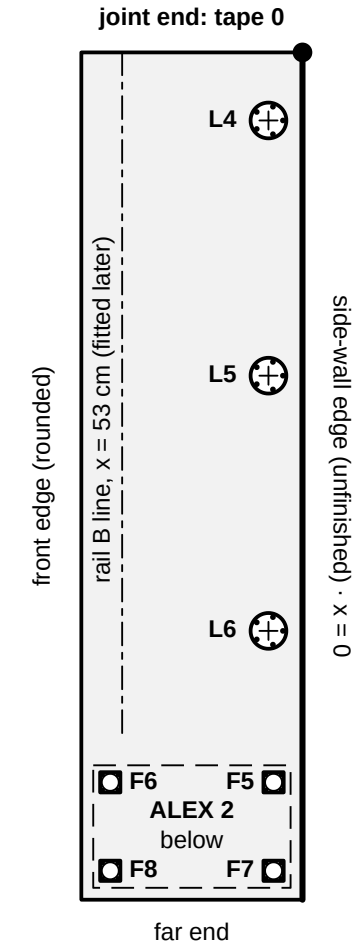


Fig. G-3 Top B face down, joint end away from you. Hook the tape on the joint end; the table gives the reading for each mark.

Mark	x	y	Tape reading
L1	15	42	42 from A's back edge
L2	65	10	65 from the side-wall end
L3	140	10	140 from the side-wall end
L4	10	85	20 from B's joint end
L5	10	160	95 from the joint end
L6	10	235	170 from the joint end
F1	214.5	8.5	214.5 from the side-wall end
F2	240.5	8.5	corner jig, far back corner
F3	214.5	56.5	214.5 from the side-wall end
F4	240.5	56.5	corner jig, far front corner
F5	8.5	279.5	214.5 from the joint end
F6	56.5	279.5	214.5 from the joint end
F7	8.5	305.5	corner jig, far wall corner
F8	56.5	305.5	corner jig, far front corner

x, y and tape readings in cm; x, y from the datum **SOURCE-DEFINED**. Feet are 5 cm in from each ALEX corner, which is 8.5 cm in from the top's edges. F1 F3 F5 F6 are also 34.5 cm from the far end.

Leg plates are ⌀105 mm and feet plates 59.5 mm square. L1's plate ends short of rail A and clear of the clamp 2 jaw; feet plates F3 and F6 stay 5 mm clear of the rails.

> Nothing here goes in the clamp zones C1 (36–44 cm from the side-wall end, the 9 cm next to the back edge) or C2 (36–44 cm from the back edge, the 9 cm next to the side-wall end).

G · Step 1: legs

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> Calipers on both tops before the first hole: expect 28 mm. If a top reads 27 mm, use screws 1 mm shorter. No screw may go deeper than 21 mm into a top.

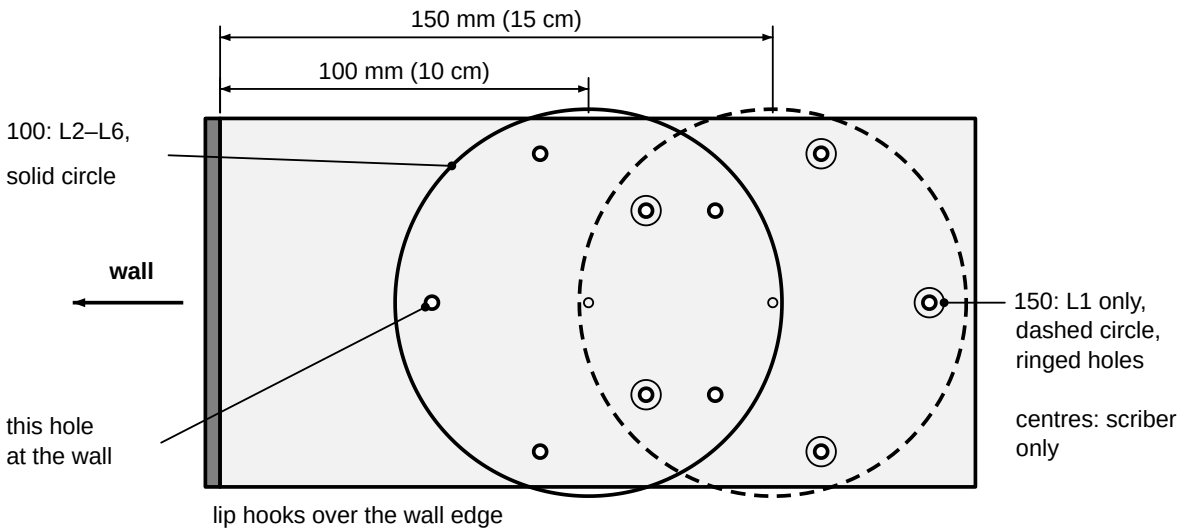


Fig. G-4 Leg jig (P18), seen from above in use, lip hooked down over the wall edge. The 100 pattern (10 cm from the edge; solid plate circle, plain holes) is for L2 to L6 and puts one hole at the wall; the 150 pattern (15 cm; dashed circle, ringed holes) is for L1 only.

Leg	Top	Hook the lip over	Tape station (cm)	Pattern
L1	A	side-wall end (x = 0)	42 from the back edge	150, ringed holes
L2 L3	A	back edge (y = 0)	65 (L2) and 140 (L3) from the side-wall end	100
L4 L5 L6	B	side-wall edge (x = 0)	20 (L4), 95 (L5), 170 (L6) from the joint end	100

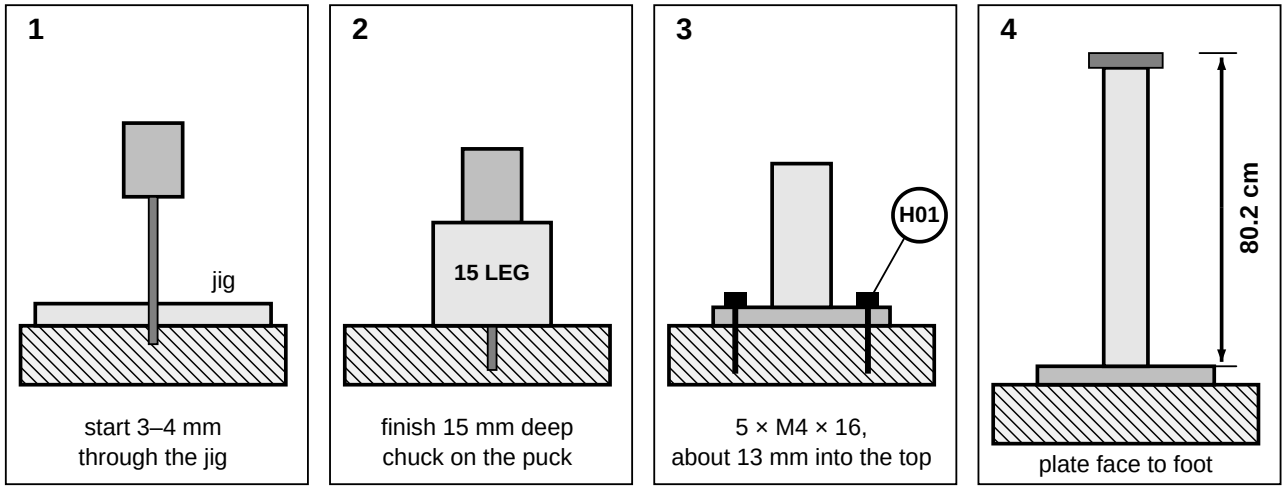


Fig. G-5 One leg, four operations (steps 1a–1d below). The drill guide's SET tower sets the 1/8 in bit first.

1

a Jig on the station, start all five pilots 3–4 mm through its holes with the 1/8 in bit. **b** Lift the jig; finish each pilot 15 mm deep with the **15 LEG** puck. **c** Screw the OLOV plate on with five **H01** M4 × 16 screws (about 13 mm into the top). **d** Pre-set the leg to **80.2 cm** from plate face to foot (or the table's number on page G-1 if you fit ALEX feet). Six legs, 30 screws.

OLOV plate ⌀105 mm, five ⌀5.5 mm holes on an ⌀85 mm circle, neighbouring holes 50 mm apart; leg range 60–90 cm. VERIFIED
105 mm plate measured 22 Oct 2026; hole pattern from a shared OLOV template, printed and test-fitted.

G · Step 2: feet plates

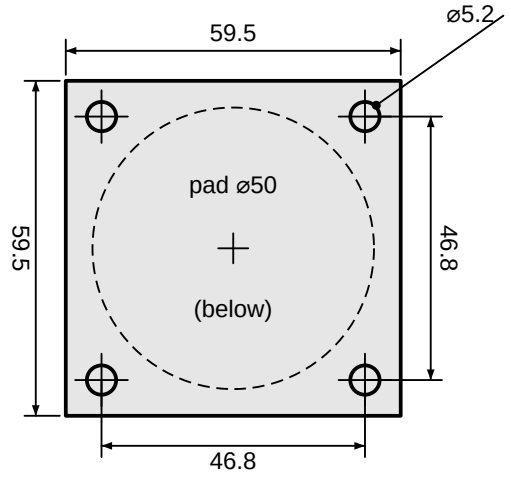


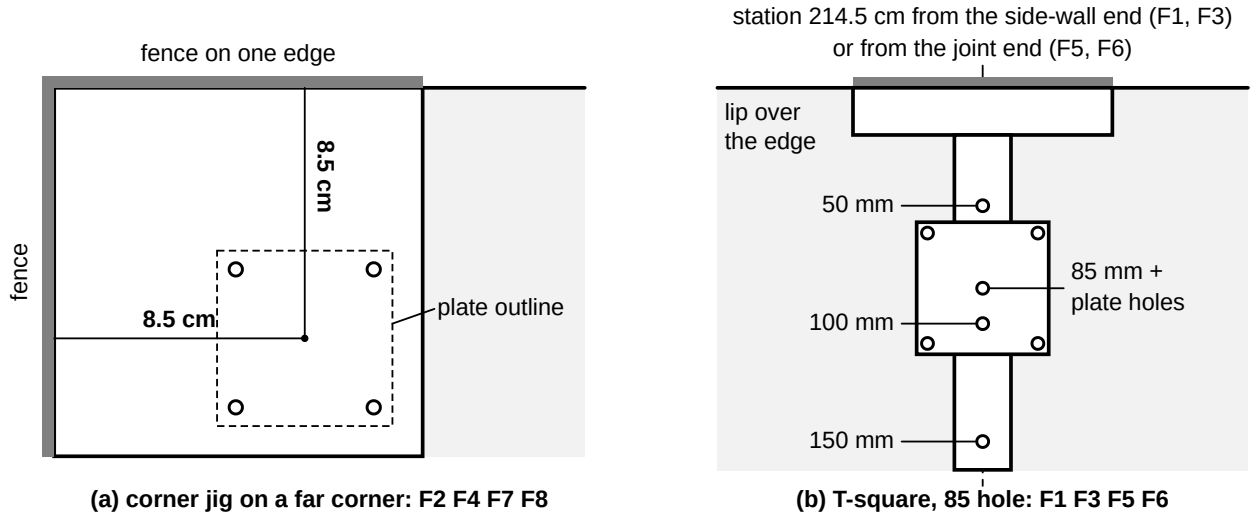
Fig. G-6 Feet plate (P09) from the screw side, mm. VERIFIED on the purchased plate, 23 Oct 2026: 41.6 mm between two neighbouring holes (inside jaws), 52.0 mm across them (outside jaws).

Feet	Tool	Placement
F2 **F4**	feet corner jig	top A's two far corners, fences on both edges
F1 **F3**	T-square, 85 hole	lip over A's back edge (F1) or front edge (F3), station 214.5 cm from the side-wall end
F7 **F8**	feet corner jig	top B's two far corners
F5 **F6**	T-square, 85 hole	lip over B's side-wall edge (F5) or front edge (F6), station 214.5 cm from the joint end

ASSUMPTION / FIELD CHECK

ASSUMPTION / FIELD CHECK

The feet's load rating is unknown; the design wants 50 kg or more each.



(a) corner jig on a far corner: F2 F4 F7 F8

(b) T-square, 85 hole: F1 F3 F5 F6

Fig. G-7 Both tools put the foot's four plate holes (1/8 in, on the 46.8 mm square) square to the edges in one placement, seen face down. Centre holes are for a scriber only. T-square offsets are in mm, as engraved on it.

2 Start the four pilots 3–4 mm through the tool's plate holes with the 1/8 in bit, lift the tool, finish 15 mm deep with the **15 LEG** puck. Screw the plate on with four **H02** #8 × 5/8 pan heads (14 mm into the top), pad pointing away from the top. Eight plates, 32 screws.

3 Wind each pad out until the printed foot height gauge (102.0 mm inside) just slides over the foot from the side with no rock. All eight then match without a ruler.

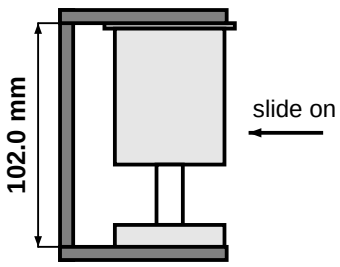


Fig. G-8 Foot height gauge.

G · Step 4: place the ALEX units

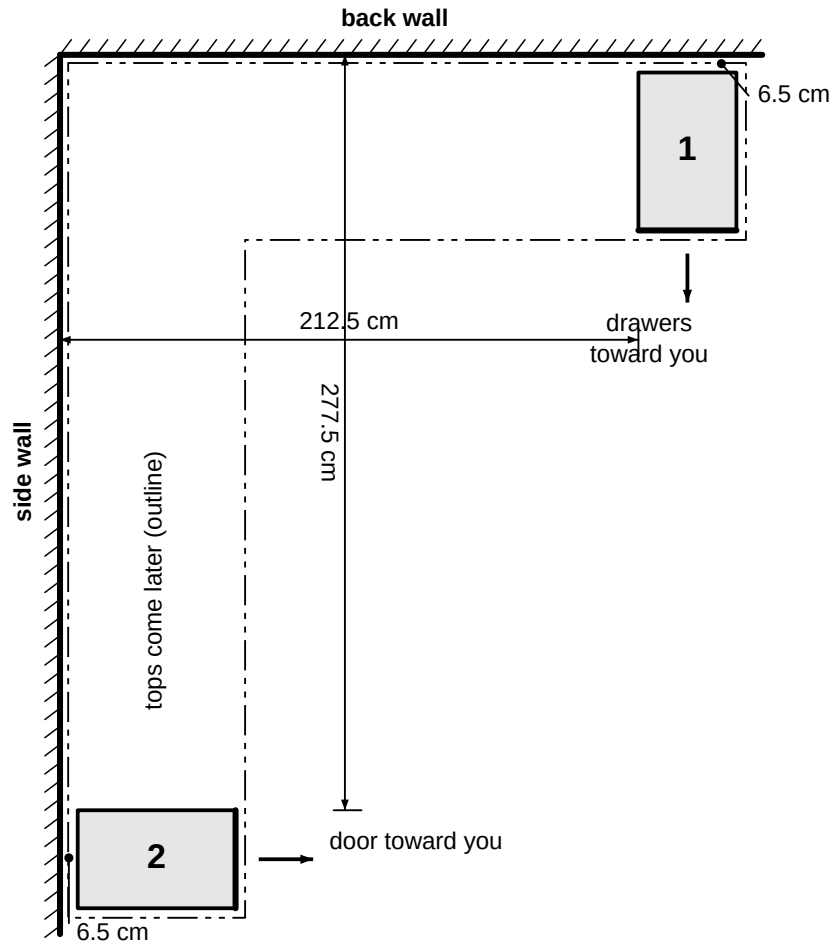


Fig. G-9 Plan from above. The tops are not up yet, so the cabinets are set from the walls (cm). Heavy line: the drawer fronts of ALEX 1 face you (+y); the door of ALEX 2 faces you (+x).

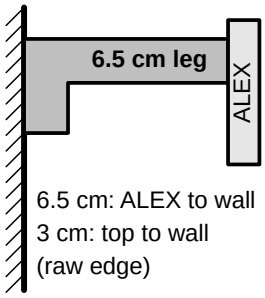


Fig. G-10 Wall gauge (P18): the 6.5 cm leg just fits between the wall and a cabinet; the 3 cm leg checks a top's raw edge.

4

Stand **ALEX 1** (P03) with its back 6.5 cm off the back wall and its near side 212.5 cm from the side wall, drawers toward you. Stand **ALEX 2** (P04) with its side 6.5 cm off the side wall and its near end 277.5 cm from the back wall, door toward you. Painter's tape on the floor round each one saves guessing. **No rings yet:** they go on in step 5, traced from the pads.

Once the tops are on, 3.5 cm of top shows all round each cabinet and the tops stand 3 cm off the walls. That is only roughly right at this point; step 5 sets it exactly by pushing the cabinet under the standing top.

Optional ALEX feet (decided on page G-1): fit them now, before the cabinets stand. Unscrew the stock feet, lay the unit on its back, hook the corner template over each bottom corner and use its inner three holes (r 14.5 mm). Pilot 5/64 in, 12 mm deep with the 12 ALEX puck; three #6 × 5/8 flat heads per foot down the wells, four feet per unit, all the same height.

! **ASSUMPTION / FIELD CHECK** ASSUMPTION / FIELD CHECK
The drawings take an ALEX as 70 cm tall, which is what makes the feet 102 mm between the ALEX top and the underside at 80.2 cm. Measure your units, with and without their stock feet, before you wind the feet in step 3.

G · Step 5: rings, by tracing the pads

A 50 mm pad in a 52 mm ring leaves about 1 mm each side, closer than a cabinet can be placed with a tape. So the pads mark their own rings: the top stands on the bare cabinet first. Two people.

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> **Pad diameter:** calipers on one pad before you print the rings. The ring is bored 52 mm for a 50 mm pad. If yours differs, change `pad_d` in `parts/structure/foot-ring.scad` and reprint.

VERIFIED ALEX top panel 15.9 mm (5/8 in), measured 23 Oct 2026: the #6 × 5/8 screws reach 13.4 mm into it and leave 2.5 mm; the 12 mm pilot leaves 3.9 mm.

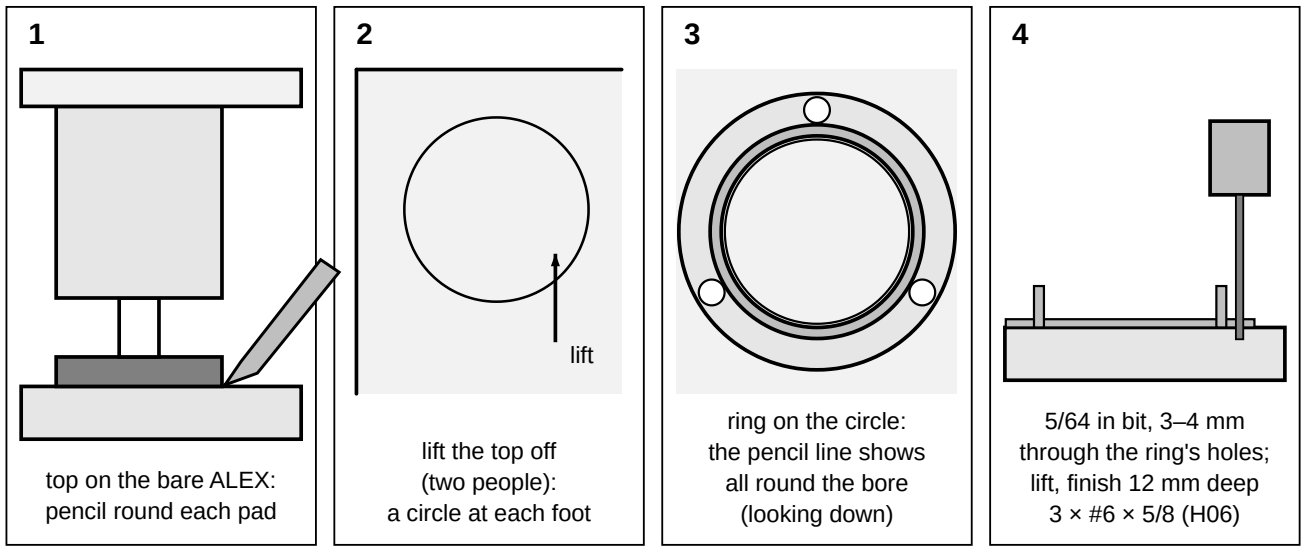


Fig. G-11 Fitting one ring (steps 5a–5d). Panels 1 and 4 are sections, mm; panels 2 and 3 are seen from above.

5

a Stand the top on its legs with its four pads on the **bare** ALEX top. Push the cabinet, never the top, until 3.5 cm of top shows all round it and the wall gauge's 6.5 cm leg just fits between the wall and the cabinet. Pencil round each pad. **b** Lift the top off onto the blanket (two people). **c** Set a ring on each circle and look down through its open bottom until the pencil line shows all the way round the bore. **d** Start the three pilots 3–4 mm through the ring's own holes with the 5/64 in bit, lift the ring, finish 12 mm deep with the **12 ALEX** puck, put the ring back and drive three **H06** #6 × 5/8 flat heads. Then stand the top up again: the pads drop into the rings. From now on move the cabinet, not the top.

G · The ring, and the check

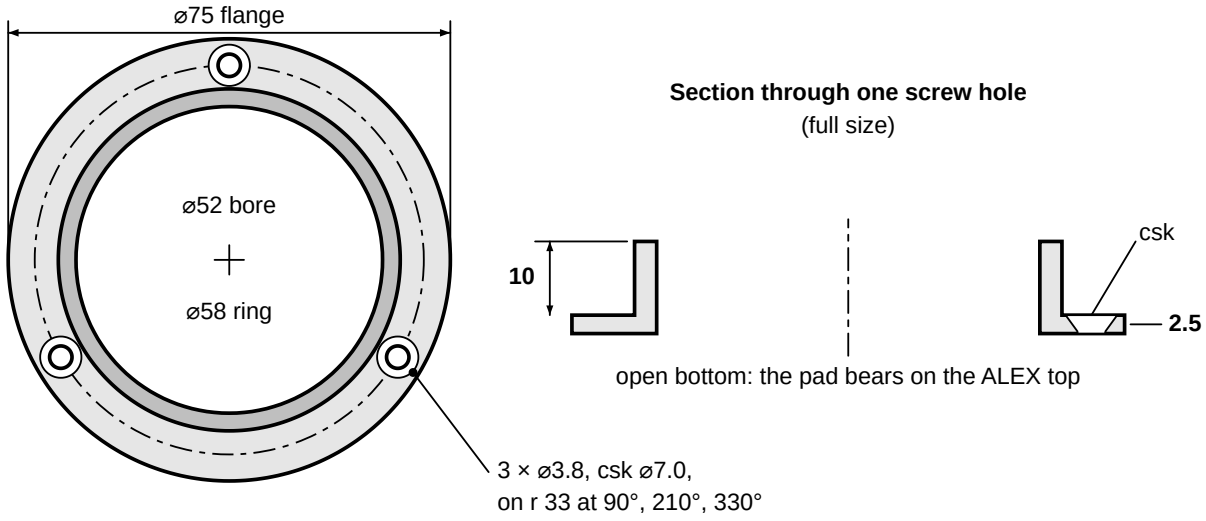


Fig. G-12 Printed locating ring (P10), from `parts/structure/foot-ring.scad`. VERIFIED from its own geometry file, mm: bore ø52, ring ø58 (3 wall) and 10 tall on a ø75 × 2.5 flange, three ø3.8 holes countersunk ø7.0 on r 33. Print flange-down, PETG or ABS.

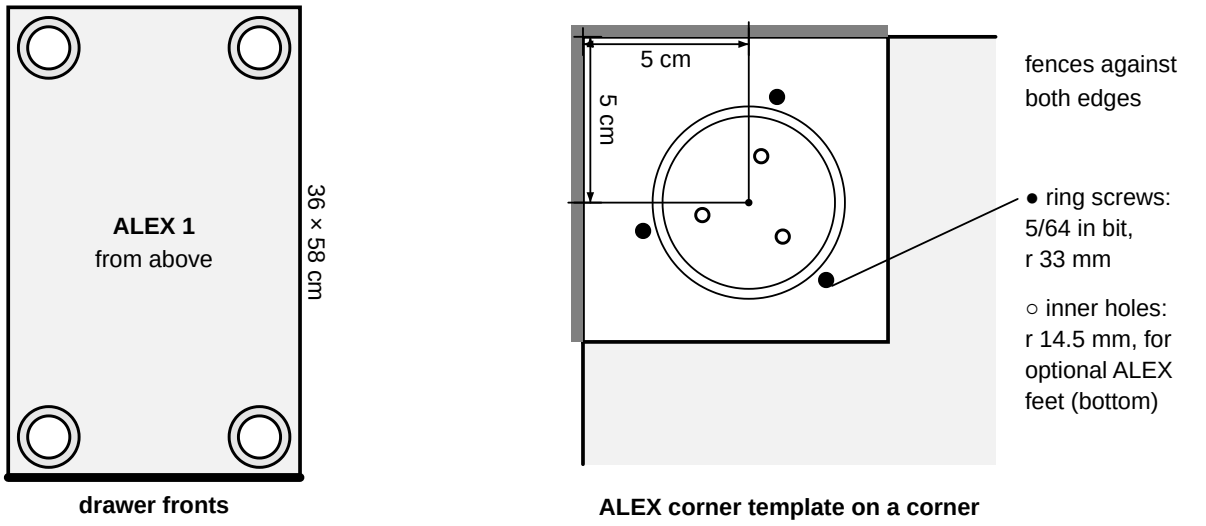


Fig. G-13 The check, after step 5. Left: ALEX 1's top with its four rings, centres 5 cm in from each corner; ALEX 2 is the same at all four corners. Right: drop the ALEX corner template over a corner, fences against both edges: its ring outline should sit over the fitted ring. It is a check, not the way to place the rings.

If a ring is clearly off its template outline, the cabinet was not 3.5 cm in from the top's edges when you traced. Lift the top off, unscrew that cabinet's rings, set the cabinet again and trace again.

G · Foot, pad and ring in section

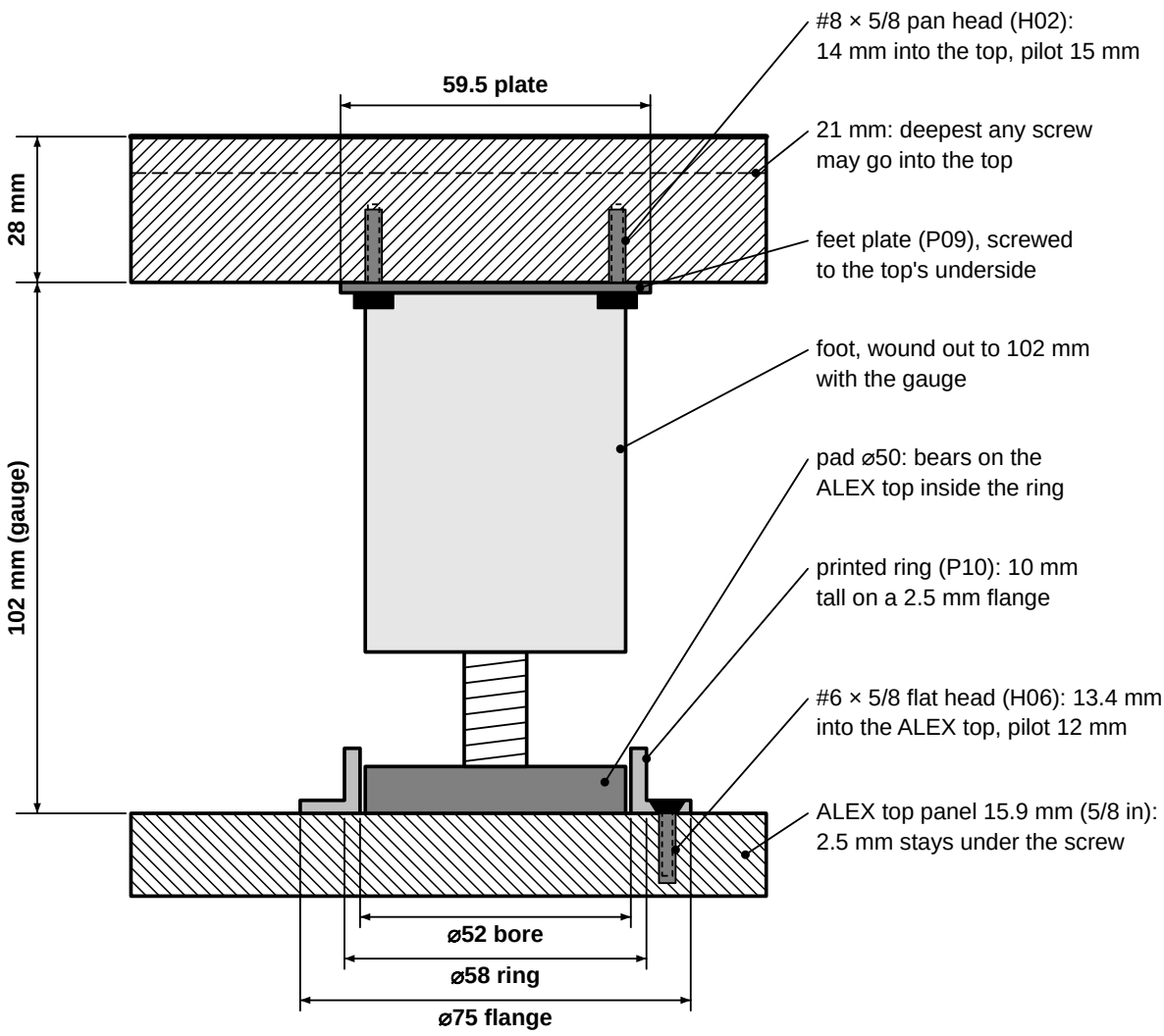


Fig. G-14 Section through one foot, its ring and one ring screw (mm). Hatched: the top above and the ALEX top panel below, both cut. The foot's internal shape is drawn schematically: only its 102 mm setting and ⌀50 mm body and pad are known. Heights above the floor: surface 83.0 cm, underside 80.2 cm, ALEX top 70 cm ASSUMPTION / FIELD CHECK.

Dimension	mm	Confidence
Top thickness	28	MEASURE ON HARDWARE expect 28 mm
Foot, plate face to pad (gauge)	102	SOURCE-DEFINED
Feet plate, hole spacing	59.5 / 46.8	VERIFIED
Pad diameter	50	MEASURE ON HARDWARE before relying on the 52 mm bore
Ring bore / ring / flange	52 / 58 / 75 × 2.5	VERIFIED geometry file
ALEX top panel thickness	15.9	VERIFIED 5/8 in, 23 Oct 2026; 2.5 mm under each screw

G · Load path

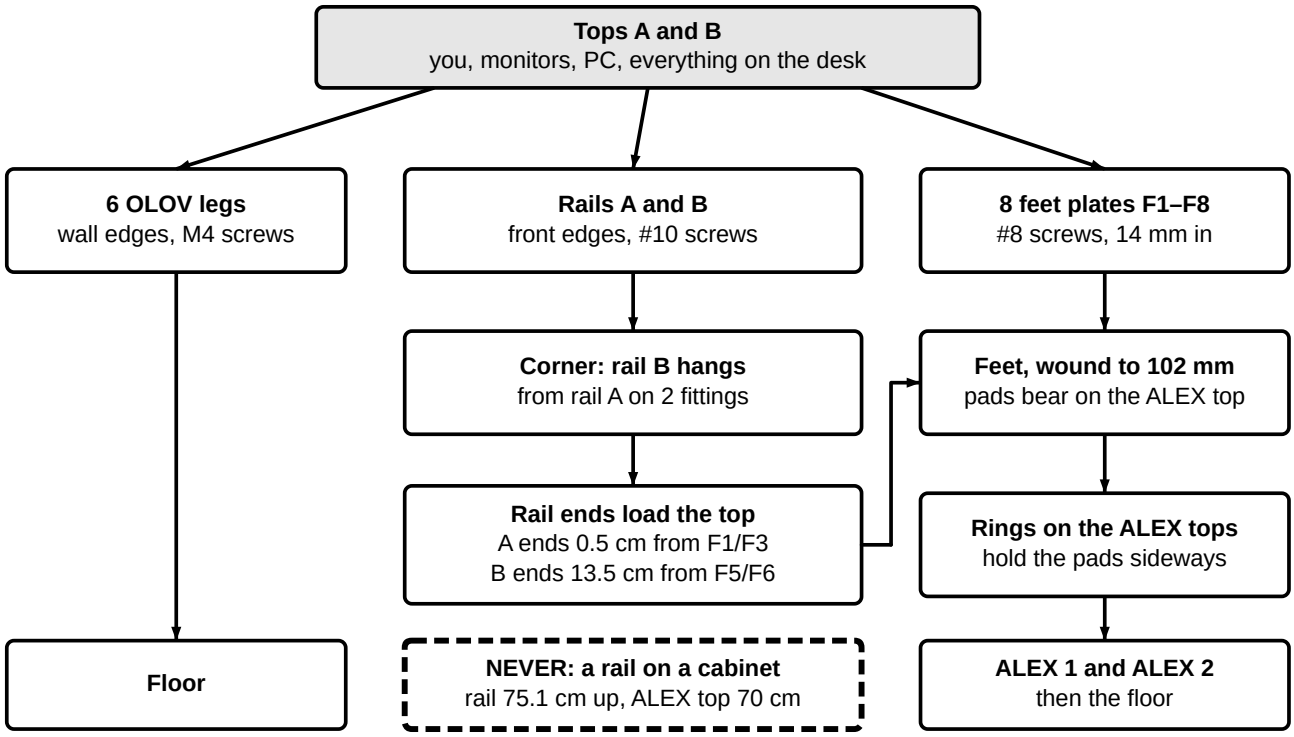
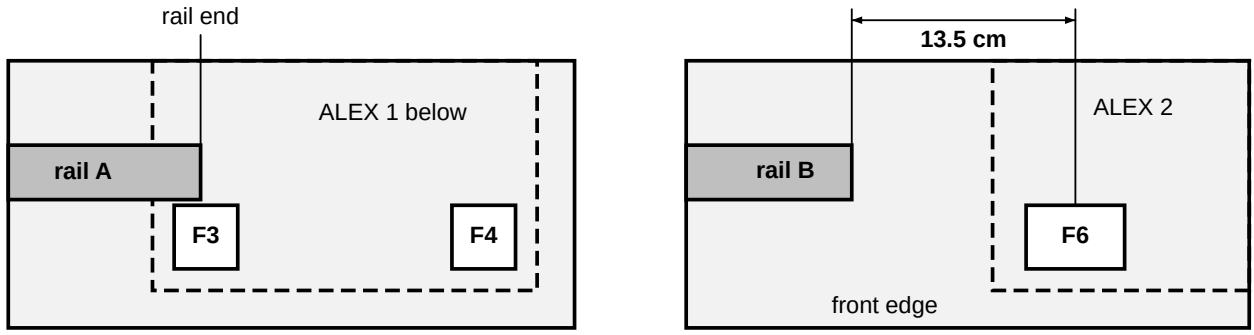


Fig. G-15 What carries what. The legs and the feet carry the tops to the floor. The rails carry the front edges; their ends load the top, and the top loads the nearest feet plates. The pads bear on the ALEX tops; the rings only stop them sliding.



(a) rail A's far end, plan
rail end x 214 cm
F1/F3 centres x 214.5 cm

(b) rail B's end, plan (y to the right)
rail end y 266 cm · ALEX 2 face y 274.5 cm
F5/F6 centres y 279.5 cm

Fig. G-16 Where the rails stop (cm from the datum; front edges at the bottom). (a) Rail A ends at x 214 cm, 0.5 cm short of the F1/F3 centres at 214.5 cm, beside F3's plate. (b) Rail B ends at y 266 cm; the top bridges the 13.5 cm to F5/F6.

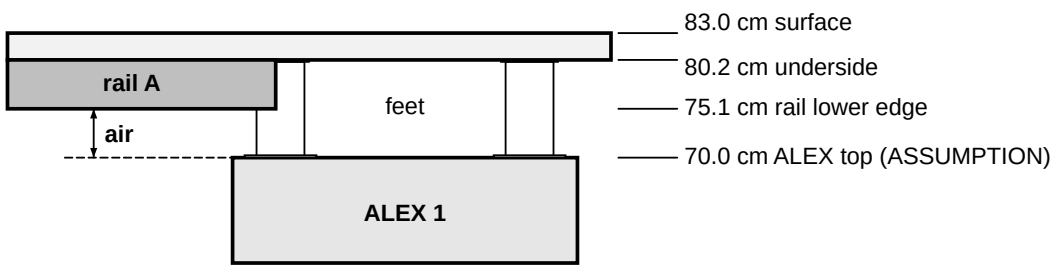


Fig. G-17 Elevation at ALEX 1, seen from the room. Rail A's end passes over the cabinet in plan but stays in the air above it: its lower edge is 75.1 cm above the floor, the ALEX top 70 cm.

G · The assembled underside

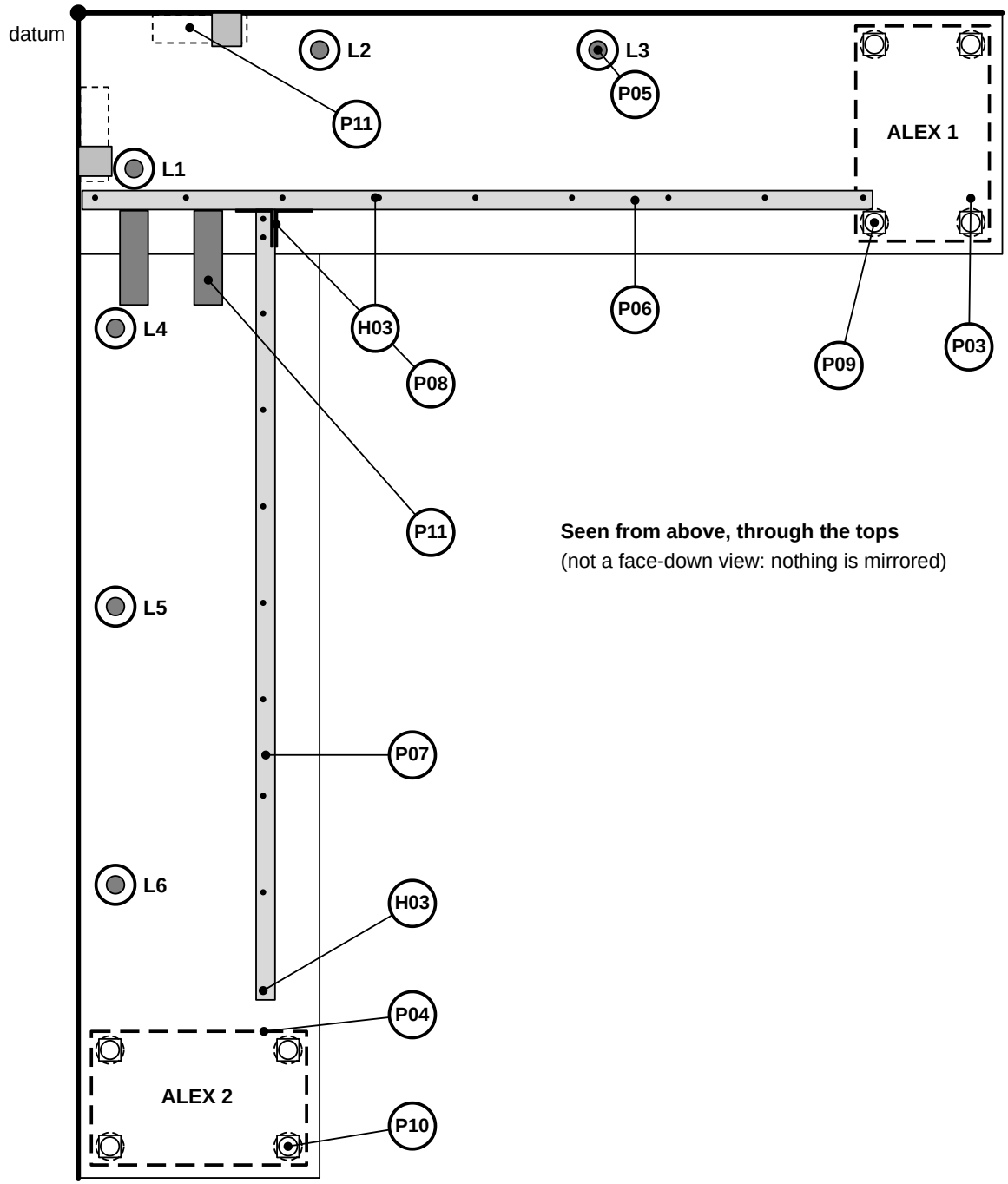


Fig. G-18 Everything under the tops once the desk is built, seen from above through the tops. Datum (0, 0) at top A's back-left corner. Rails, corner fittings and joining plates are fitted in later chapters; they are shown here to check against.

No.	Part	No.	Part
P03	ALEX 1, drawer unit 36 × 58 × 70 cm	**P08**	Gusseted 90-degree strut fitting x 2
P04	ALEX 2, storage unit with door 36 × 58 × 70 cm	**P09**	Round adjustable foot 4 in x 8
P05	OLOV leg x 6, 105 mm plate	**P10**	Printed locating ring x 8
P06	Rail A, steel angle 213 cm	**P11**	Mending plate 10 x 3 x 0.12 in x 4 (two joining, two clamp pads)
P07	Rail B, steel angle 213 cm	**H03**	#10 x 1 in pan head, Robertson (rails into the tops), 19

H · Joining and positioning the countertops

Two people. Both tops are face down with legs, feet and rails fitted, every OLOV pre-set to 80.2 cm, every pad set with the foot height gauge (102 mm). The ALEX units stand near their places with no rings on yet.

Parts	Hardware	Tools
P01 top A · P02 top B · P03 ALEX 1 · P04 ALEX 2 · P05 OLOV leg · P06 rail A · P07 rail B · P08 strut fitting × 2 · P09 foot · P10 ring × 8 · P11 mending plate × 2 (joining)	H06 #6 × 5/8 in flat head × 24 (rings) · H03 #10 × 1 in × 2 (rail B holes 1, 2) · H04 #10 × 5/8 in × 8 · H05 #10 washer × 8 · H08 3/8-16 × 1-1/2 in bolt × 6 · H09 3/8 washer × 12 · H10 3/8-16 nylon nut × 6	Drill guide (SET tower, pucks 12 ALEX, 25 RAIL and 12 PLATE), 5/64 and 1/8 in bits, sharp pencil, wall gauge, Robertson #2 on a 3 in extension, level, straightedge, tape, torque wrench or 15 cm wrench

1

Stand top A on the bare ALEX 1. The ALEX units have no rings yet. With a helper, turn top A over and stand it on its legs with its four pads on the bare ALEX top: lower the ALEX end first, then the leg end. Push the ALEX unit, never the top, until 3.5 cm of top shows all round the cabinet and the wall gauge's ALEX 65 leg (6.5 cm) just fits between the wall and the ALEX.

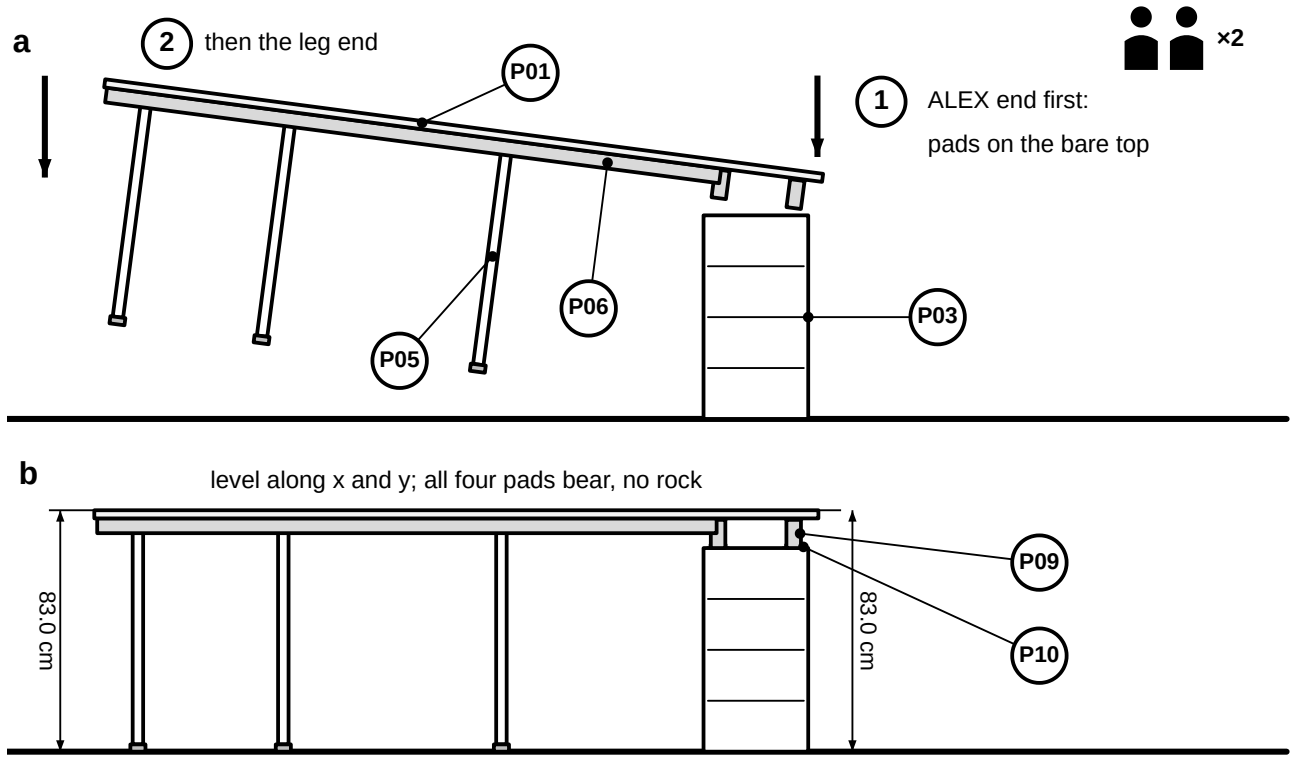


Fig. H-1 · Top A from the front (1:26). a: step 1, ALEX end first, the pads land on the bare ALEX top; b: step 3, back on its legs with the pads in their rings, level, 83.0 cm at both ends. Rail A hangs under the top; it never touches ALEX 1.

!

Push the cabinet, never the top.
Top A is only positioned by moving ALEX 1 under it. Once the rings are on (step 2) and the pads are in them, top A cannot be pushed at all: if anything is off, lift the ALEX end a few millimetres and move the ALEX unit.

2

Fit the rings by tracing.

1. With top A standing where step 1 left it, draw round each of the four pads on the ALEX top with a sharp pencil.
2. Two people: lift top A off and lay it face down on the blanket.
3. Set a ring (P10) on each pencil circle, flange down. Look down through its open bottom and move it until the pencil line shows all the way round inside the ⌀52 mm bore.
4. Hold it there. Start the three 5/64 in pilots through the ring's own holes, 3–4 mm deep, lift the ring, and finish each pilot 12 mm deep with the drill guide's **12 ALEX** puck.
5. Put the ring back and drive three H06 #6 × 5/8 in flat-head screws.

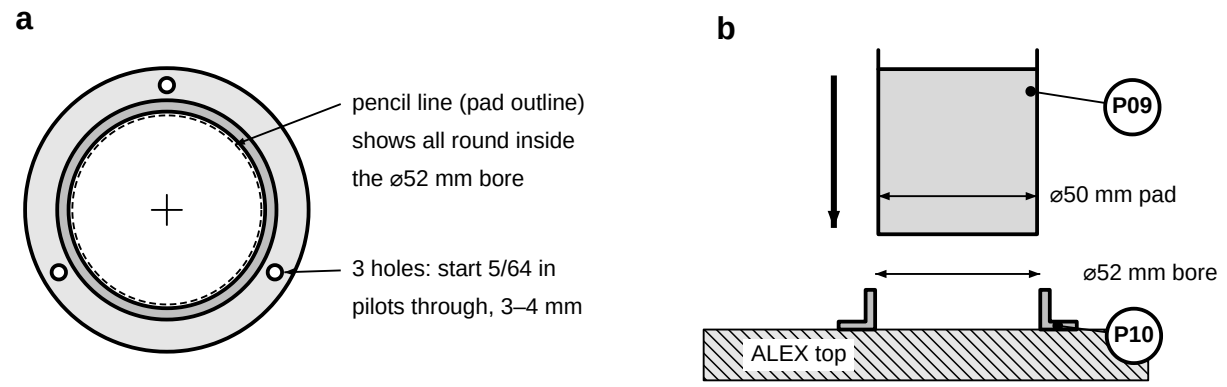


Fig. H-2 · a: a ring on its pencil circle, from above (1:2); b: the pad drops into the ring, section (1:2.4). Ring bore 52 mm VERIFIED (from its geometry file); pad 50 mm MEASURE ON HARDWARE. The ALEX corner template is a check on the ring positions, not the method.

3

Stand top A again and level it. Two people, ALEX end first as in step 1: the pads drop into the rings. Level along x and along y by turning the OLOV feet and the round feet until it reads flat and all four pads bear with no rock. Target: surface 83.0 cm above the floor at every corner. SOURCE-DEFINED

!

Once the pads are in the rings, top A cannot be pushed.
If it is not 3 cm off a wall, lift the ALEX end a few millimetres and slide the ALEX unit, not the top. The wall gauge's EKB 30 leg (3 cm), or a 3 cm offcut, against the wall is the easiest gauge.

83.0 cm assumes ALEX units 70.0 cm tall ASSUMPTION / FIELD CHECK until measured. With the optional ALEX feet the surface is 85.0, 87.0 or 89.0 cm (2, 4 or 6 cm feet).

4

Top B: rings on ALEX 2, then slide it home.

1. Wind top B's OLOV legs 2–3 mm short, so rail B's corner end (it sticks out 12 cm past B's end) can later slide under top A.
2. Turn top B over onto its legs with its pads on the bare ALEX 2, ALEX end first, rail B's end still clear of top A. Push ALEX 2, never the top, until 3.5 cm of top shows all round the cabinet and the wall gauge's ALEX 65 leg (6.5 cm) just fits between the side wall and the ALEX.
3. Fit the four rings on ALEX 2 by tracing, exactly as step 2. Stand top B again: the pads drop into the rings.
4. Push top B (ALEX 2 slides with it, the pads stay in their rings) until rail B's corner end touches the outer face of rail A's vertical leg and B's end meets A's front edge (y = 65 cm), 3 cm off the side wall.
5. Wind B's legs up until the two surfaces are flush.

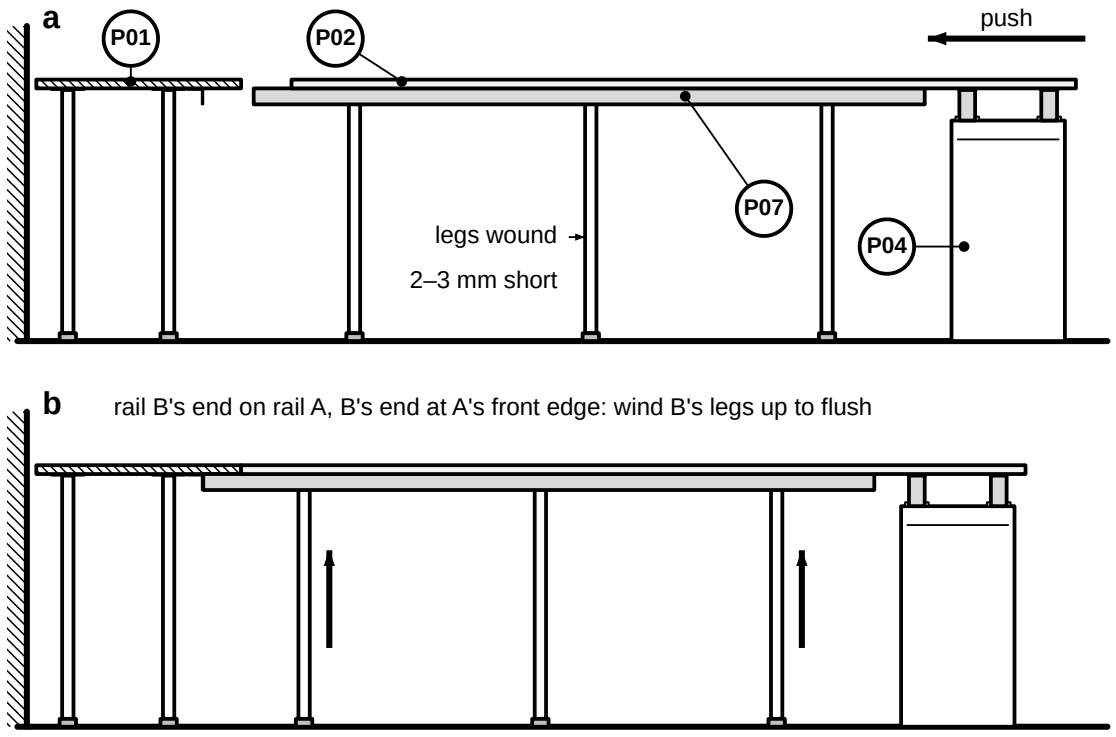


Fig. H-3 · Top B seen from the knee space, looking at the side wall (1:24); top A cut beyond B's inside edge, ALEX 1 not shown. a: rings on, legs short, push; b: home, legs wound up. Neither rail touches an ALEX.

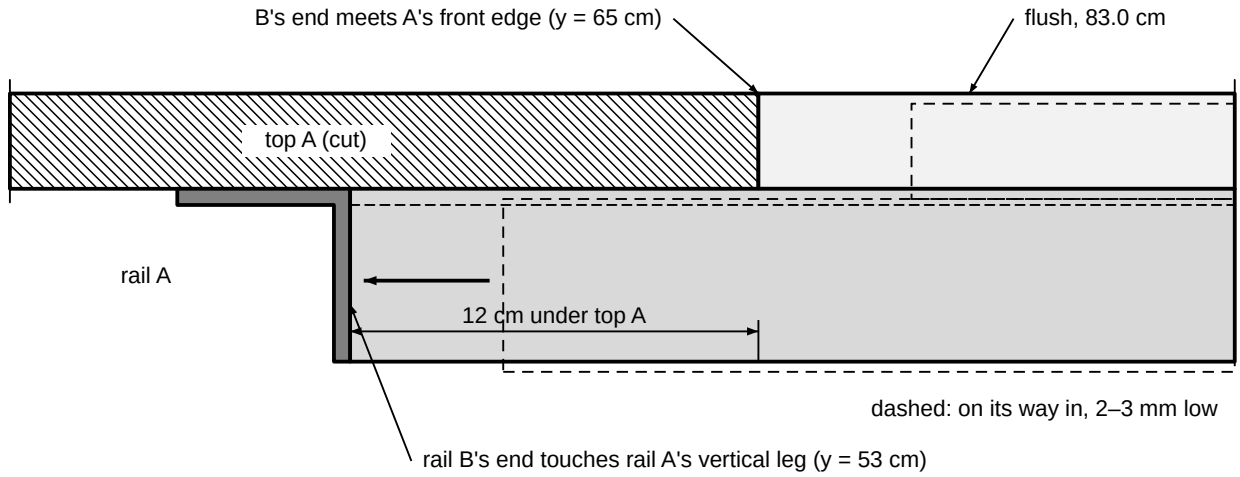


Fig. H-4 · The joint in section, same view (1:2.2). Dashed: top B on its way in, 2–3 mm low. Solid: home. Rail B's flat leg bears on top A's underside for its first 12 cm; its holes 1 and 2 are drilled in step 7.

!

Never force the corner.
If the corner will not meet or come flush, adjust leg height, not the joint. Never hammer, clamp or lever the tops together.

5 **Check under the corner, then level the joint.** Rail B's end must touch the face of rail A's vertical leg with no gap, and rail B's outer face must line up with the 52.0 cm mark on rail A. If not, lift top B a little and move it. Then bring the two surfaces level at the joint by turning the OLOV feet and the round feet, never by packing anything under the joining plates. A fingertip across the joint finds no step.

6 **Joining plates P1 and P2, from below.** Screw each plate on the outline you drew: y 53.3–78.7 cm, long axis along y, centred on x = 15 cm and x = 35 cm. They are not centred on the joint: each starts 3 mm clear of rail A. Pilot every hole 1/8 in, 12 mm deep, with the drill guide's **12 PLATE** puck held up overhead; drive H04 #10 × 5/8 in with an H05 washer. Two screws in each top. SOURCE-DEFINED

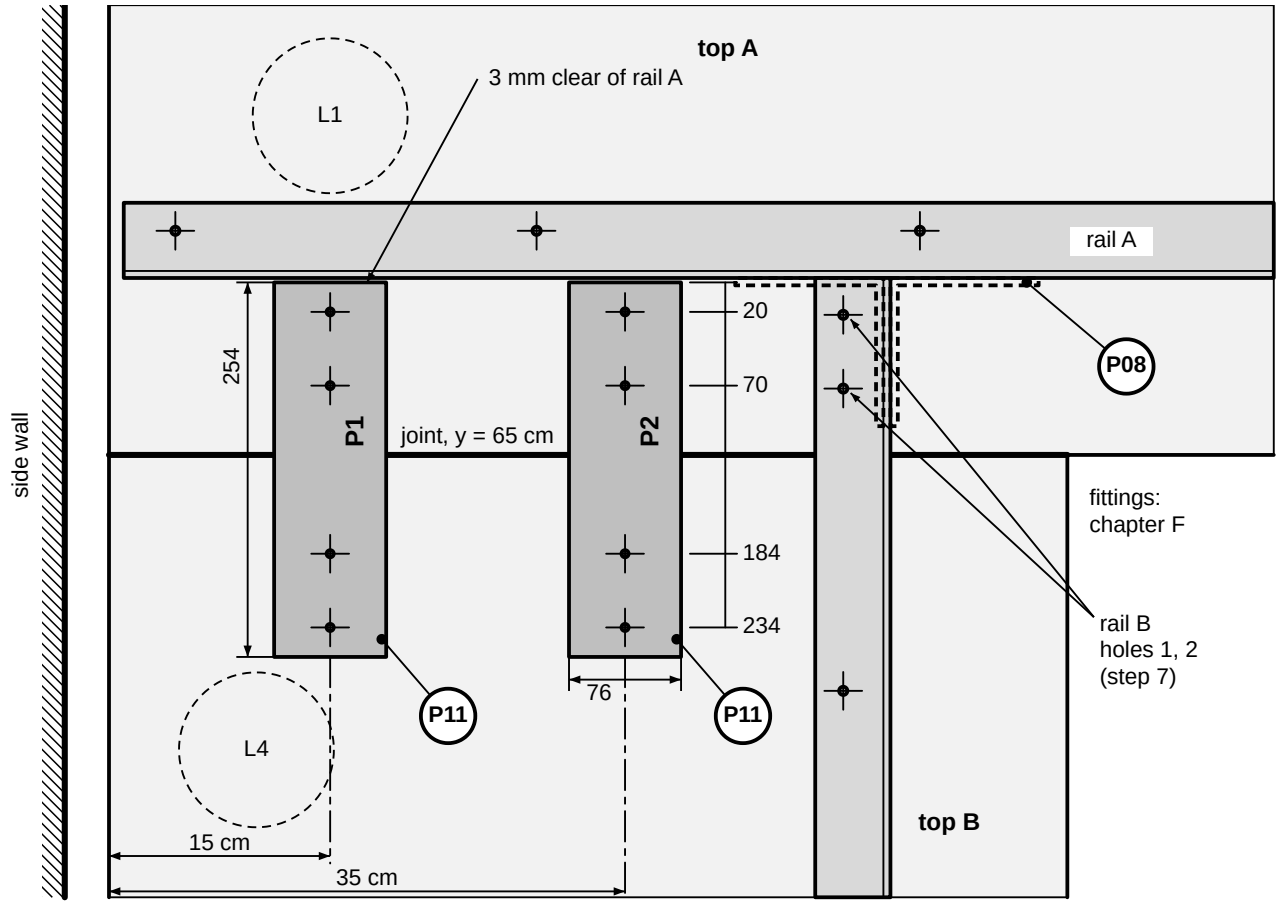


Fig. H-5 · The corner from above, looking through the tops (1:5). Datum: top A's back-left corner, x along the back wall, y along the side wall; positions in cm, plate sizes and hole spacings in mm. Plate holes at 20, 70, 184 and 234 mm from the plate end nearest rail A. Dashed: OLOV plates L1 and L4, and the two fittings (orientation in chapter F).

Plate	Centre x (cm)	Span y (cm)	Screws in top A, y (cm)	Screws in top B, y (cm)	Pilot (mm)
P1	15	53.3–78.7	55.3, 60.3	71.7, 76.7	⌀3.2 (1/8 in) × 12 deep, puck 12 PLATE
P2	35	53.3–78.7	55.3, 60.3	71.7, 76.7	same

The plates align the tops; they do not carry the corner. Every plate hole is more than 4 cm from an edge, so all four get a screw.

7 **Rail B meets top A.** Through rail B's holes 1 and 2 (2.5 cm and 7.5 cm from its corner end, both under top A) drill a 1/8 in pilot with the drill guide's **25 RAIL** puck standing on the flat leg; it stops the bit 20 mm into the wood. Drive H03 #10 × 1 in with the Robertson #2 bit on the 3 in extension until the head seats. SOURCE-DEFINED

8

Bolt on the two fittings, then tighten. Orientation is in chapter F: outside fitting heel at the outside corner, inside fitting heel at the inside corner with its gusset pointing at the floor.

1. Rail A: four H08 bolts from the fitting side, H09 washer under each head, washer and H10 nylon-insert nut on the inside of the leg. Finger-tight.
2. Rail B: two H08 bolts through outside fitting, leg and inside fitting, washer each end, H10 nut. Finger-tight.
3. Only when the surfaces are flush (step 5) and rail B's end touches rail A: tighten all six to **25 N·m (18 ft·lb)**; without a torque wrench, one hand's full pull on a 15 cm wrench, then stop. SOURCE-DEFINED

Tight, not snug: the fittings' 21 × 12 mm slots leave 2.5 mm of play across and 11.5 mm along each bolt, and the 7/16 in rail holes 1.6 mm, so only the clamping force holds the corner. Fitting legs 100.5 × 41 mm, 5 mm thick; slot centres 29.15 mm and 78.45 mm from the heel VERIFIED. Rail bolt holes, drilled on the bench: rail A 54.92 cm and 59.85 cm (outside fitting) and 48.59 cm and 43.66 cm (inside fitting) from the wall end; rail B 2.92 cm and 7.85 cm from the corner end VERIFIED.

9

Checks. Do every one; if one fails, fix it with the feet or the legs, never at the joint.

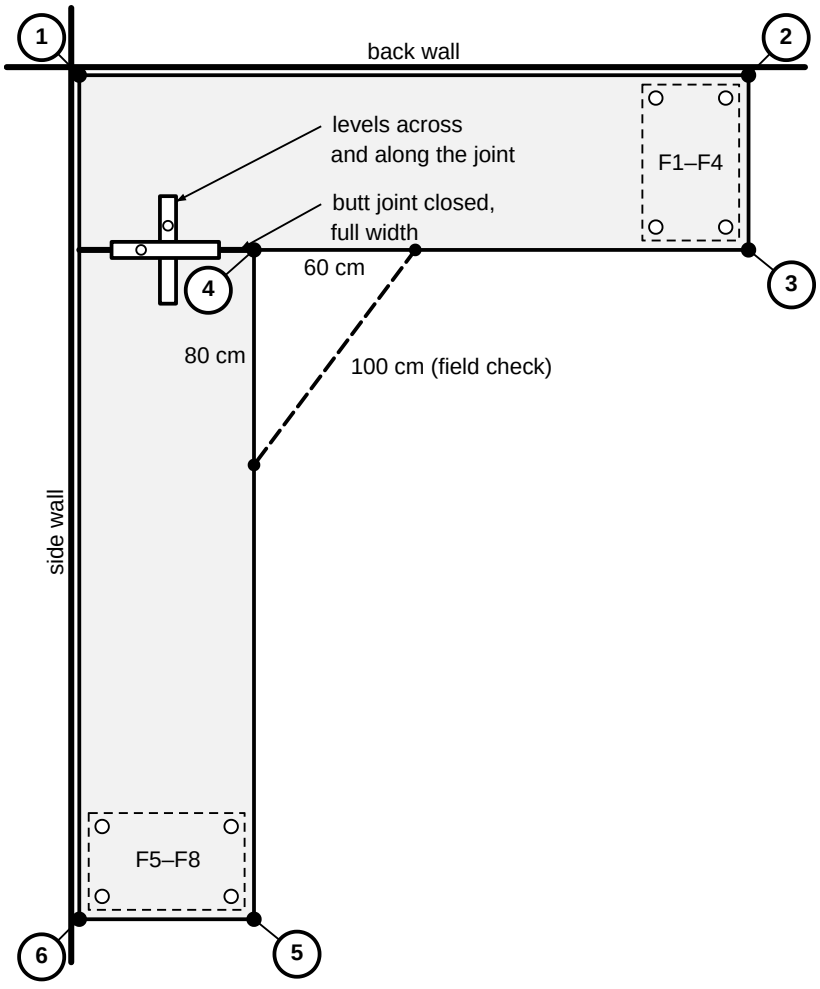


Fig. H-6 · The checks in plan (1:28). Datum: top A's back-left corner; all cm.

- **Square:** the butt joint is closed across its full 65 cm width (no light), rail B's end touches rail A, rail B's outer face is on rail A's 52.0 cm mark.
- **Level:** a level across the joint and along it reads flat both ways; a fingertip finds no step.
- **Height:** surface 83.0 cm (± 3 mm) at all six outer corners, 1–6. SOURCE-DEFINED
- **Pads:** all eight bear. Turn each of F1–F8 by hand in its ring: none spins free or rocks.

!

FIELD CHECK, not in the build manual ASSUMPTION / FIELD CHECK
Optional 3-4-5 triangle at the inside corner: hook the tape on the inside corner and mark 60 cm along A's front edge and 80 cm along B's inside edge; the tape between the two marks reads 100 cm when the L is square. Write down what it reads. If it is off, check the butt joint for a gap first, then move ALEX 2 with top B on it. Never force the corner.

Load test, sideways push and settling record: final checks. Never stand or sit on the corner.

8b · Wall-gap trim (optional)

> **Do this after chapter I.**
> The clamp pieces wrap round the VIVO clamp backs, so both clamps must already be fitted and tight. The trim goes on before the final push to the walls.

Parts P19 (corner, two 80 mm fillers, two clamp pieces, 240 segments, connector tiles), H07 #4 × 1/2 in flat head, the drill guide's **10 EDGE** puck with a 5/64 in bit, the wall gauge.

10 **Measure, set, print.** With the desk in place against both walls, measure the gap between each top's raw back edge and the wall, in mm, at five or six points along each wall. Take the smallest reading, look it up below, and set `gap` in `parts/accessories/wall-gap-common.scad` to the value beside it; then print the trim. MEASURE ON HARDWARE

Smallest gap (mm)	27	28	29	30	31	32	33
Set gap to	26	27	28	29 (as saved)	30	31	32

Wall	From the wall corner outward: desk positions in cm, piece lengths in mm	Slotted segments (cm)
back	corner (to 20), 80 mm filler (20–28), clamp piece (28–52), 8 × 240 mm, 1 × 50 mm (to 249)	3rd (100–124), 8th (220–244)
side	corner (to 20), 80 mm filler (20–28), clamp piece (28–52), 10 × 240 mm, 1 × 220 mm (to 314)	3rd (100–124)

The corner piece covers each wall's first 20 cm of desk. 24 connector tiles: 11 joints on the back wall, 13 on the side wall. Cut list from parts/README.md.

11 **Pull the desk out, 40 cm from both walls.** Two people, one at each ALEX unit: slide the whole desk away from both walls. The ALEX units slide on their feet and the pads stay in their rings. Nothing comes apart.

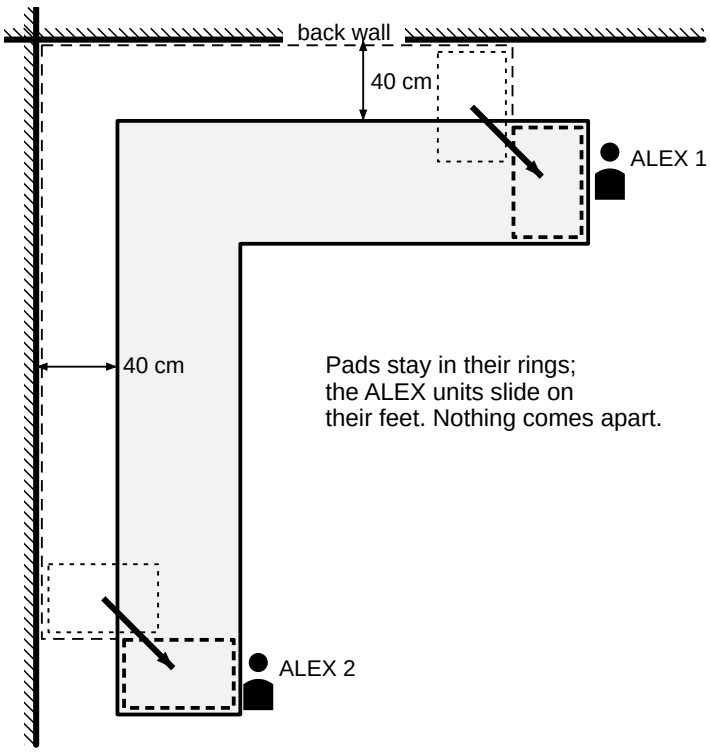
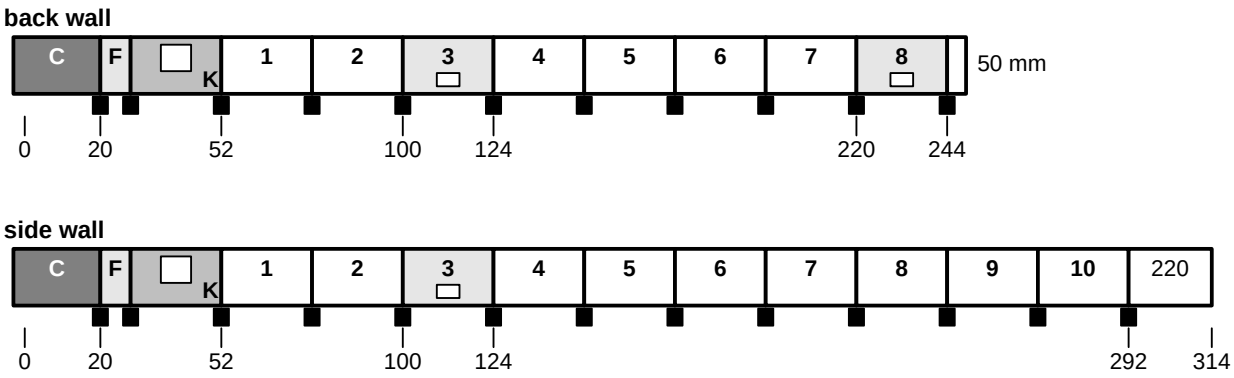


Fig. H-7 · Plan (1:40). Dashed: the desk in place, 3 cm off both walls. Grey: pulled 40 cm off both, so the raw edges are open to work on.

12 **Build each chain from the corner outward.** Corner piece, 80 mm filler, clamp piece with its notch round the clamp back, then the 240 mm segments (slotted ones where the cut list says), ending with the short piece. Butt every joint and push a connector tile up into the pockets.



Desk positions in cm. C corner · F 80 mm filler · K clamp piece (white: notch round the clamp back)
1, 2 … 240 mm segments, grey with a slot = slotted · ■ connector tile under every joint

Fig. H-8 · Chain order along both walls, desk positions in cm from the corner, piece lengths in mm (1:20). The notch (white, in K) wraps the clamp back, 80 mm long and 17.5 mm proud VERIFIED.

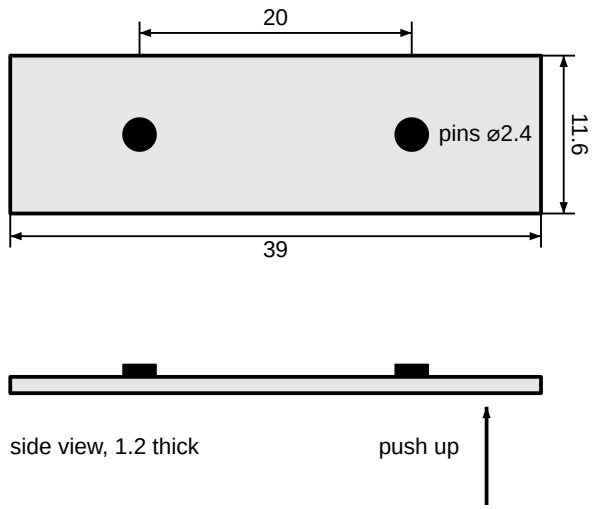


Fig. H-9 · Connector tile (1.8:1, mm), one under every joint, pins up. It bridges the 20 × 12 mm pockets of two butted pieces.

Every piece, the corner included, has a pocket on its underside at each end, over the gap. Butt two pieces, then push a tile up into both pockets until its pins seat: the joint cannot step, bow or drift, and nothing shows on top.

If a wall is out and the notch misses the clamp, change the 80 mm filler's `seg` and reprint it, nothing else.

13 **Pilot and screw.** Through each lip hole, pilot 5/64 in, 10 mm deep, drill horizontal, with the drill guide's **10 EDGE** puck held against the raw edge. Drive an H07 #4 × 1/2 in flat head into each: two per segment, one per corner leg, one each side of a clamp notch.

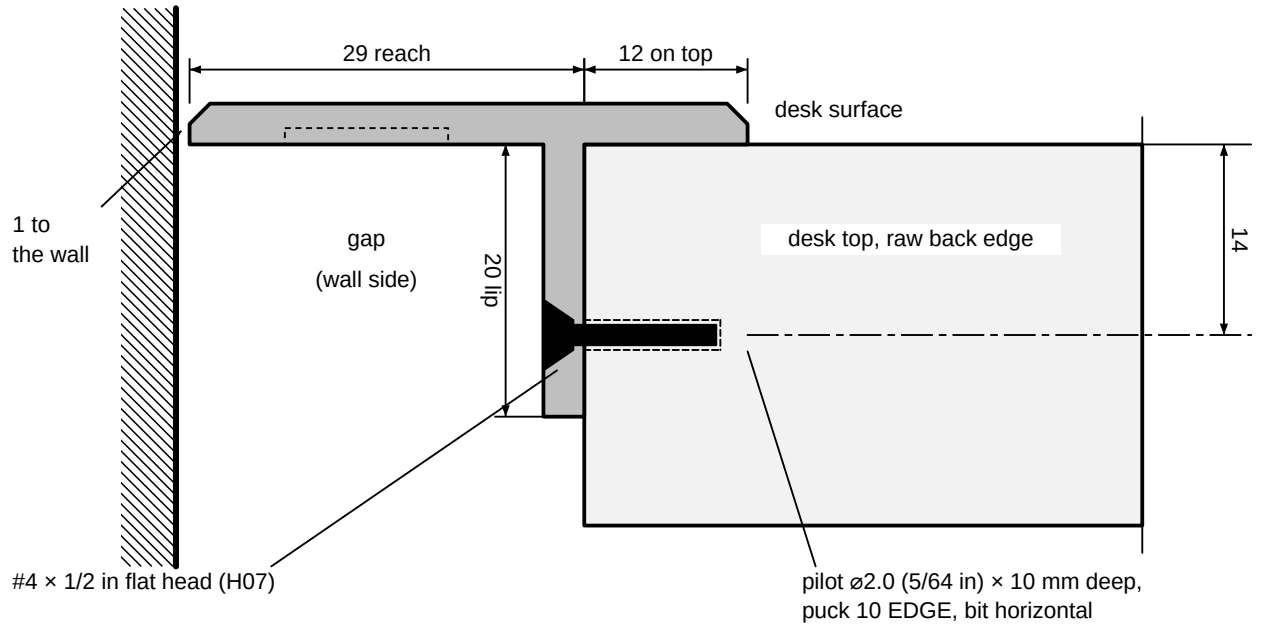


Fig. H-10 · Trim on the raw back edge, section (1.8:1, mm). 12 mm rests on the surface, the 20 mm lip drops into the gap, the screw sits 14 mm below the surface: nothing shows on top and nothing goes into the top face. Profile from parts/accessories/wall-gap-common.scad **SOURCE-DEFINED**; top 28 mm thick **MEASURE ON HARDWARE**.

14 **Slide the desk back.** Two people again, one at each ALEX, until the wall gauge's ALEX 65 leg (6.5 cm) just fits between the wall and each ALEX. The trim's outer edge stops 1 mm short of the wall.

Retighten every screw after two weeks.

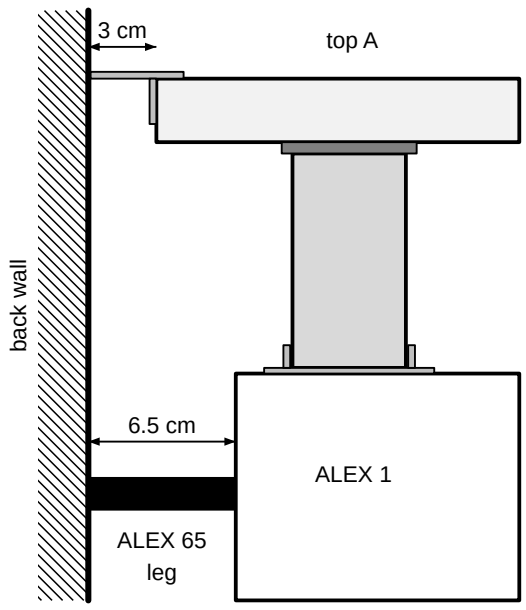


Fig. H-11 · Wall gauge at ALEX 1, section along the back wall (1:3.3).

I · Monitor mount and accessories

The VIVO STAND-V300CU (P12) has two independent C-clamps and no base bar. Both clamp onto top A's own wall edges, never on the joint: clamp 1 centred at **x = 40.0 cm** on the back edge, clamp 2 centred at **y = 40.0 cm** on the side-wall end. The crossbar runs on the diagonal between the poles. Each lower jaw bears on a spare steel mending plate (P11), because particleboard crushes under a bare steel jaw over months. Datum: top A's back-left corner (0, 0); x along the back wall, y along the side wall.

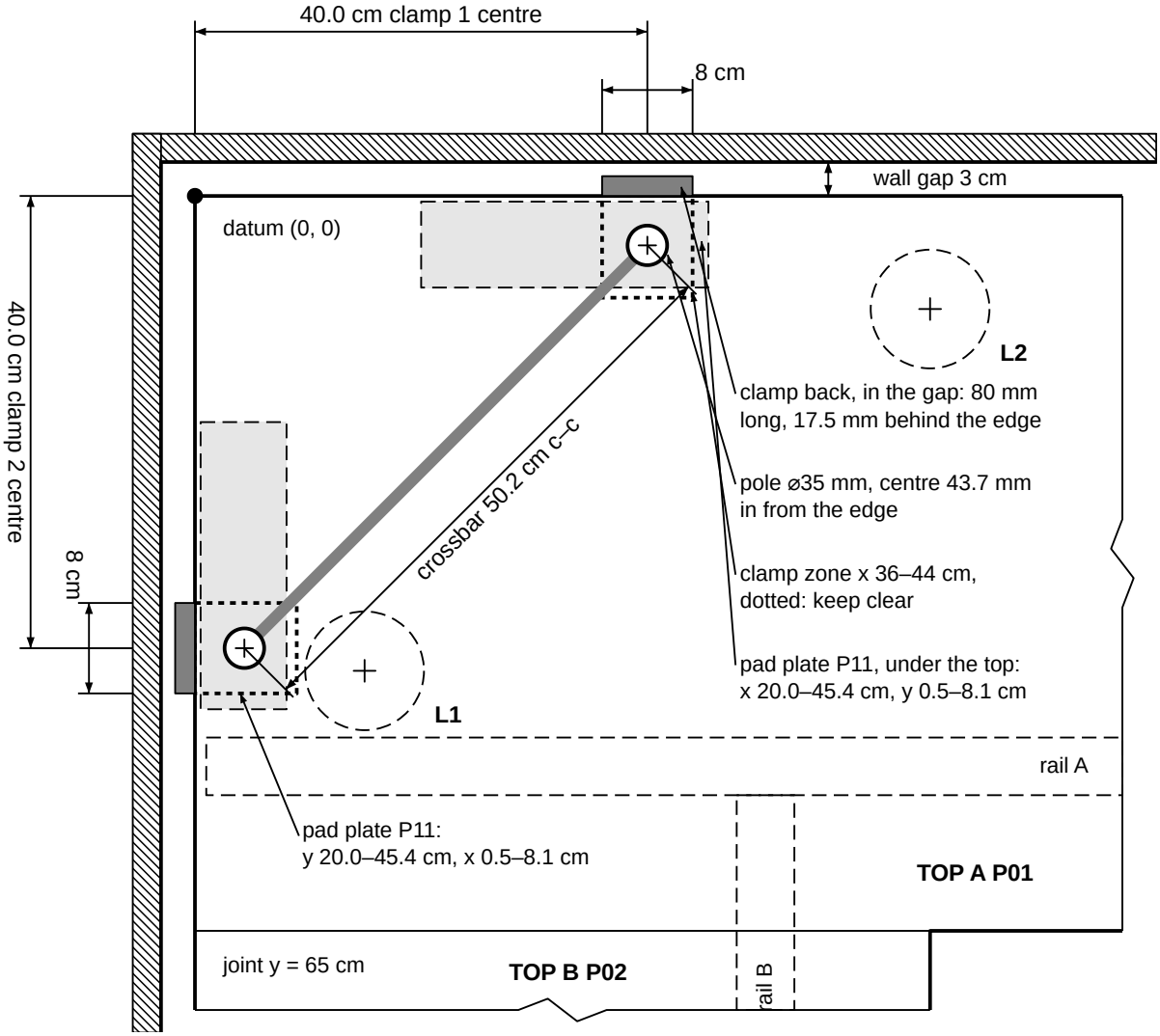


Fig. I-1 Plan of the corner from above; tape readings in cm, caliper sizes marked mm. Dashed: under the top (pad plates P11, legs L1 and L2, rails). Dotted: the clamp zones, 36–44 cm along the edge, keep clear. Dark: the clamp backs in the 3 cm wall gap. Black circles: poles, centre 43.7 mm in from the edge.

Balloon	Part	Balloon	Part
P01	Top A, EKBACKEN 249 × 65 cm, 28 mm thick	P11	Mending plate 10 × 3 × 0.12 in, two used here as clamp pads
P02	Top B, EKBACKEN 249 × 65 cm, 28 mm thick	P12	VIVO STAND-V300CU (clamps, poles, crossbar, arms)

What is measured at the clamp

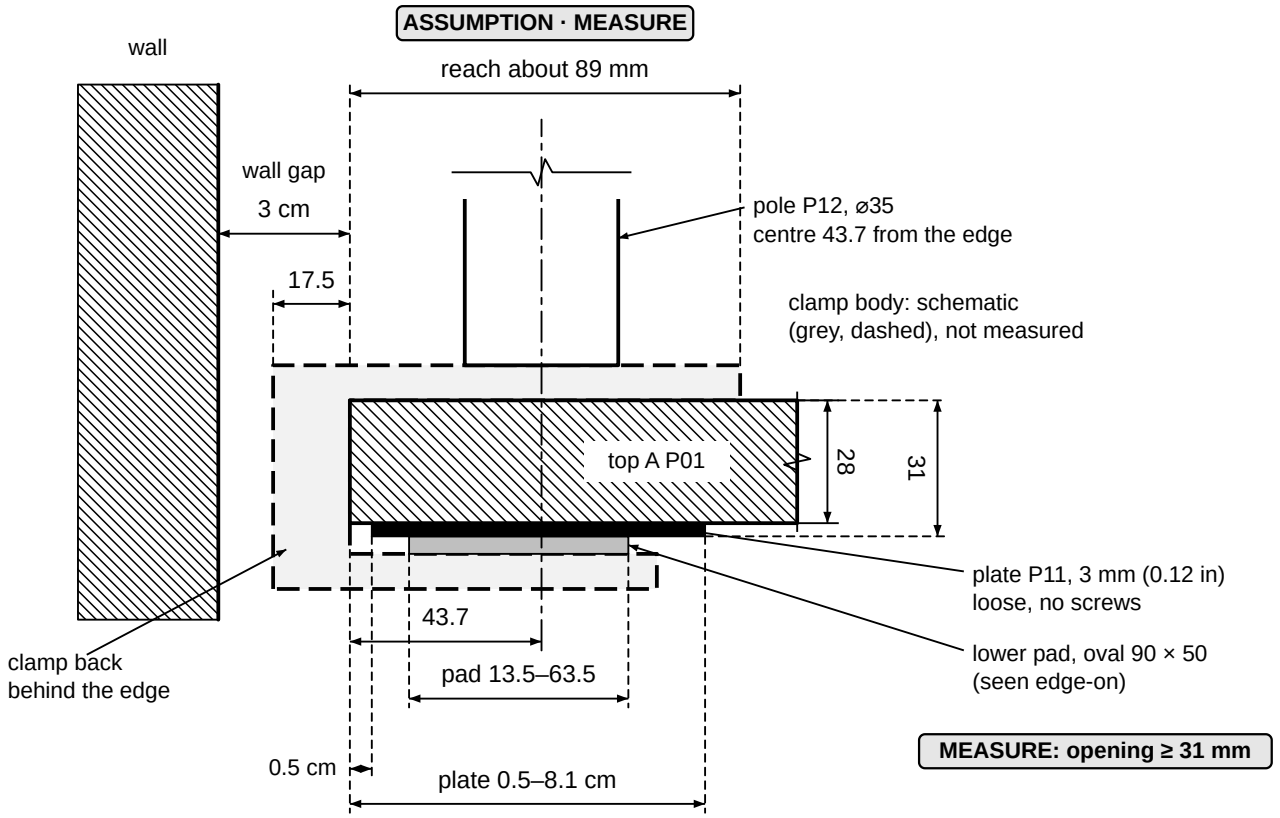


Fig. I-2 Section through clamp 1 at x = 40.0 cm, looking along the back wall; mm unless marked cm. The clamp body is drawn as a plain C (grey, dashed) because its shape is not measured; only the dimensioned values are. The section through clamp 2 at y = 40.0 cm is the same, turned 90°.

Value	Size	Confidence
Crossbar, centre to centre of its end collars (19¾ in)	50.2 cm	VERIFIED
Pole diameter; pole centre in from the desk edge	35 mm; 43.7 mm	VERIFIED
Clamp back: length along the edge; projection behind the edge (it sits in the 3 cm wall gap)	80 mm; 17.5 mm	VERIFIED
Lower pad: oval 90 × 50 mm, 13.5 to 63.5 mm in from the edge, so it lands on the plate (0.5–8.1 cm from the edge)	13.5–63.5 mm	VERIFIED
Clamp centre from the corner: ideal 39.9 cm; built at 40.0 cm, the crossbar sets the last millimetre	40.0 cm	SOURCE-DEFINED
Clamp zones x 36–44 cm, y 0–9 cm (clamp 1) and y 36–44 cm, x 0–9 cm (clamp 2); pad plates x 20.0–45.4 cm, y 0.5–8.1 cm and y 20.0–45.4 cm, x 0.5–8.1 cm	Fig. I-1	SOURCE-DEFINED
Top 28 mm plus plate 3 mm (0.12 in): under 60 mm, so VIVO's Option 1 holes (2 and 4 on the lower clamp)	31 mm	SOURCE-DEFINED
Top arm reach from the edge	about 89 mm	ASSUMPTION / FIELD CHECK
Maximum jaw opening (must be 31 mm or more)	≥ 31 mm	MEASURE ON HARDWARE

Hang the mount

1 **Lay the pad plates.** Lay a spare mending plate P11 flat on the underside, on each pad outline drawn when top A was marked, long side along the edge: x 20.0–45.4 cm, y 0.5–8.1 cm at clamp 1; y 20.0–45.4 cm, x 0.5–8.1 cm at clamp 2. Loose, no screws: the jaw holds it. SOURCE-DEFINED

> ! **MEASURE ON HARDWARE before hanging the clamps**
> Top arm reach (about 89 mm) and maximum opening (needs 31 mm or more: 28 mm top + 3 mm plate) are not measured. Open the lower clamp fully on the Option 1 holes and measure the gap with calipers; if it is under 31 mm, stop. The clamp zones are drawn 9 cm deep from the edge; if the reach measures more, keep the zone clear to the measured reach.

2 **Hang both lower clamps, finger-tight only.** Clamp 1 centred at x = 40.0 cm on A's back edge, clamp 2 centred at y = 40.0 cm on A's side-wall end (Fig. I-1). Use the VIVO sheet's Option 1 holes (2 and 4): top and plate together are 31 mm, under 60 mm. The clamp backs sit in the 3 cm wall gap; the lower pad lands on the plate, 13.5–63.5 mm from the edge (Fig. I-2).

3 **Fit the poles and the crossbar as the spacing guide** (VIVO sheet step 3). The crossbar measured 50.2 cm centre to centre, so it fixes the pole spacing. **Only now tighten both clamps**, with the crossbar on. VERIFIED

4 **Refit the crossbar at monitor height** (VIVO sheet step 6): 40–43 cm above the top (123.0–126.0 cm from the floor).

5 **Hang the three screens, all landscape**, and set the heights in Fig. I-3. The G5 goes on the centre bracket (VESA 75), screen centred on the diagonal at (36.5, 36.5) cm, 20 cm in front of the crossbar. The Acers go on the side arms, toed in 15°; each Acer's near end ends up about 12 cm from the G5's curved end. The side arms run along the wall behind each panel, then forward into the VESA plate.

Weights against the mount's ratings

Screen	On the mount	Rating	Confidence
Samsung Odyssey G5 32 in P13 (1000R, VESA 75), centre	about 5 kg	centre bracket 48.4 lb curved (59.4 lb flat); top edge at most 26.5 in (67.3 cm) above the top	SOURCE-DEFINED
Acer XZ320QU 31.5 in P14 (1500R, VESA 100), each side	about 5 kg (5.6 kg with its stand)	17.6 lb (8 kg) per side arm	SOURCE-DEFINED

> The clamps load top A's edges near legs L1 and L2, not the corner tip. Do not add a soft pad under a jaw: the steel plate is the pad.

Heights above the top

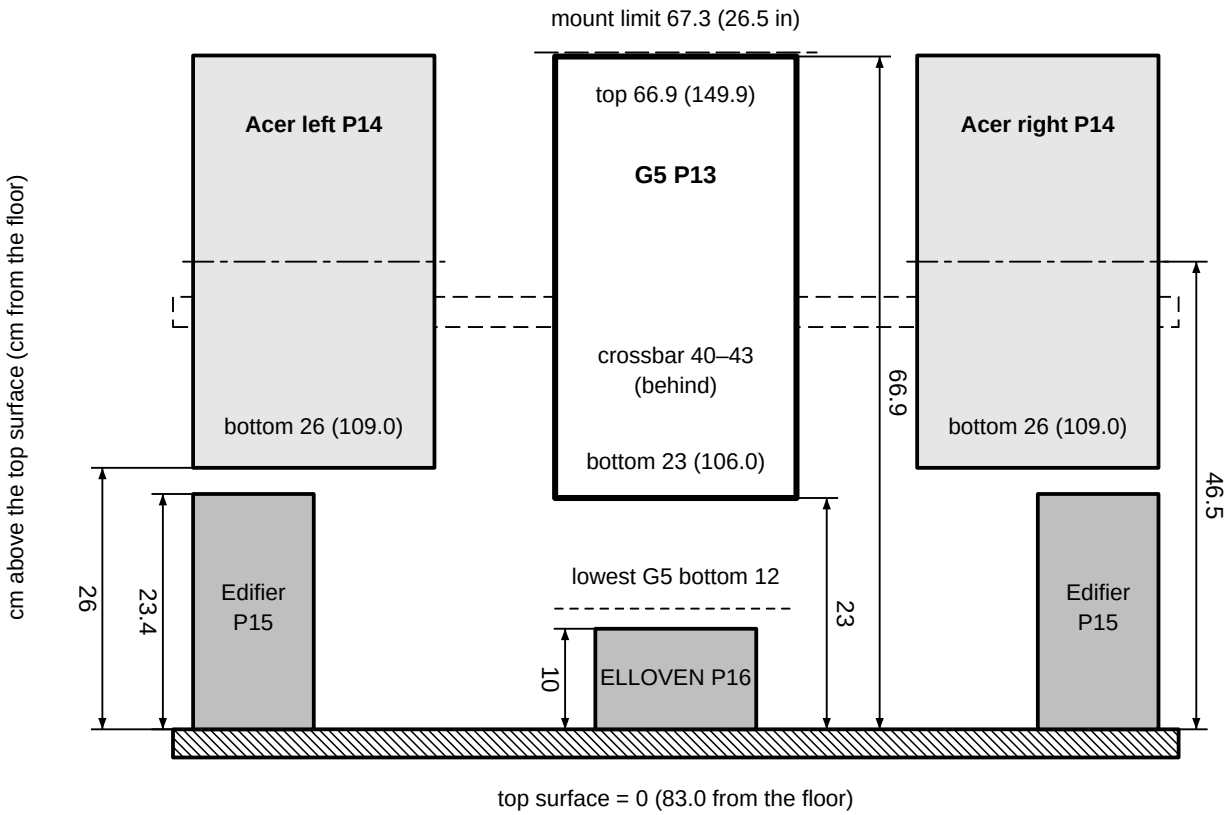


Fig. I-3 The three screens seen from the seat, unfolded flat. Heights to scale, in cm above the top surface (in brackets, cm from the floor); widths are schematic. The Edifiers stand under the outer half of each Acer.

Item (all cm)	Above the top	From the floor	Note
Top surface	0	83.0	underside 80.2, top 28 mm thick
ELLOVEN P16 top	10	93.0	13 under the G5
G5 bottom edge	23	106.0	SOURCE-DEFINED
Edifier P15 top	23.4	106.4	2.6 under each Acer
Acer bottom edges	26	109.0	panel centre 46.5 above the top
VIVO crossbar	40–43	123.0–126.0	refitted at monitor height
G5 top edge	66.9	149.9	0.4 under the mount's 67.3 (26.5 in) limit

IF THE ASSEMBLED MOUNT IS LOWER
The G5's top edge at 66.9 cm is only 0.4 cm under the 67.3 cm limit. If the assembled mount will not reach, lower the G5: a bottom edge anywhere down to 12 cm above the top still clears the ELLOVEN (10 cm tall).

Everything else on the desk

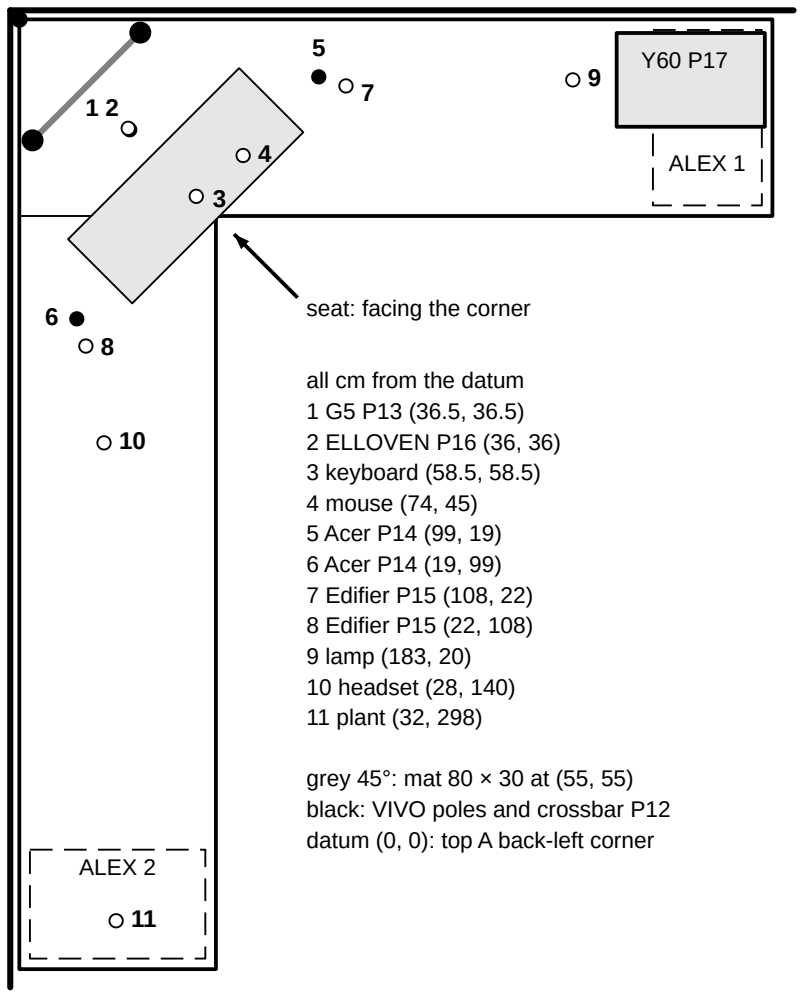


Fig. I-4 Plan of the desk, centres in cm from the datum, top A's back-left corner. Dashed: ALEX 1 and ALEX 2.

ELLOVEN at (36, 36) cm, turned 45°, drawer toward you: its back corners clear the clamp plates and poles by over 2 cm. At (33, 33) cm it would sit on the clamps.

The mat's back edge touches the ELLOVEN's front feet; its front edge reaches the inside corner of the L.

The Y60 (P17) stands on its I/O end on the printed pedestal over ALEX 1: long axis along the back wall, side glass toward you, front glass up. Keep it there, not mid-wing.

Item	Centre (x, y), cm	How
G5 P13	(36.5, 36.5)	on the diagonal, 20 cm in front of the crossbar
ELLOVEN P16	(36, 36)	turned 45°, drawer toward you
Deskmat 80 × 30 cm	(55, 55)	turned 45°
NuPhy Air75 V2; mouse	(58.5, 58.5); (74, 45)	keyboard turned 45°, on the mat
Acer P14, right; left	(99, 19); (19, 99)	along each wall, toed 15° toward you
Edifier P15 ×2	(108, 22); (22, 108)	grilles toward you, toed with the Acers
HYTE Y60 P17	x 197.5–246.5, y 4.5–35.5	on its pedestal, over ALEX 1
ELYONA lamp	(183, 20)	beside the Y60, arm toward the keyboard
Headset stand; plant	(28, 140); (32, 298)	on top B, hook toward you; plant over ALEX 2

All positions **SOURCE-DEFINED** (docs/build-manual.md step 8).

J · Final inspection and load testing

Ten checks, in the build manual's order, before you call the desk done. Each is an ACCEPTANCE CHECK: a pass or fail you can see or measure. Several of them test a STRUCTURAL ESTIMATE from the design review that has not been measured on this desk; the check says which. Write the results in the boxes and in the sign-off table at the end of this chapter. Datum for every drawing: top A's back-left corner (0, 0).

Check 1 · Surface height at the six corners ACCEPTANCE CHECK

1

Measure from the floor to the top face at each of the six corners of the L (Fig. J-1). Every reading **83.0 cm ± 0.3 cm** (82.7 to 83.3 cm). With the optional ALEX feet fitted, read instead: 2 cm feet 85.0 cm, 4 cm feet 87.0 cm, 6 cm feet 89.0 cm, each ± 0.3 cm. If a corner is out, turn that corner's OLOV foot or round foot, then measure all six again.

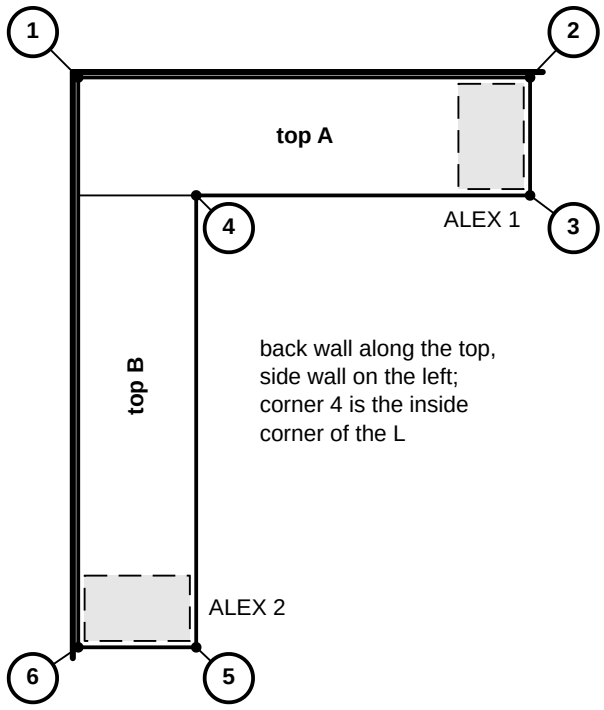


Fig. J-1a The six corners, numbered for the record.

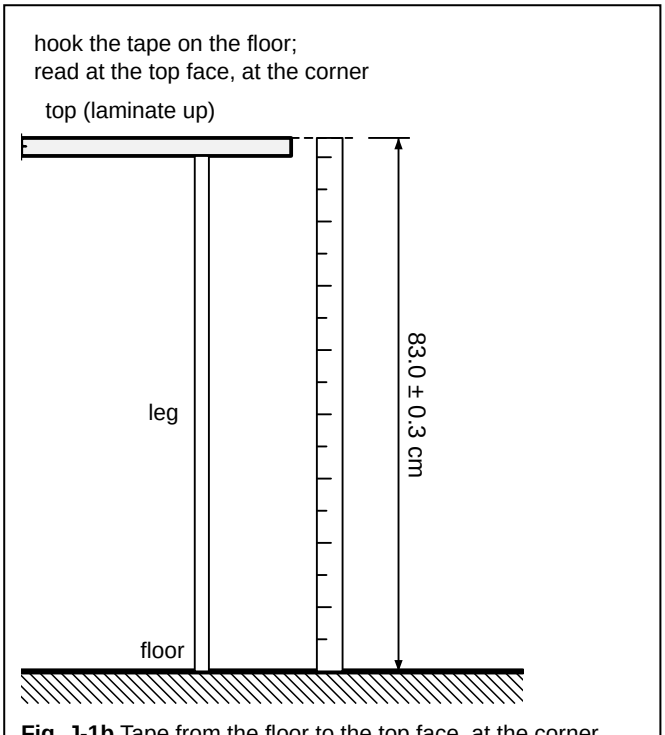


Fig. J-1b Tape from the floor to the top face, at the corner.

Corner	1	2	3	4	5	6
Reading (cm)						

Check 2 · Level and flush across the joint ACCEPTANCE CHECK

2

A level across the joint reads flat both ways: across it (1) and along it (2). A fingertip drawn across the joint (3) finds no step. Level the two tops with the OLOV feet and the round feet, never by packing anything under the joining plates.

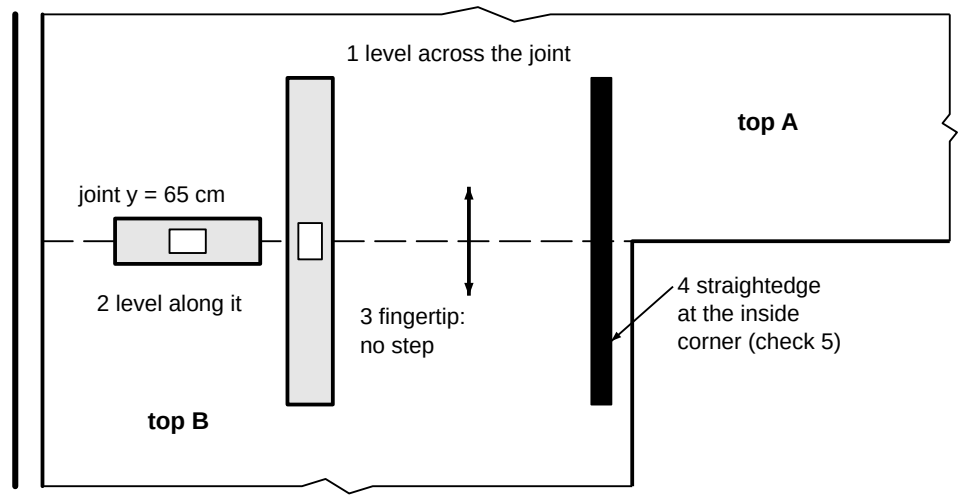


Fig. J-2 Plan of the joint at y = 65 cm. 1, 2: level positions; 3: fingertip; 4: where the straightedge of check 5 goes, at the inside corner.

For the remaining checks you need



Tape
floor to top face, checks 1 and 8



Level, straightedge
checks 2, 3 and 5; a rule to read the gap



20 kg
a case of water plus a few books, check 3



Wrench
torque wrench at 25 N·m, or a 15 cm wrench, checks 5 and 10

Do the checks in this order, with the screens and everything else in place.

Check 3 · Corner load test ACCEPTANCE CHECK

NEVER STAND OR SIT ON THE DESK
The test load is a known weight: 20 kg of water and books. Do not stand or sit on the corner; that is not what the beam was sized for.

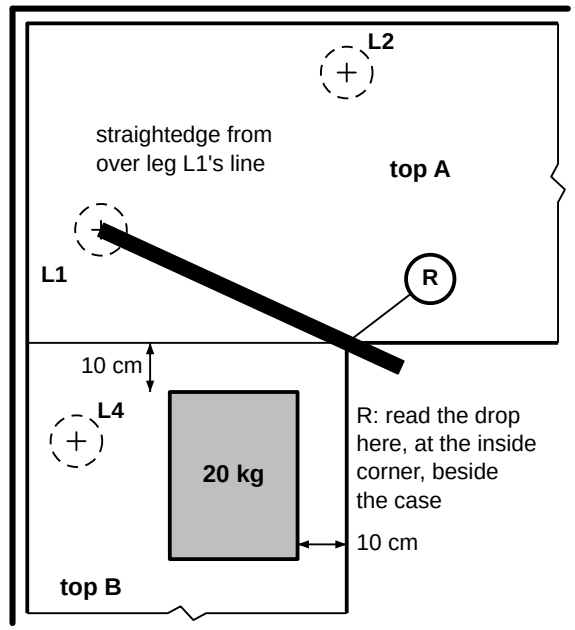


Fig. J-3a Plan. Case faces 10 cm in from top B's front edge and from the joint (case size schematic). R: reading point.

3

Set **20 kg** (a case of water plus a few books) on top B's inside corner, **10 cm in from both edges**.

Rest a straightedge from leg L1's line across the corner. Read the drop at the corner (R): **no more than 2 mm**. The estimate is about 1 mm. STRUCTURAL ESTIMATE

Lift the weight off. The corner returns to where it was, nothing creaks, no screw or bolt has moved.

Tests: sag at the inside corner (rail A carrying rail B's corner reaction) and corner-joint slip (the six fitting bolts holding by friction). STRUCTURAL ESTIMATE

Drop under 20 kg (mm)	Returned?	Creak / movement?

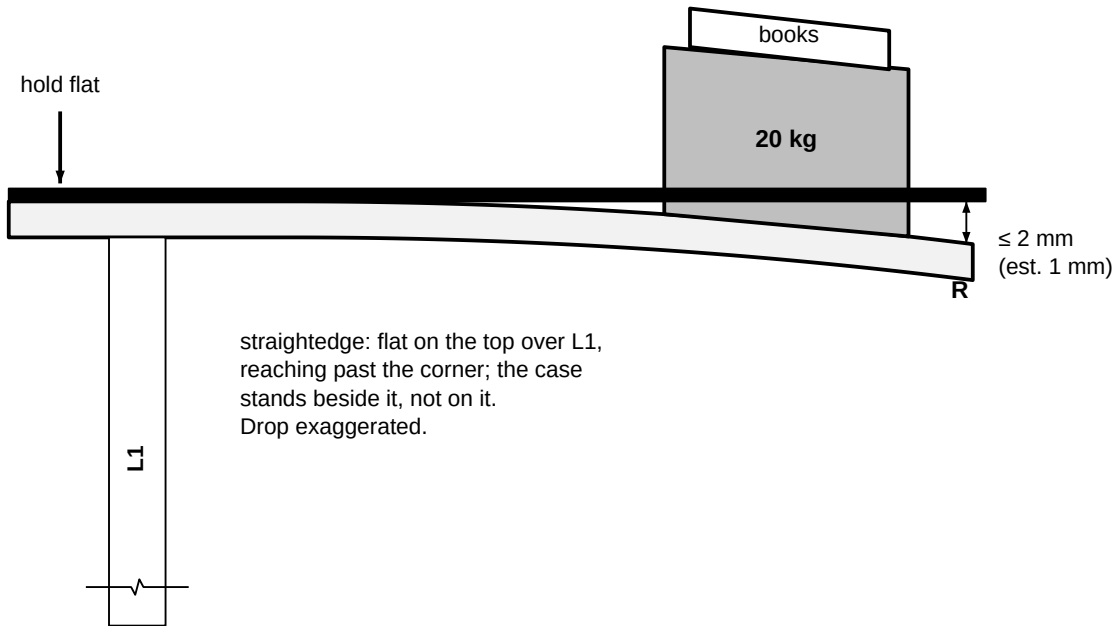


Fig. J-3b Elevation along the straightedge, from leg L1 to the corner. The case stands beside the straightedge, not under it.

Check 4 · All eight pads bear ACCEPTANCE CHECK

4

At each of the eight feet F1–F8, try to turn the pad in its ring by hand. None spins free or rocks. A pad that spins or rocks is not bearing: see chapter K, "Top moves on its ALEX".

Tests: the racking estimate (eight pads captive in their rings hold the frame against a sideways push) and the ALEX load share. STRUCTURAL ESTIMATE

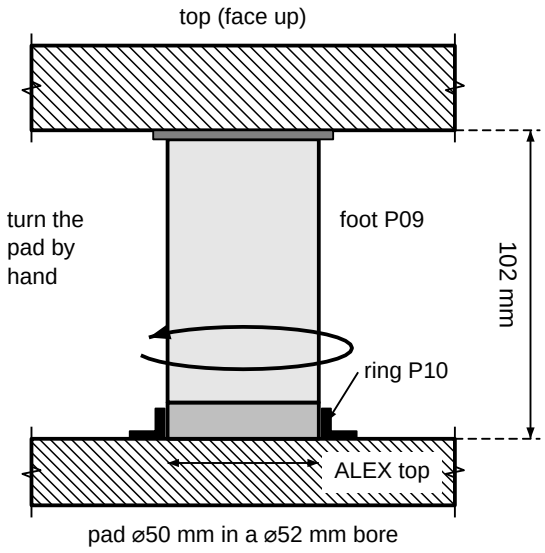


Fig. J-4 Foot P09 on its pad in ring P10 on the ALEX top.

Check 5 · Straightedge at the joint, now and later ACCEPTANCE CHECK

5

Lay a straightedge across the joint at the inside corner (Fig. J-2, position 4) and **write down any step**. Check it again after two weeks and after the first month. If the corner has settled, retighten the six fitting bolts (25 N·m, chapter K).

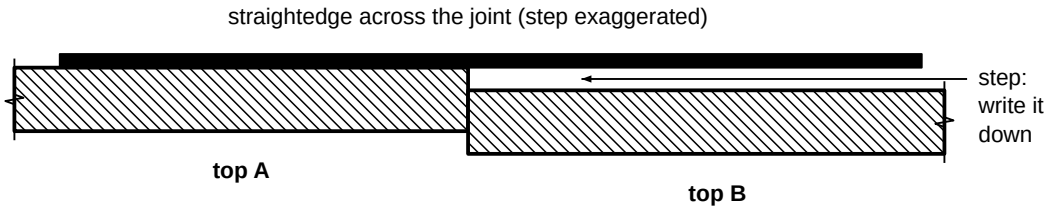


Fig. J-5 Section across the joint at the inside corner. Step drawn exaggerated.

Step at the joint	Day 0	Two weeks	One month	Bolts retightened?
mm, and which top is higher				

Tests: corner-joint slip, an estimate, not a measured capacity. STRUCTURAL ESTIMATE

Check 6 · Sideways push ACCEPTANCE CHECK

6

Push the desk sideways along each wall. No sway at the top beyond a few millimetres, and no top moves on its ALEX.

Tests: racking. The OLOV legs give almost no racking resistance; the two ALEX units, locked to the tops by the pads in their rings, do. STRUCTURAL ESTIMATE

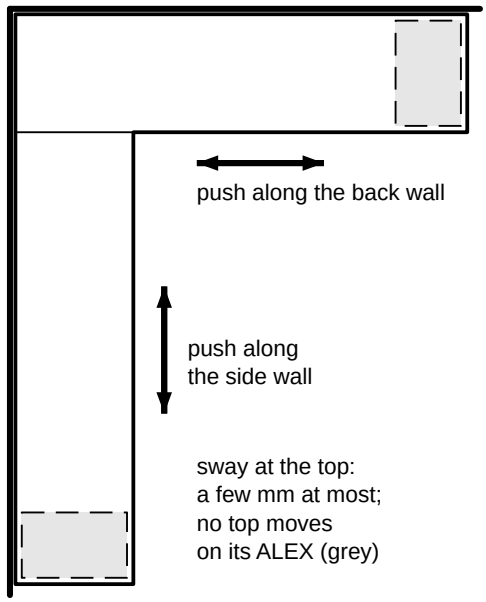


Fig. J-6 Push along the back wall, then along the side wall.

Check 7 · Screw tips ACCEPTANCE CHECK

7

No screw tip visible, or felt as a bump, on either laminate face. Run a hand over both tops, above every leg, foot plate, rail screw and joining plate.

Check 8 · Knees clear ACCEPTANCE CHECK

8

Knees clear. Sit at the corner and slide in: nothing within reach of a knee or thigh. The rails' lower edge is 75.1 cm from the floor, 12 cm back from the front edge.

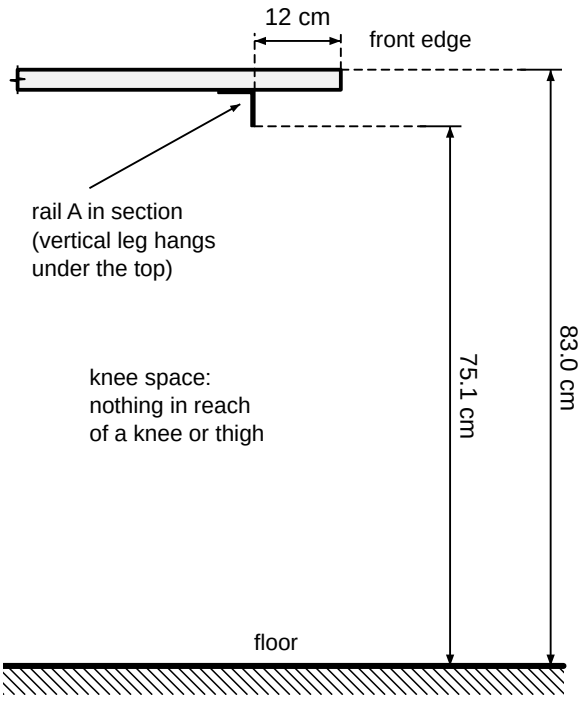


Fig. J-7 Section through top A's front at rail A.

Check 9 · ALEX fronts ACCEPTANCE CHECK

9

Both ALEX fronts open fully (ALEX 1's drawers, ALEX 2's door) without touching a foot or a leg.

Check 10 · Retighten ACCEPTANCE CHECK

10

Retighten every screw after two weeks, then repeat check 5.

Sign-off

#	Acceptance check	Pass when	Structural estimate it tests	Result
1	Surface height, six corners	83.0 cm ± 0.3 cm each (85.0 / 87.0 / 89.0 cm with 2 / 4 / 6 cm ALEX feet)	—	
2	Level and fingertip across the joint	flat both ways; no step felt	—	
3	Corner load test, 20 kg	drop ≤ 2 mm; returns; no creak; nothing moved	sag at the inside corner (estimate about 1 mm); corner joint slip	
4	All eight pads bear	none spins free or rocks	racking; ALEX load share	
5	Straightedge at the joint	step written down; re-checked at two weeks and one month	corner joint slip (settling)	
6	Sideways push along each wall	sway a few mm at most; no top moves on its ALEX	racking	
7	Screw tips	none visible or felt on either laminate face	—	
8	Knees clear	nothing in reach; rail lower edge 75.1 cm from the floor, 12 cm back	—	
9	ALEX fronts	open fully, touch no foot or leg	—	
10	Retighten every screw	done after two weeks	—	

ACCEPTANCE CHECK pass or fail on this desk. STRUCTURAL ESTIMATE a number from the design review that the check confirms or refutes. If a check fails, see chapter K.

Checked by	Date	Two-week re-check	One-month re-check

K · Maintenance and troubleshooting

Schedule

When	Do	See
Day 0	Write down the step at the joint (straightedge at the inside corner).	Check 5, Fig. J-5
Two weeks	Retighten every screw. Straightedge at the joint again; if the corner has settled, retighten the six fitting bolts to 25 N·m. Look at each printed ring and its three screws.	Checks 5 and 10; troubleshooting
One month	Straightedge at the joint again; retighten the six fitting bolts if it settled.	Check 5
Any time	If the corner ever drops a millimetre, retighten the six fitting bolts.	Corner bolts, below

> **NEVER DEEPER THAN 21 MM INTO A TOP**
> The tops are 28 mm particleboard with a thin laminate skin; a screw reaching the skin lifts it. Replace a screw only with the same size. If a top measured 27 mm, shorten every screw into it by 1 mm (build manual step 0). SOURCE-DEFINED

Corner bolts

Bolts six 3/8-16 × 1-1/2 in hex bolts, grade 5 (H08), washer under head and nut (H09), nylon-insert nuts (H10): four through rail A's vertical leg, two through rail B's leg with a fitting on each side

Torque 25 N·m (18 ft·lb); without a torque wrench, pull as hard as one hand can on a 15 cm wrench, then stop

Why a friction joint: each fitting leg has two 21 × 12 mm slots, centred 29.15 mm and 78.45 mm from the heel VERIFIED ; they leave each bolt 2.5 mm of play across and 11.5 mm along, and the rails' 7/16 in holes 1.6 mm, so only the clamping force keeps rail B's corner from settling

Order tighten only with the two surfaces flush and rail B's end touching rail A

The mending plates across the joint align the tops; they do not carry the corner. SOURCE-DEFINED

Troubleshooting

Symptom	Cause	Fix
Corner has dropped; step at the joint grew	The six fitting bolts lost clamping force (friction joint); supports settling	Bring the surfaces flush with the OLOV feet and the round feet, never by packing anything under the joining plates. Check rail B's end still touches rail A. Retighten all six bolts to 25 N·m. Write the new step down (Fig. J-5).
Top moves on its ALEX, or a pad spins or rocks	A pad is out of its ring or not bearing	Lift that ALEX end a few millimetres and slide the ALEX unit, not the top, until the pads drop into their rings. Each foot set with the 102 mm foot height gauge, no rock. Level along x and y; re-check 83.0 cm at the corners (Figs. J-1, J-4).
Printed ring cracked or its screws loose	Ring wall overloaded by a sideways push	Inspect at two weeks. A crack: reprint the ring (parts/structure/foot-ring.stl), PETG or ABS, flange-down: 52 mm bore, 3 mm wall, 75 × 2.5 mm flange. Refit with #6 × 5/8 in flat-head screws (H06), 5/64 in pilots 12 mm deep (12 ALEX puck). Never run the desk with a ring missing.
Leg moves or wobbles	OLOV plate screws loose	Retighten the five M4 × 16 mm screws (H01). A screw that spins: repair the hole (below), re-pilot 1/8 in, 15 mm deep (15 LEG puck). Re-set the OLOV to 80.2 cm, or the ALEX-feet table's number.
Monitor clamp creeps	Clamp loosened over time	Retighten the clamp with the crossbar fitted. Keep the steel plate (P11) flat on its outline under the jaw. Never add a soft pad under the jaw.
Screw spins (stripped)	Hole worn in the particleboard	Back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot.
A mark is off by more than 2 mm	Marking error	Redraw it. A stray 1/8 in hole in the underside does no harm.
A fitting bolt will not go in	Holes not quite in line	Open the hole in the rail with a round file; never force it.

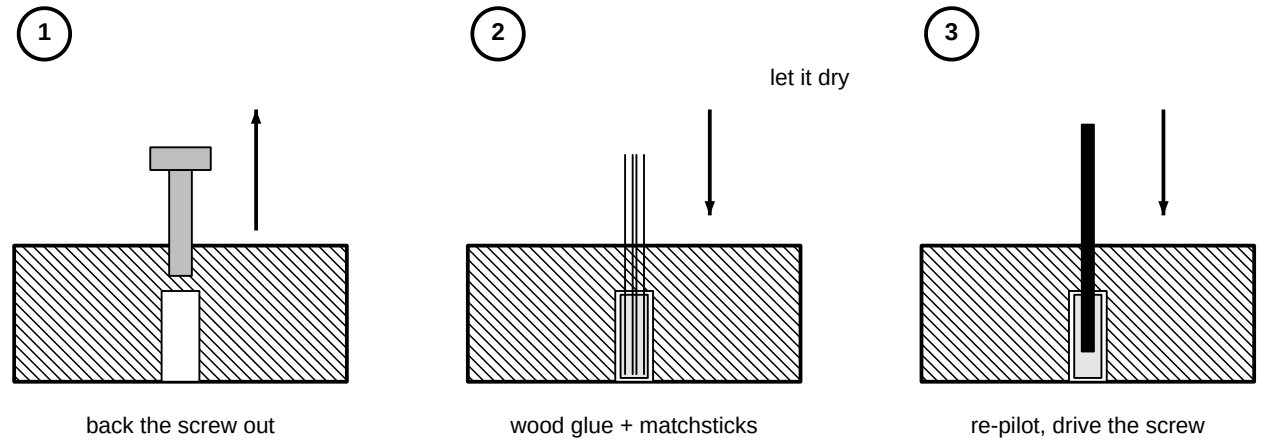


Fig. K-1 A screw that spins: back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot with the step's own bit and depth.

Every fix above is from the build manual and the build guide SOURCE-DEFINED.

Appendix · Outstanding measurements

These values were not measured on the purchased parts when this manual was built. Each is marked MEASURE ON HARDWARE at its point of use. Measure it before the step that uses it; if it differs, the step's own warning panel says what changes.

Measurement	Expected	Used in
ALEX bottom panel thickness (only for the optional ALEX feet)	not yet known	Optional ALEX feet: three #6 × 5/8 in screws each. (The top panel is measured: 15.9 mm, VERIFIED.)
ALEX height, with and without its stock feet	70.0 cm (docs assume)	Underside 80.2 cm; OLOV setting 80.2 cm; surface 83.0 cm (check 1)
Feet pad diameter (calipers)	50 mm; rings bored 52 mm	Printed rings P10 before printing; check 4
EKBACKEN thickness (calipers)	28 mm	Every screw into a top (never deeper than 21 mm; if the top is 27 mm, shorten every screw by 1 mm); clamp opening 31 mm (chapter I)
VIVO top arm reach and maximum opening	reach about 89 mm; opening 31 mm or more	Chapter I: clamp zones, Fig. I-2
Delivered rail lengths and angle thickness	213 cm layout (bought as 84 in, 213.4 cm); 4.8 mm	Build manual step 1 hole positions; rail pilot depth 25 mm from the steel face
Room: wall to wall, and baseboard thickness	252.0 cm and 317.0 cm	Build manual step 0 (252 × 317 cm of floor, including the 3 cm wall gaps); step 8b wall-gap trim
Feet load rating	unknown; the design wants 50 kg or more each	Load path through the eight feet; check 3 corner load test

Confidence labels

VERIFIED	measured on the purchased part, or taken from its own geometry file.
SOURCE-DEFINED	defined by the project documents (docs/build-manual.md). A source-defined dimension is never called verified.
ASSUMPTION / FIELD CHECK	engineering estimate or field check.
MEASURE ON HARDWARE	not yet measured: measure it before the step that uses it.

Quality control

Every number in this manual comes from manual/data.json (extracted from guide.html's D object), manual/facts.json or docs/build-manual.md. Source: HEAD of https://github.com/scotthowson/desk-setup on 23 Oct 2026 (supersedes a2e99ea with measured facts). If you change a measured value, change it in the docs first, then rebuild the manual.